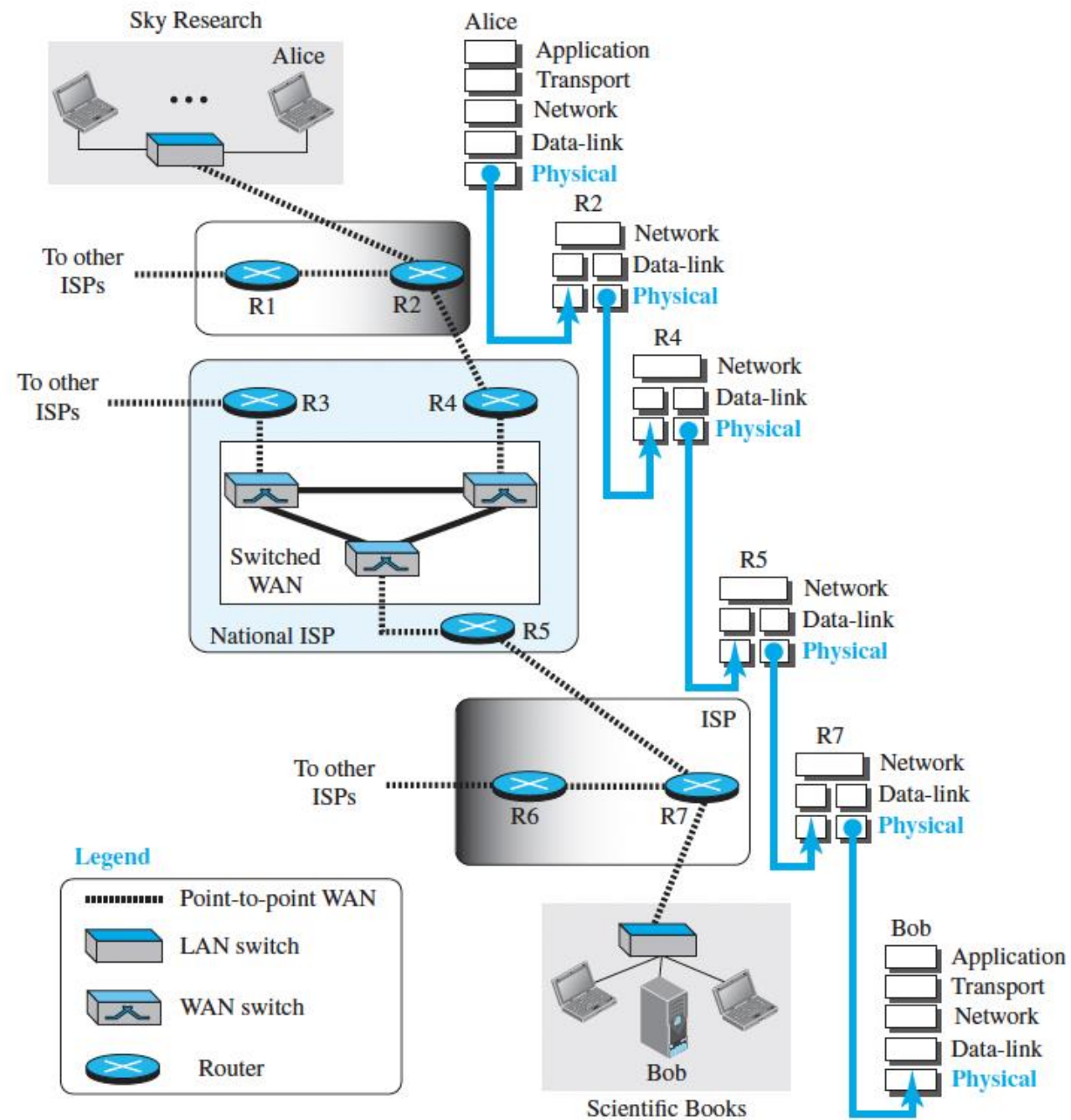


PHYSICAL LAYER

Unit 2

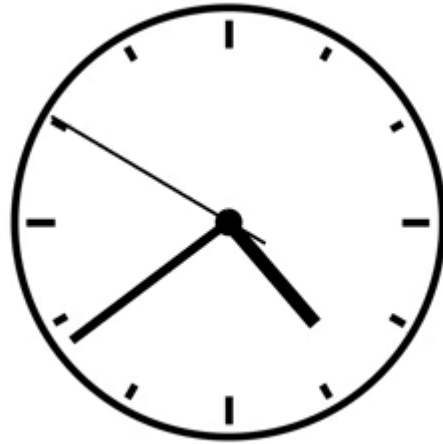


Data and Signals

- Communication at the physical layer means exchanging *signals*.
- Data need to be transmitted and received, but the media have to change data to signals.
- Both data and the signals that represent them can be either **analog** or **digital** in form.

Analog and Digital Data

- **Analog data** - information that is continuous
- **Digital data** - information that has discrete states



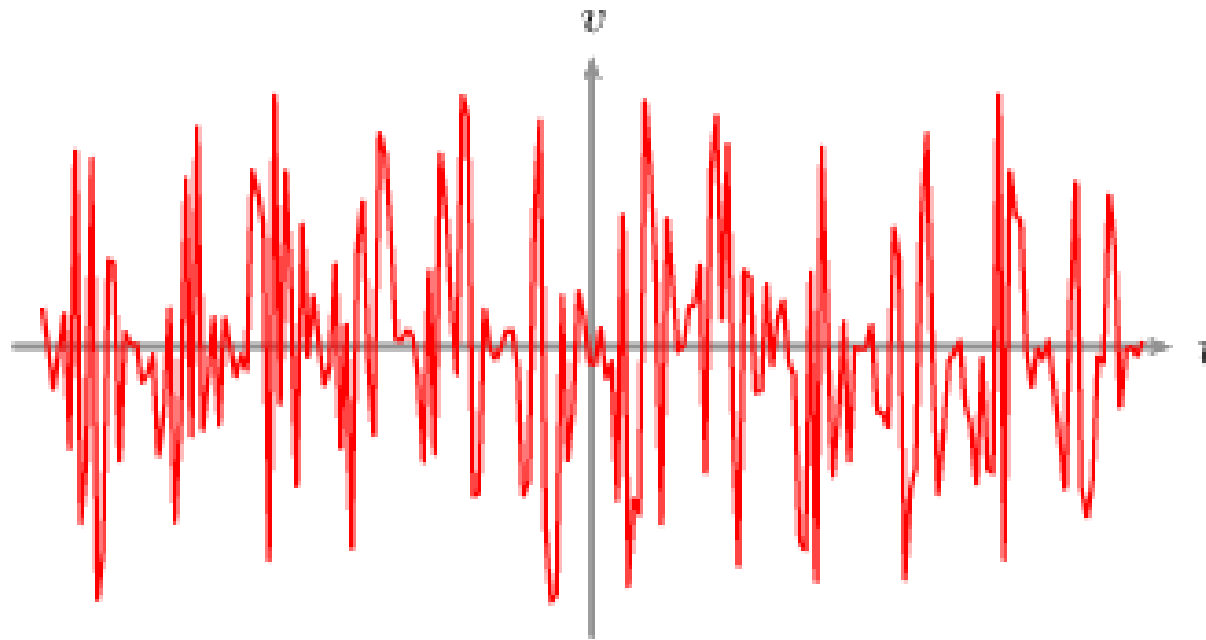
Analog and Digital Data



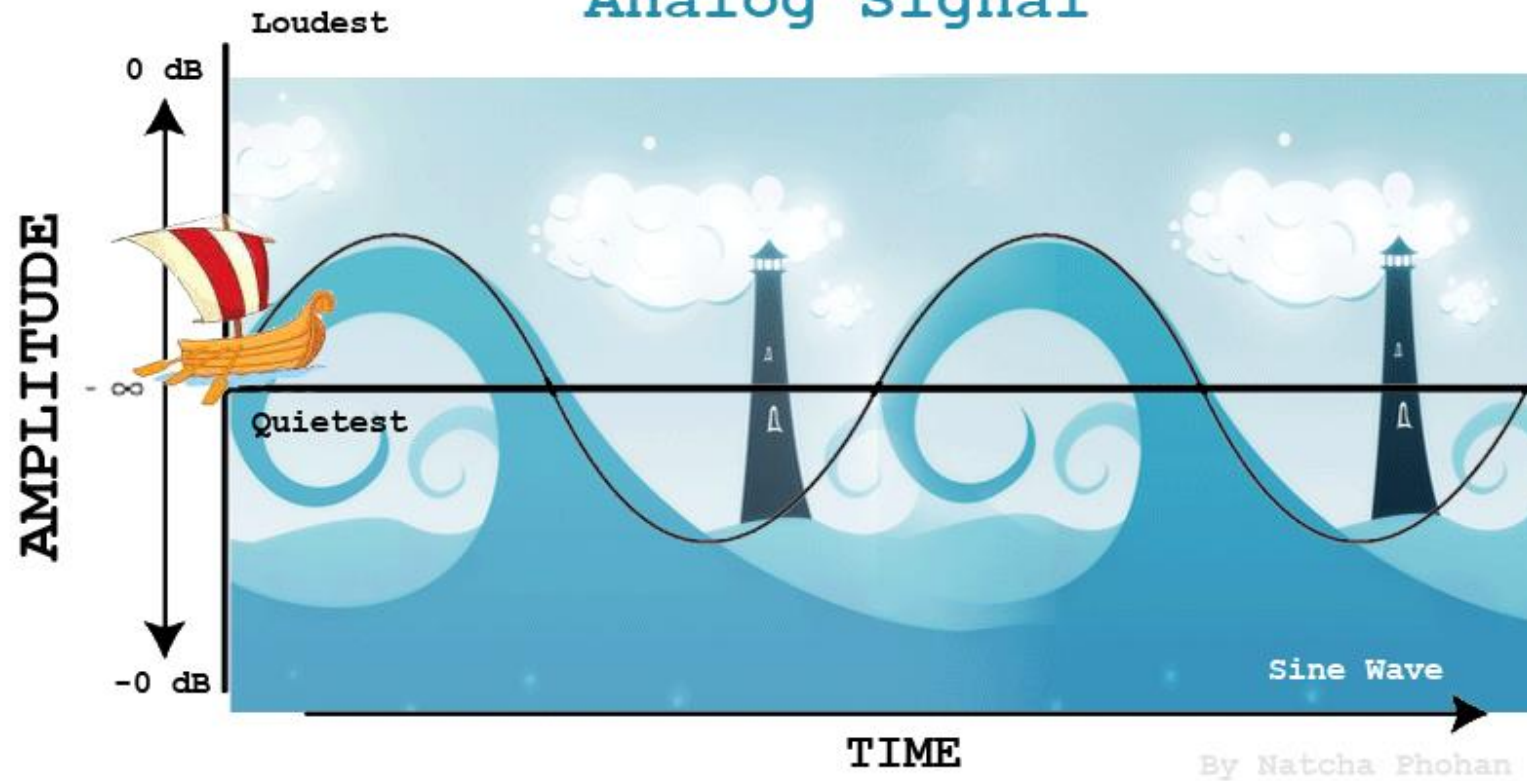


Analog Signals

- An **analog signal** has infinitely many levels of intensity over a period of time.

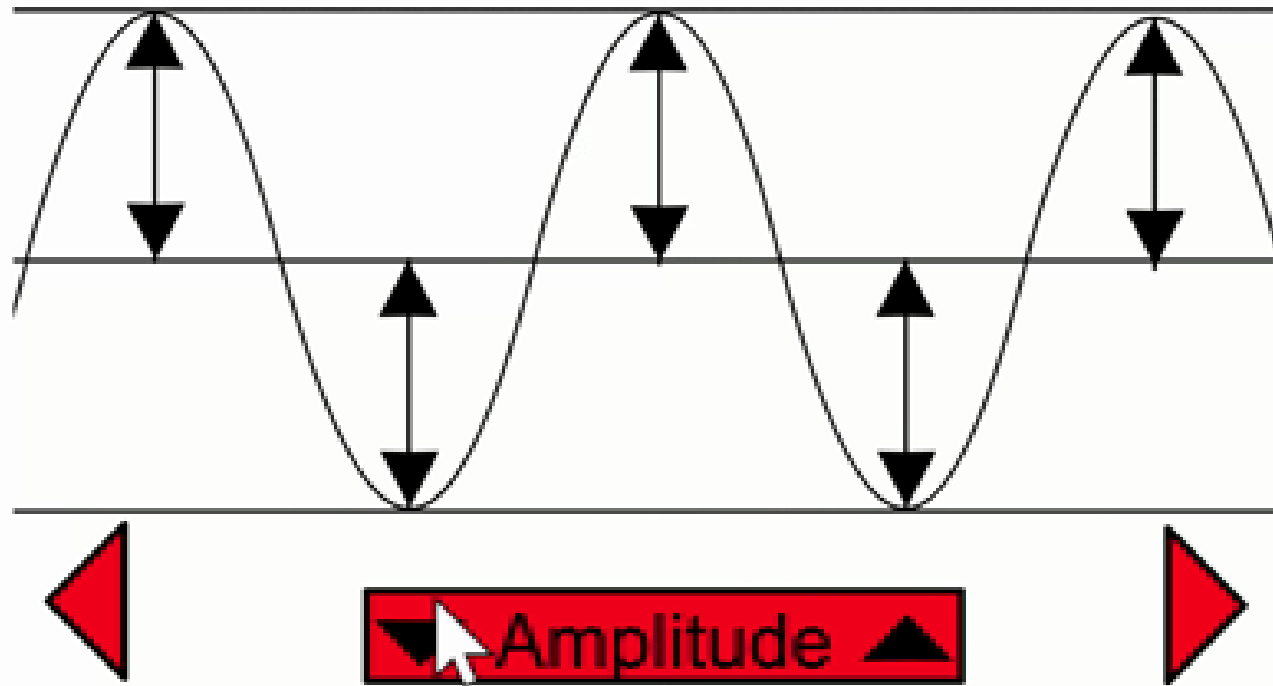


Analog Signal

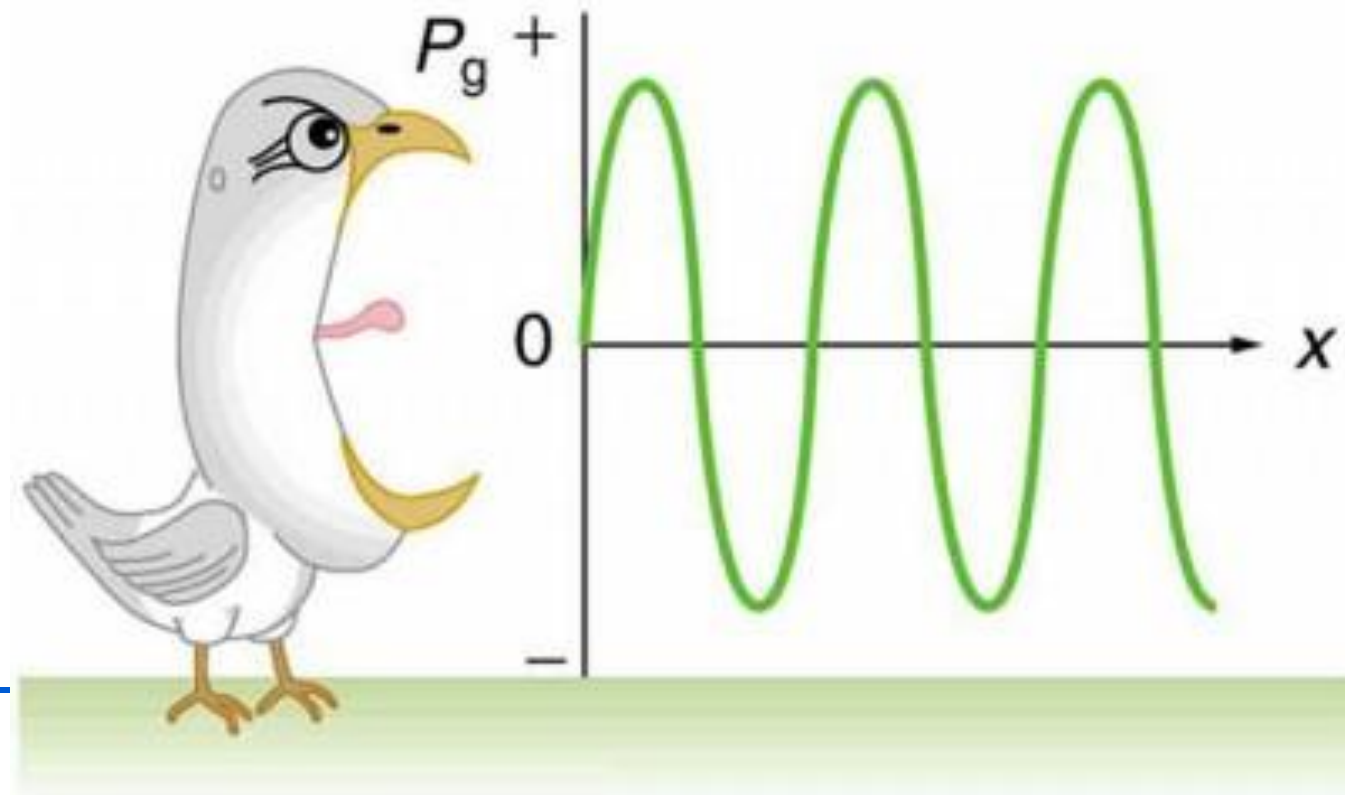
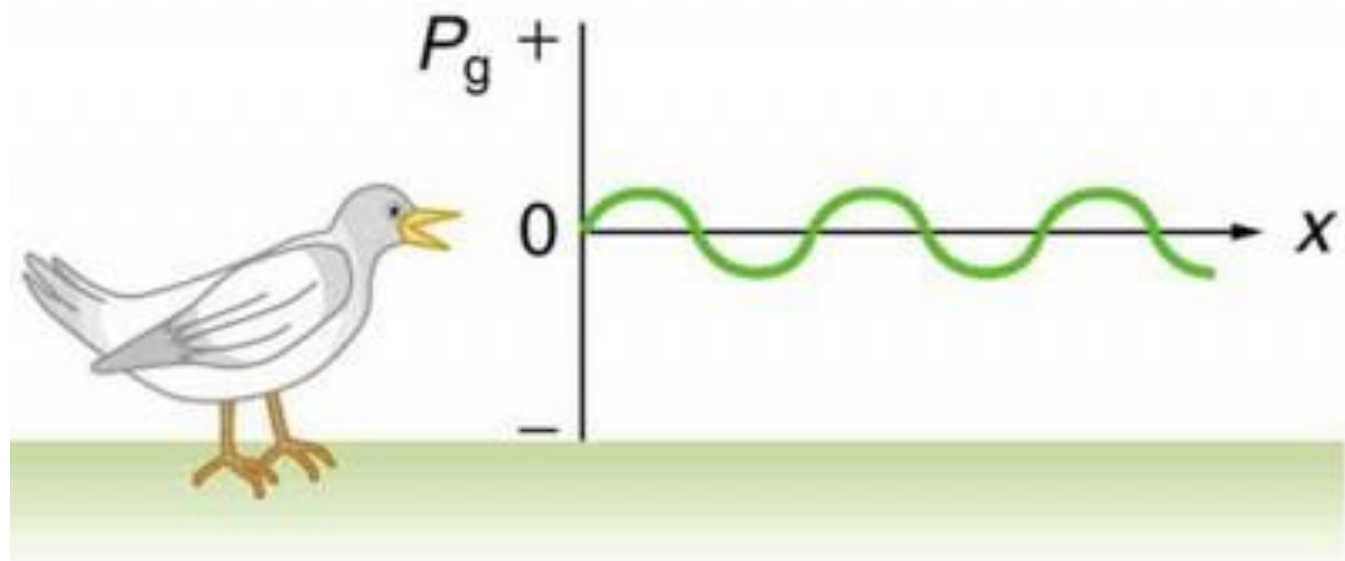


Amplitude

Amplitude

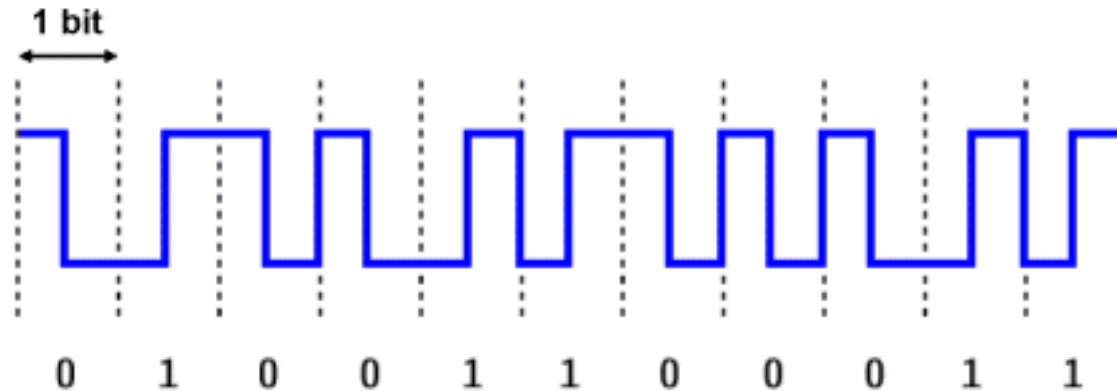


The distance between the peak or trough of the wave and the equilibrium position, or the position of the medium at rest.



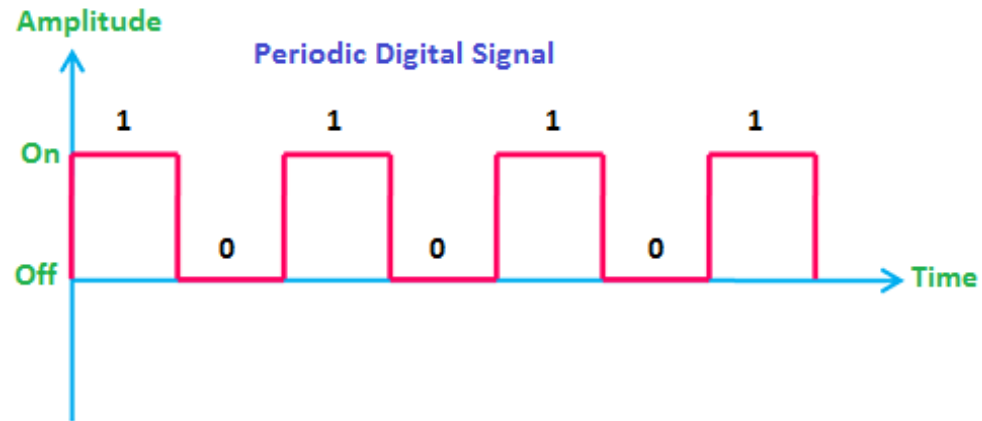
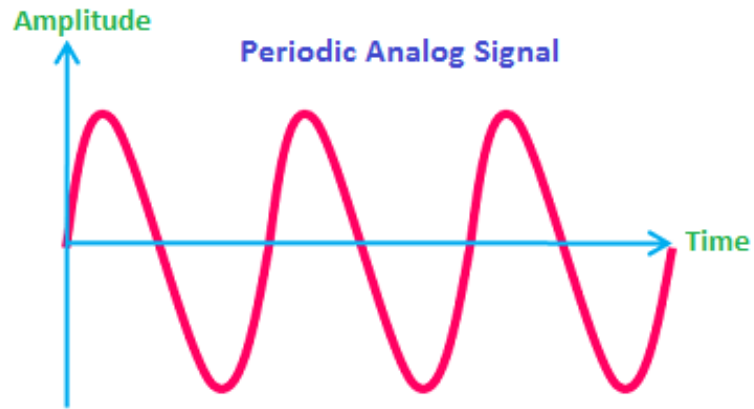
Digital Signals

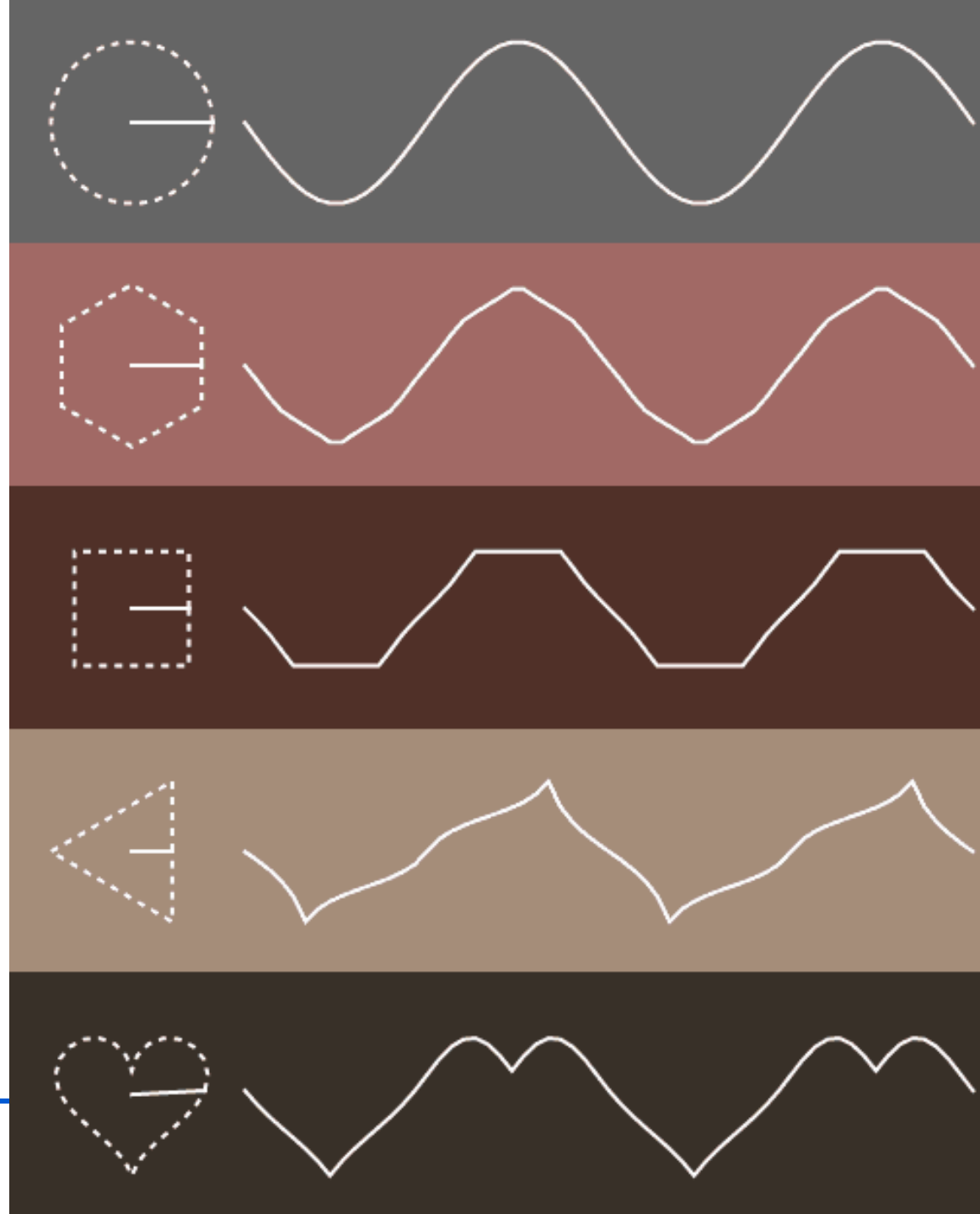
- A **digital signal** can have only a limited number of defined values, it is often as simple as 1 and 0.



Periodic and Nonperiodic Signals

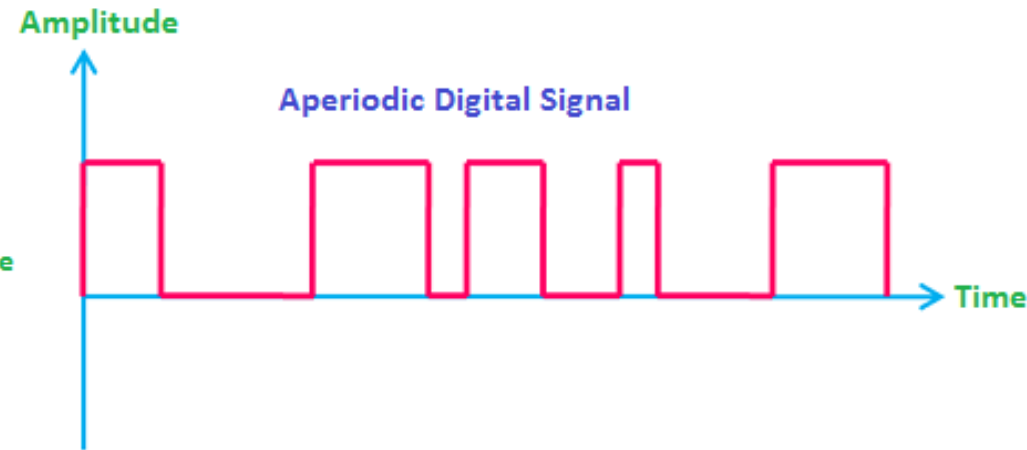
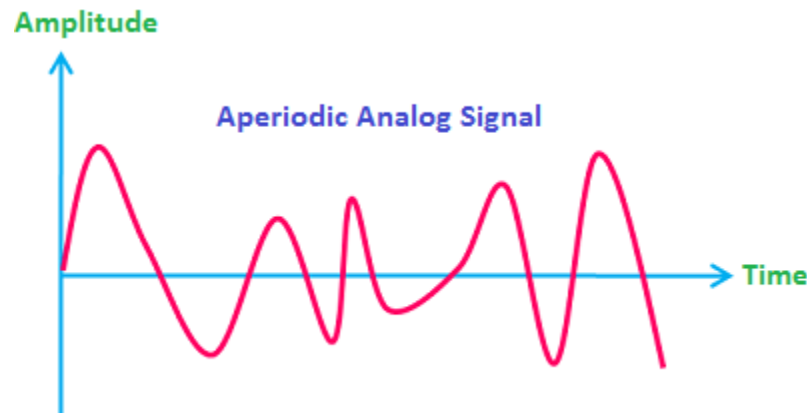
- **Periodic signal** completes a pattern within a measurable time frame, called a **period**, and repeats that pattern over subsequent identical periods.
 - The completion of one full pattern is called a **cycle**.



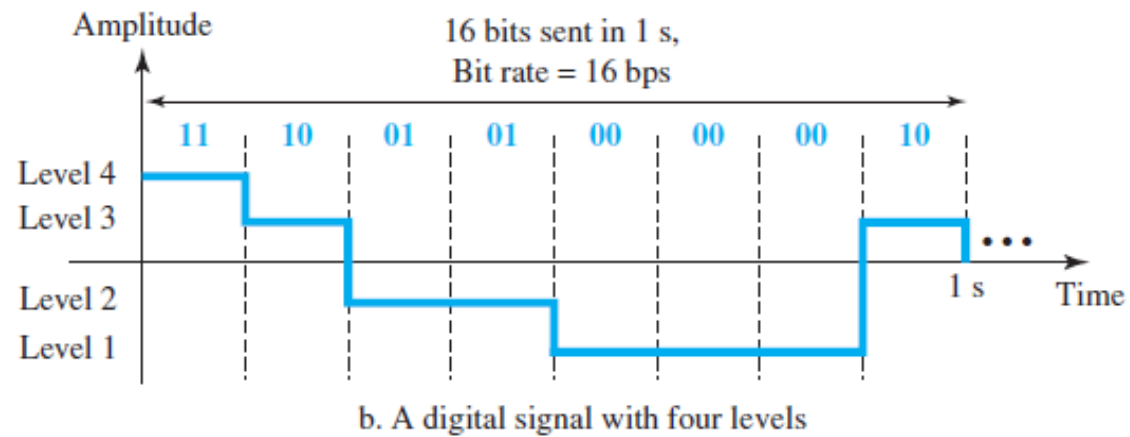
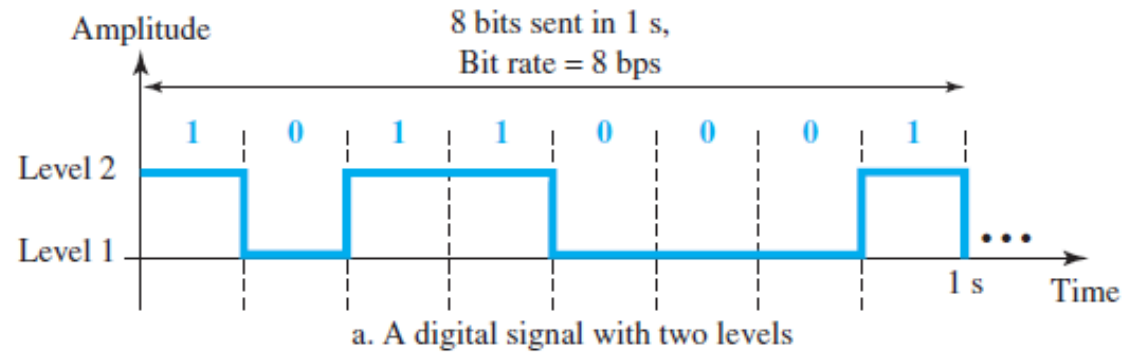


Periodic and Nonperiodic Signals

- **Nonperiodic signal** changes without exhibiting a pattern or cycle that repeats over time.
- In data communications, **periodic analog signals** and **nonperiodic digital signals** are commonly used.



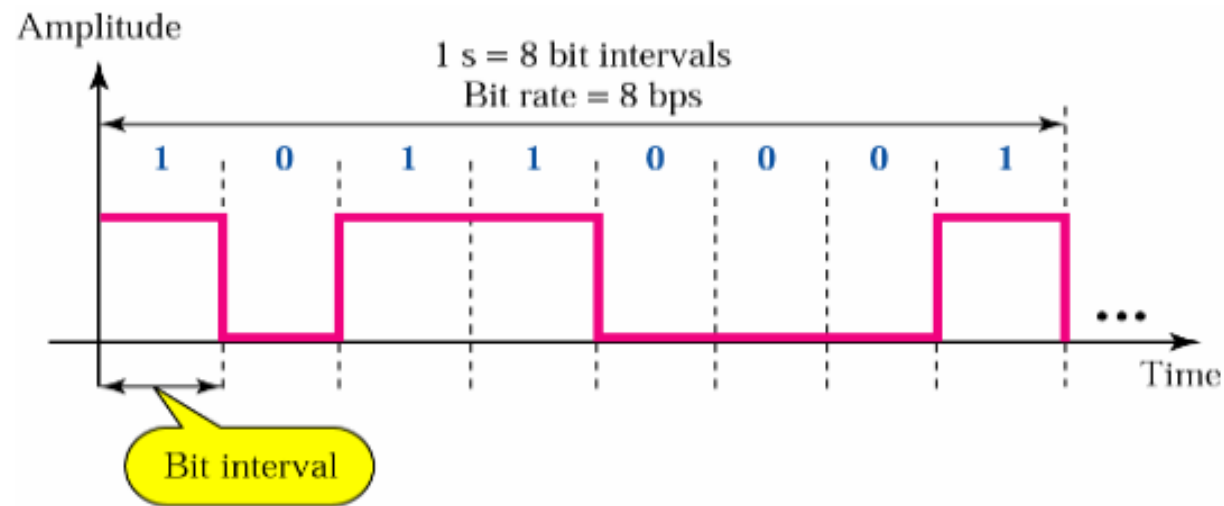
Digital Signals



If a signal has L levels, each level needs $\log_2 L$ bits.

Bit Rate

- Number of bits sent in 1 second, expressed in **bits per second (bps)**.

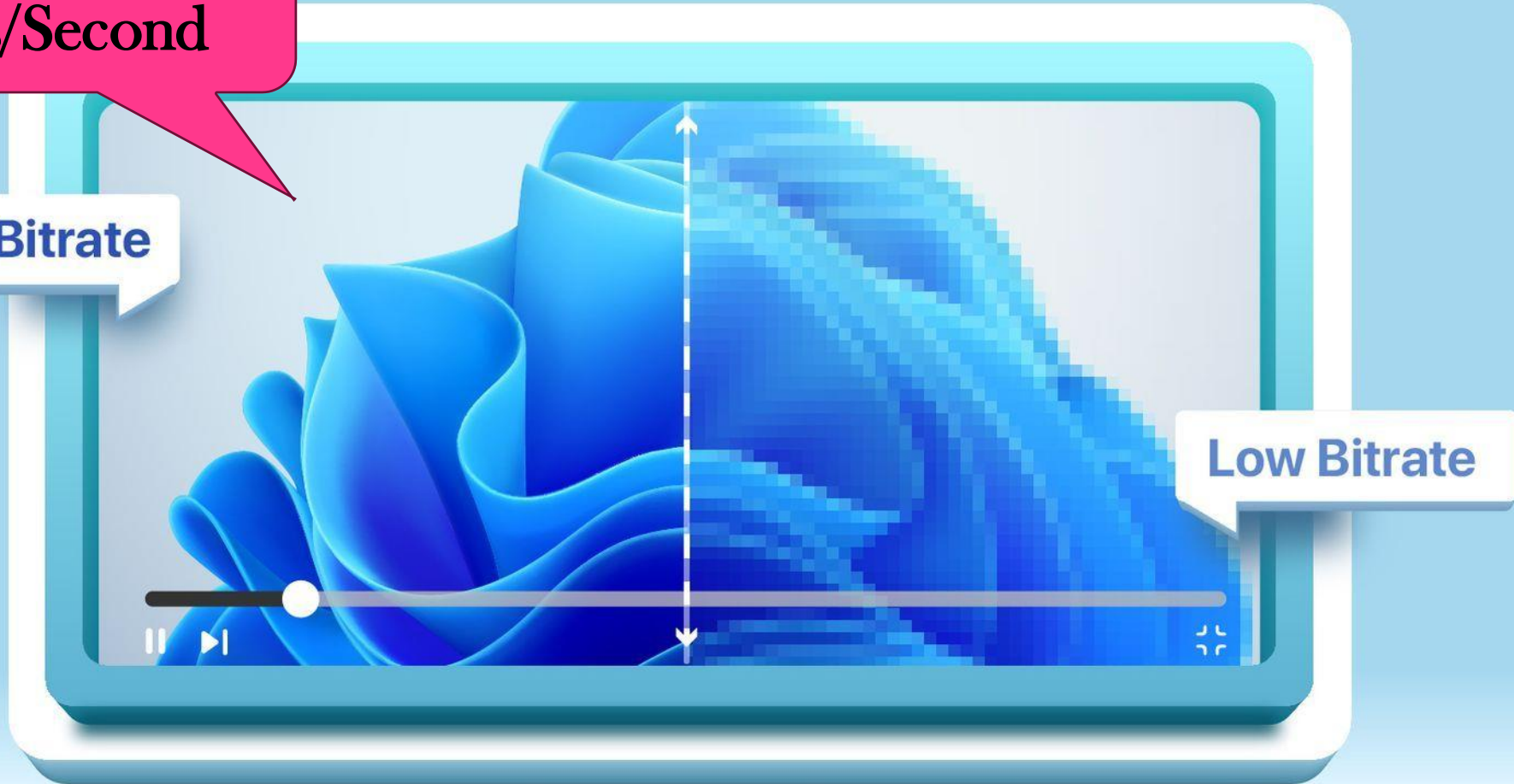


Video Bitrate

65000000
Bits/Second

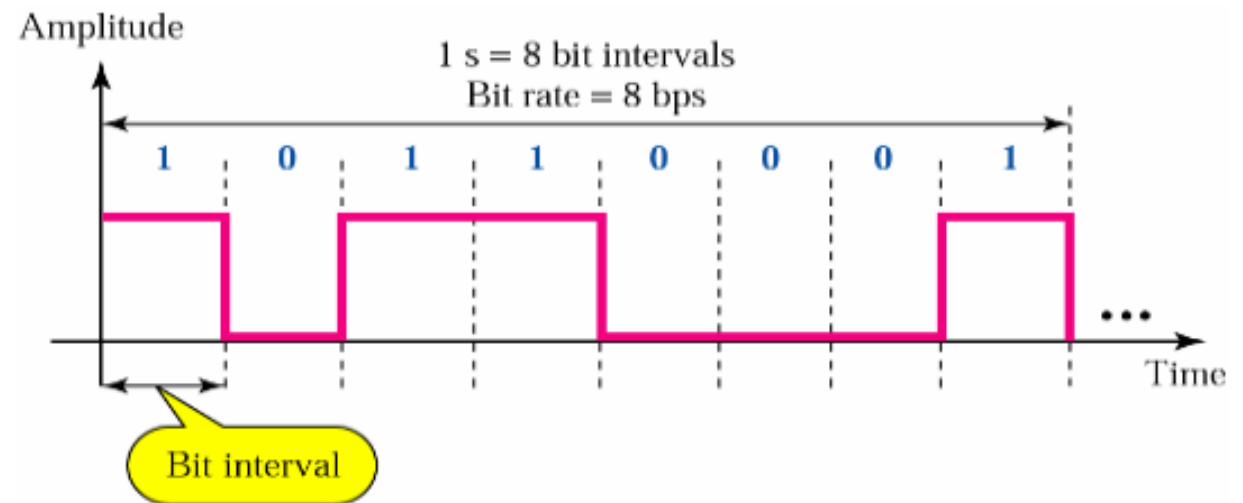
High Bitrate

Low Bitrate



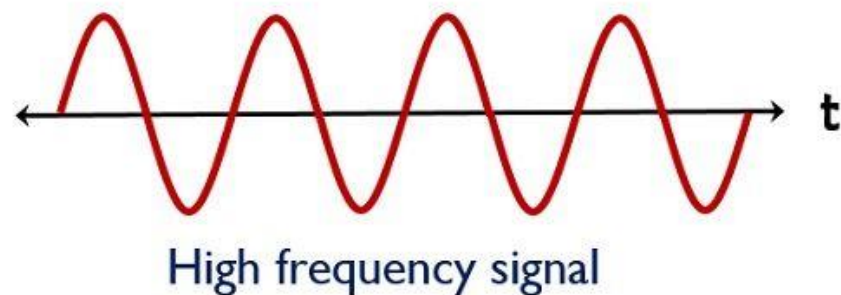
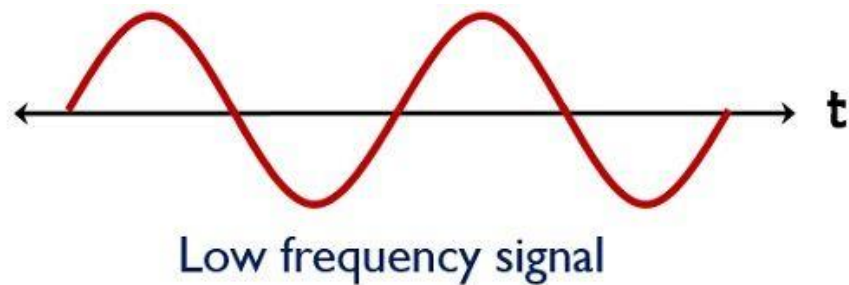
Bit Length

- The distance one bit occupies on the transmission medium.
- Bit Interval (Amount of time required to transmit one bit) $\times$ Propagation Speed
- Analogous to Distance = Time $\times$ Speed
- Bit Rate = 8 bps
- Bit Interval = $1/8 = 0.125$ s
- Example: If the propagation speed is 2×10^8 m/s
- Bit Length = $0.125 \times 2 \times 10^8$
- = 25,000,000 m (or) 25,000 km



Terms

- Frequency: The number of waves that pass a fixed point in unit time
- Bandwidth: The data transfer capacity of a network in bits per second (Bps). In terms of frequency, measured in Hertz (hz)

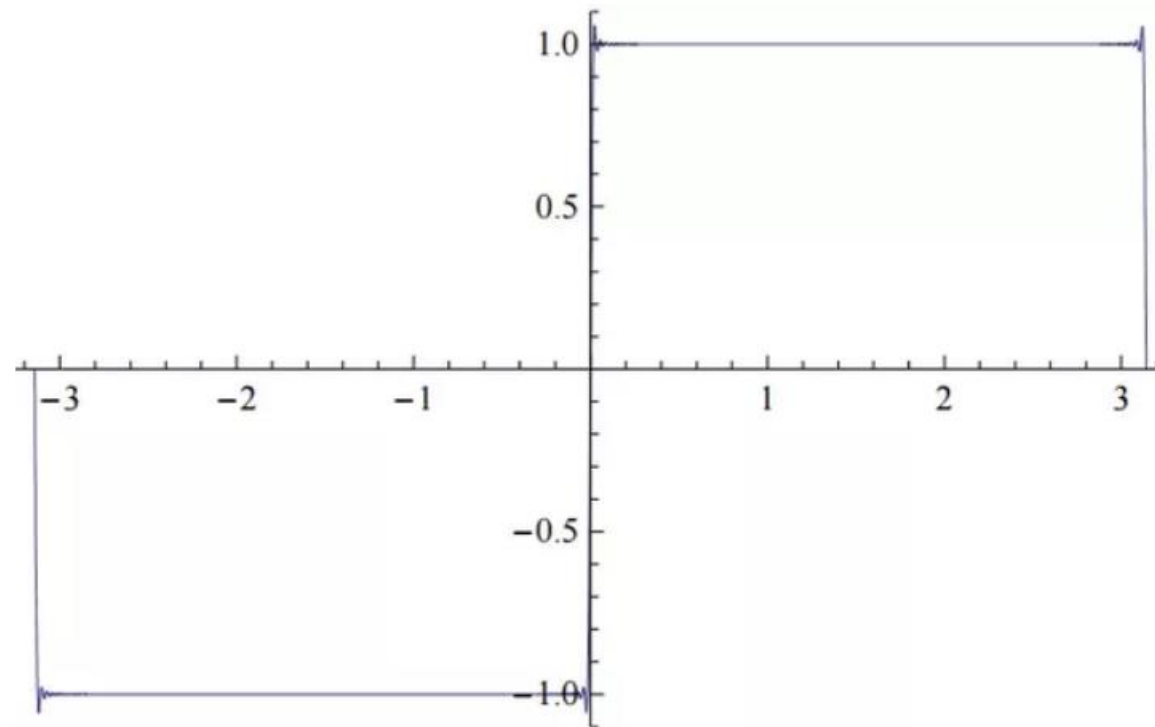


Circuit Globe

Unit 2

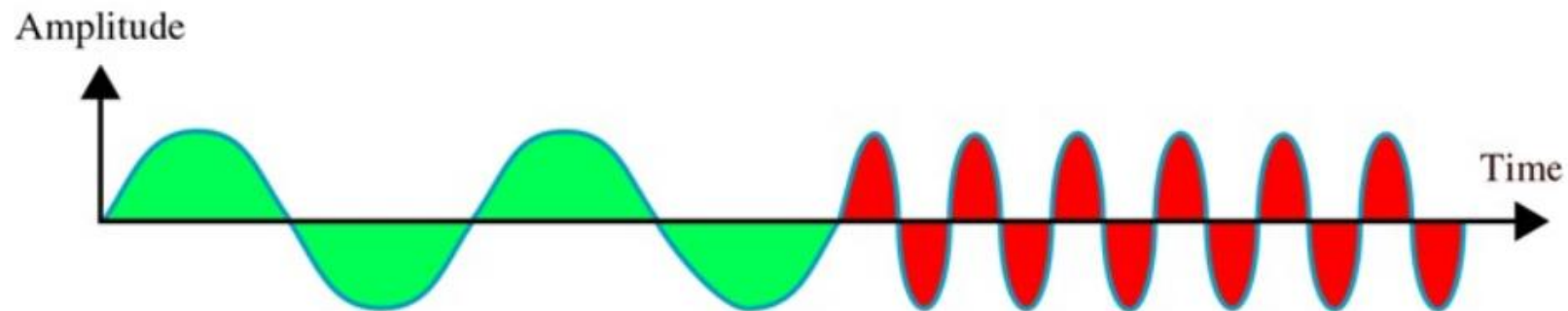


Digital Signal as Composite Analog Signal

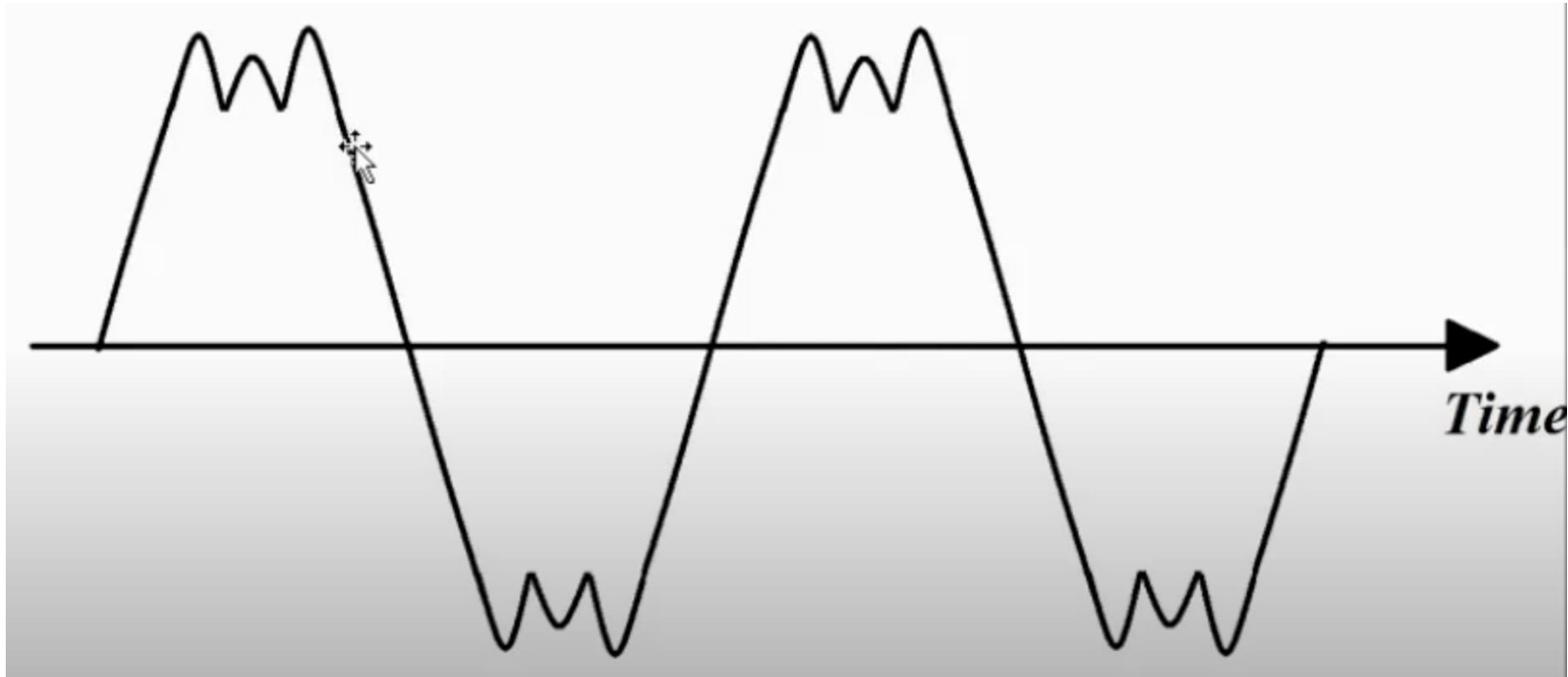


Digital Signal as Composite Analog Signal

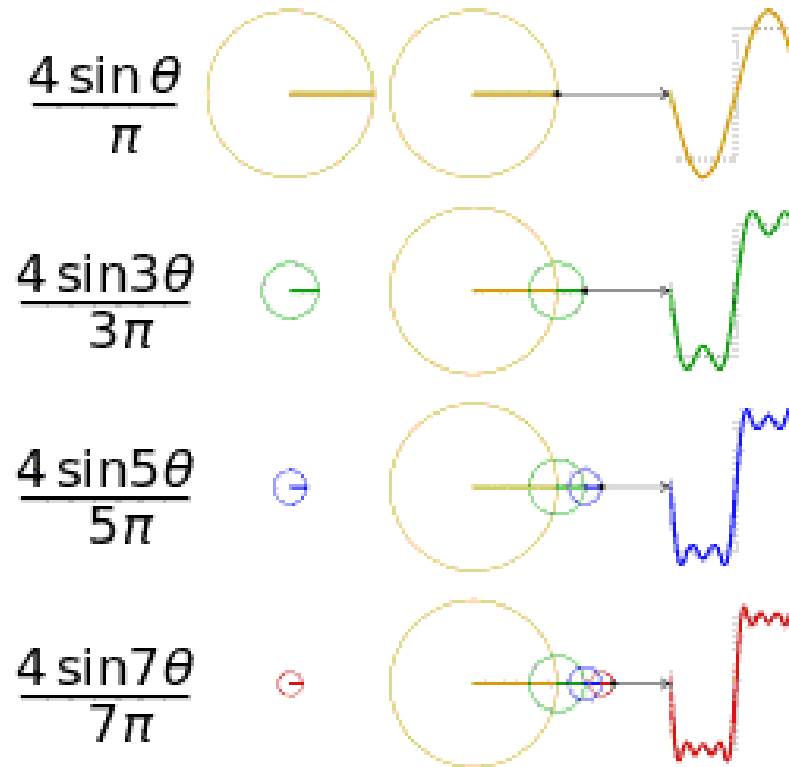
- Frequency
 - Rate of change with respect to time
 - Change in short span of time is high frequency
 - Change in long span of time is low frequency



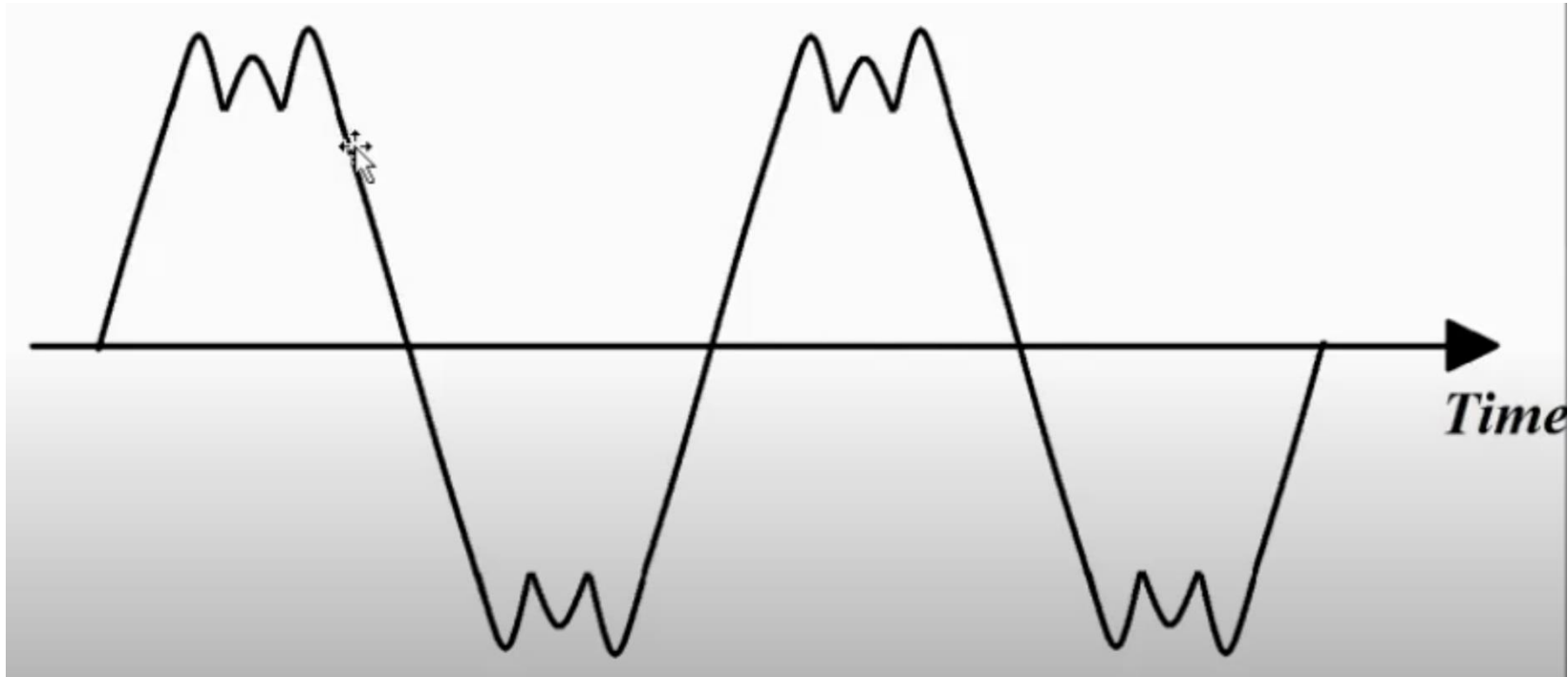
Composite Periodic Signal



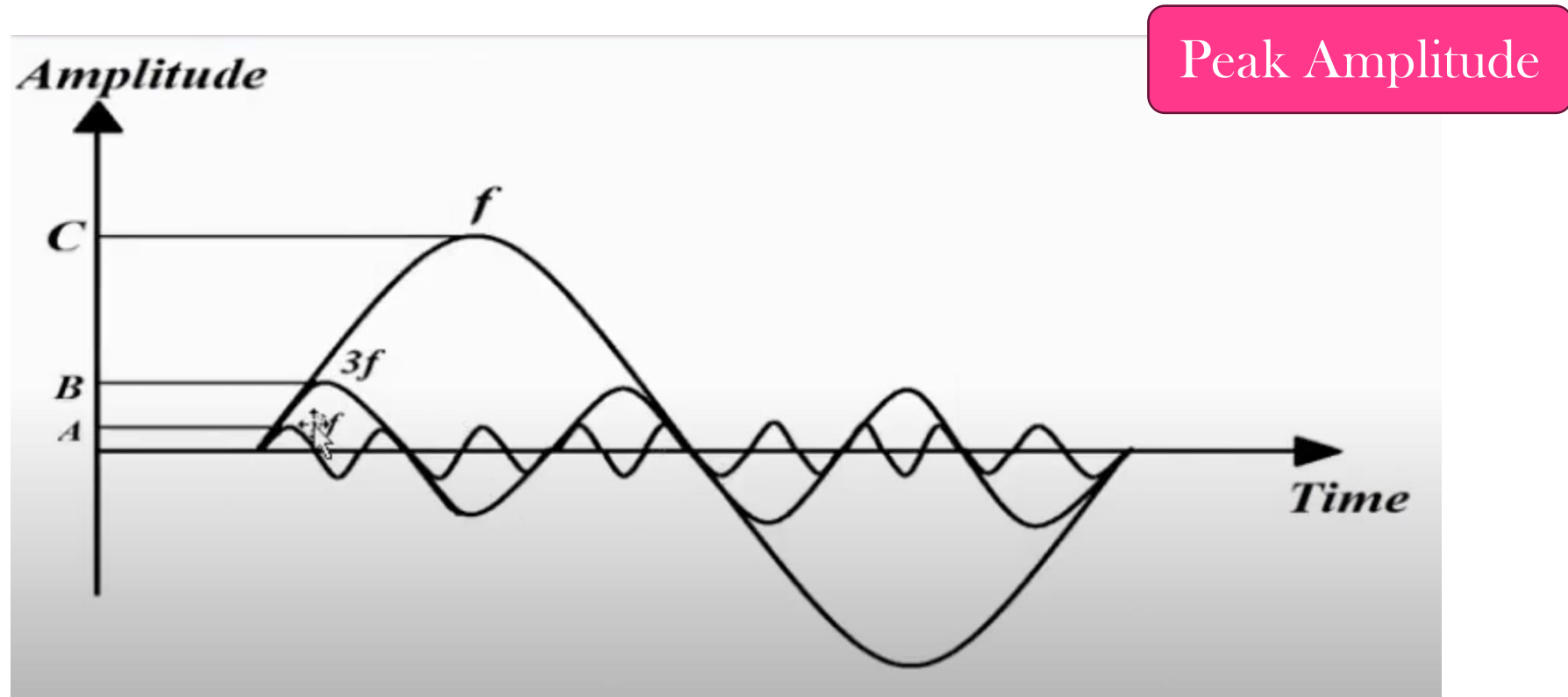
Digital Signal as Composite Analog Signal



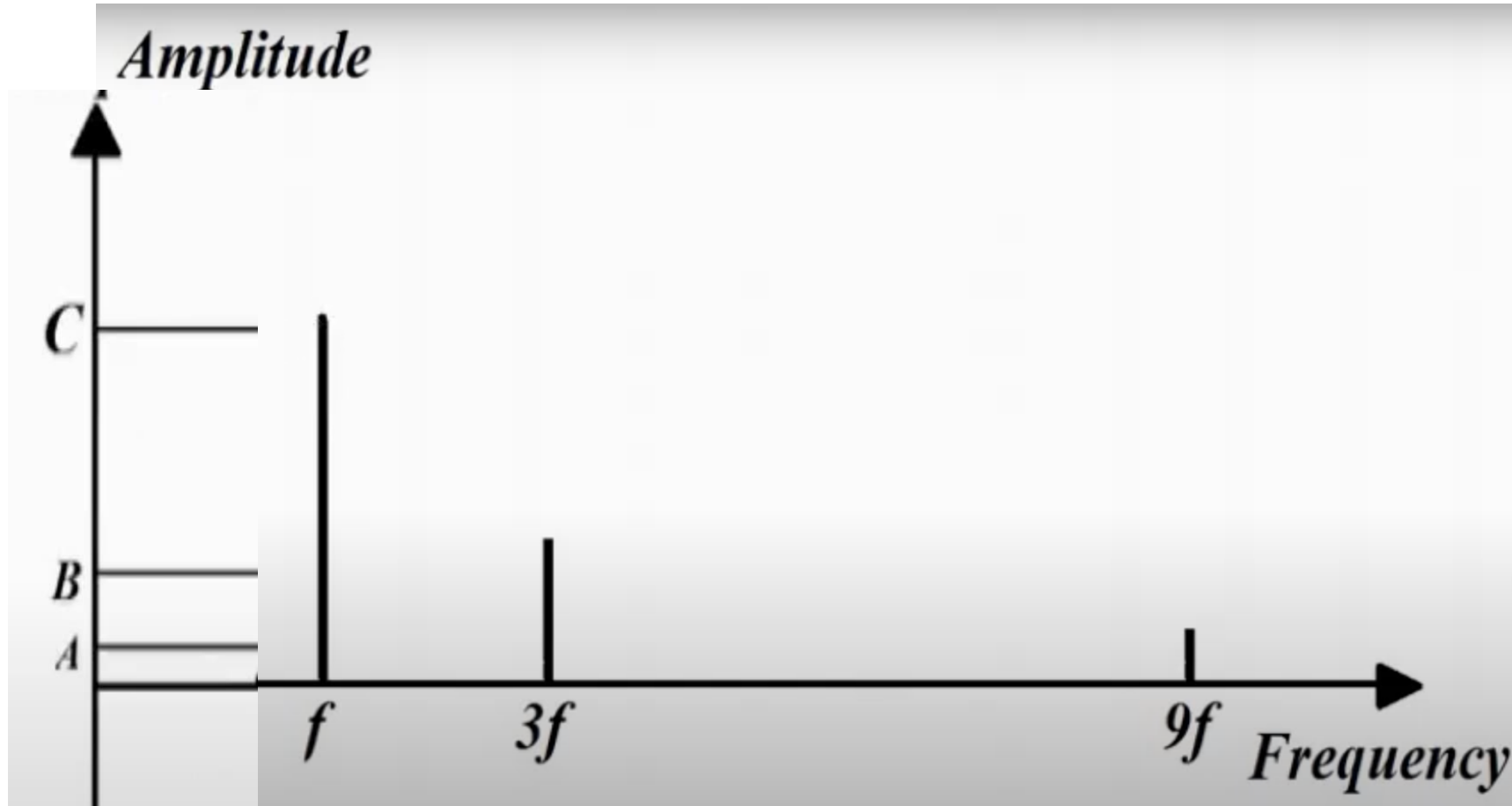
Composite Periodic Signal



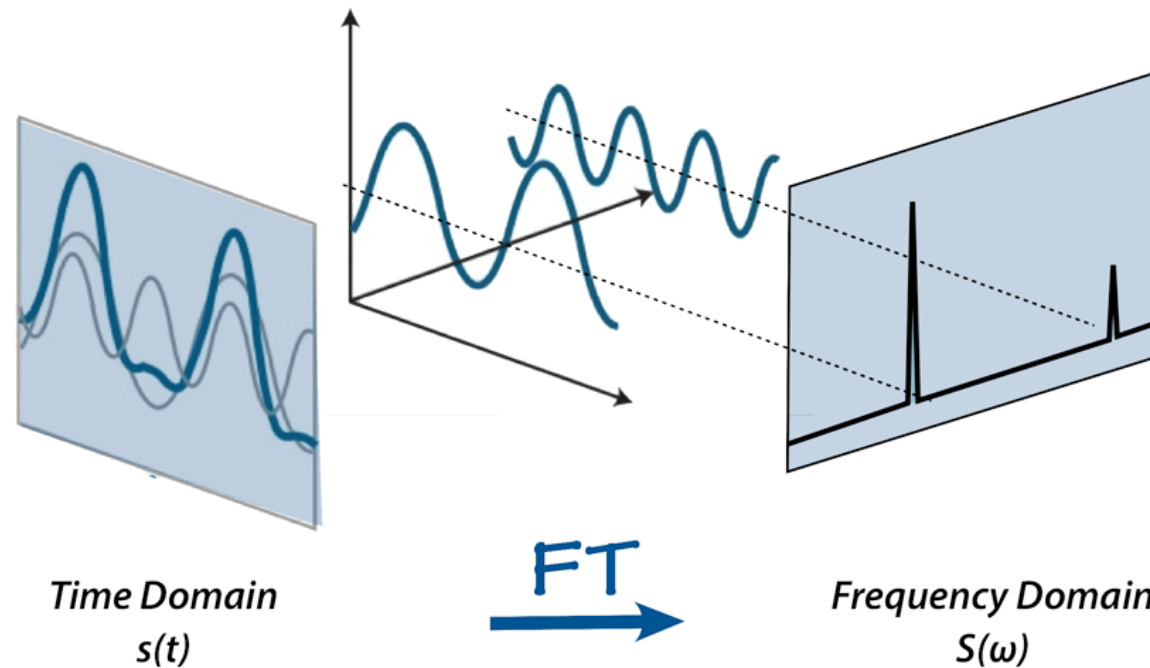
Decomposition of Composite Periodic Signal



In Frequency Domain



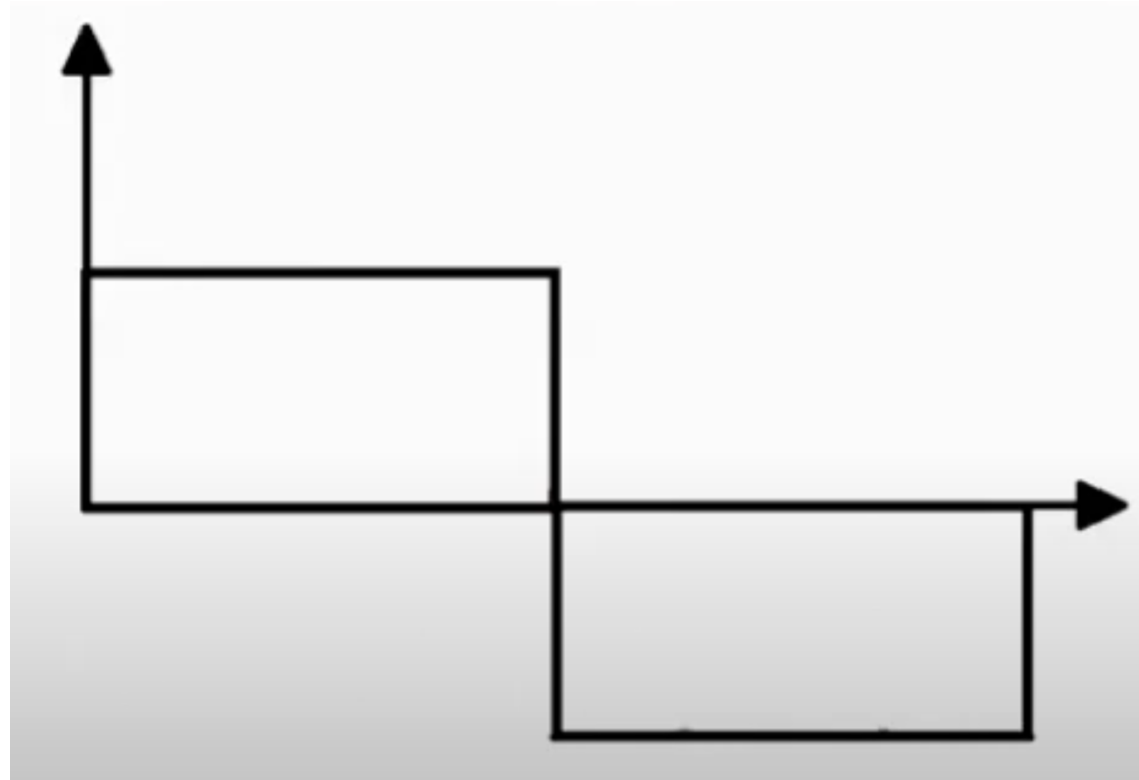
Digital Signal as Composite Analog Signal



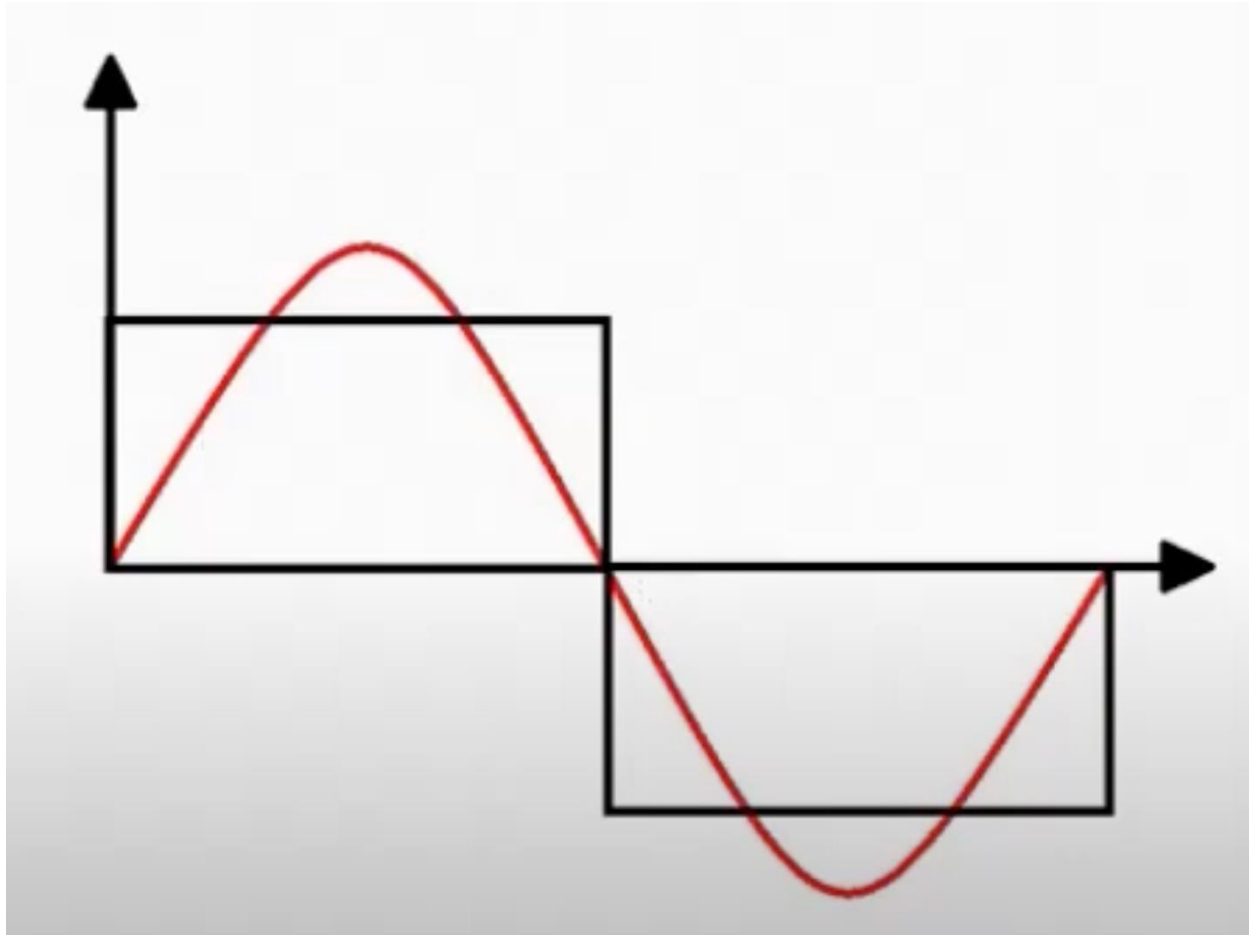
Instantaneous amplitude
with respect to time

Maximum amplitude with
respect to frequency

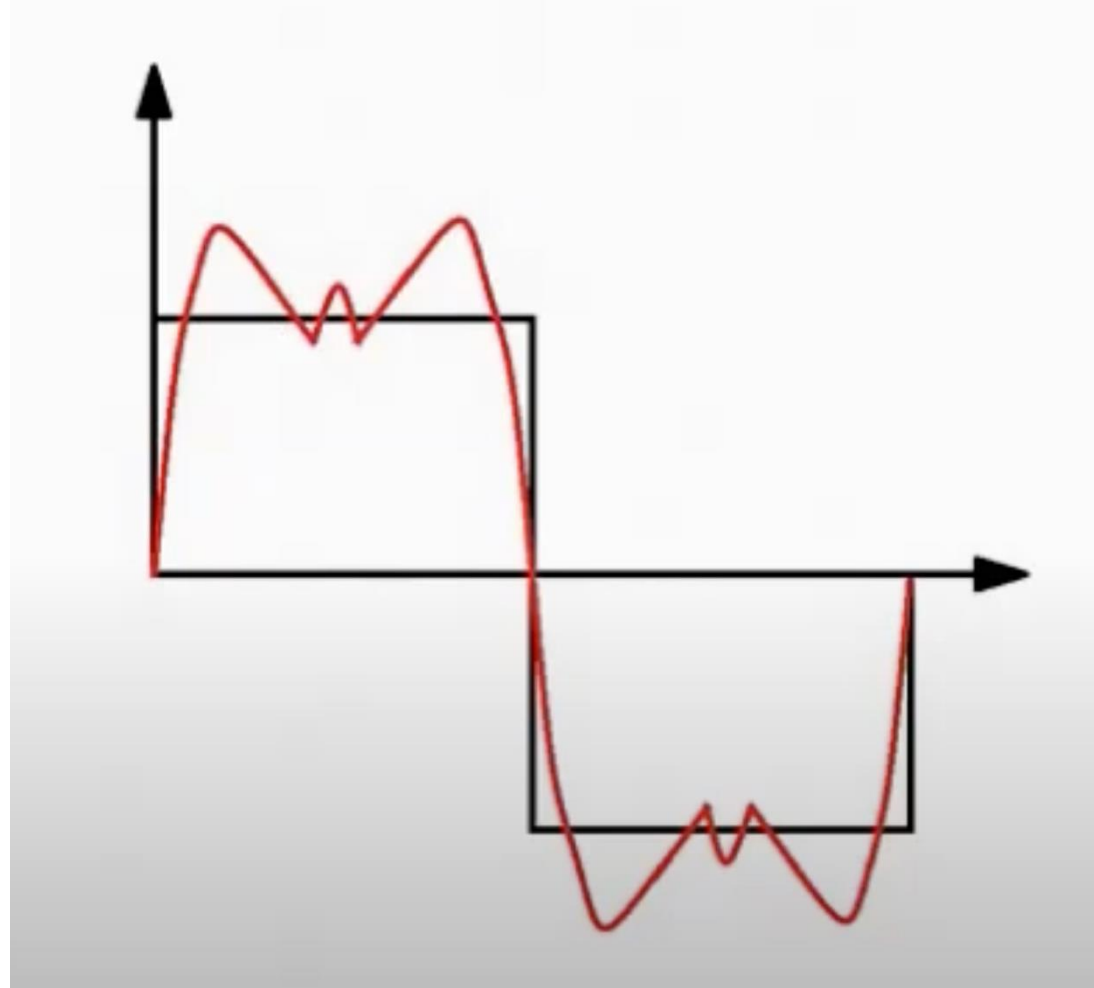
Consider this Digital Signal

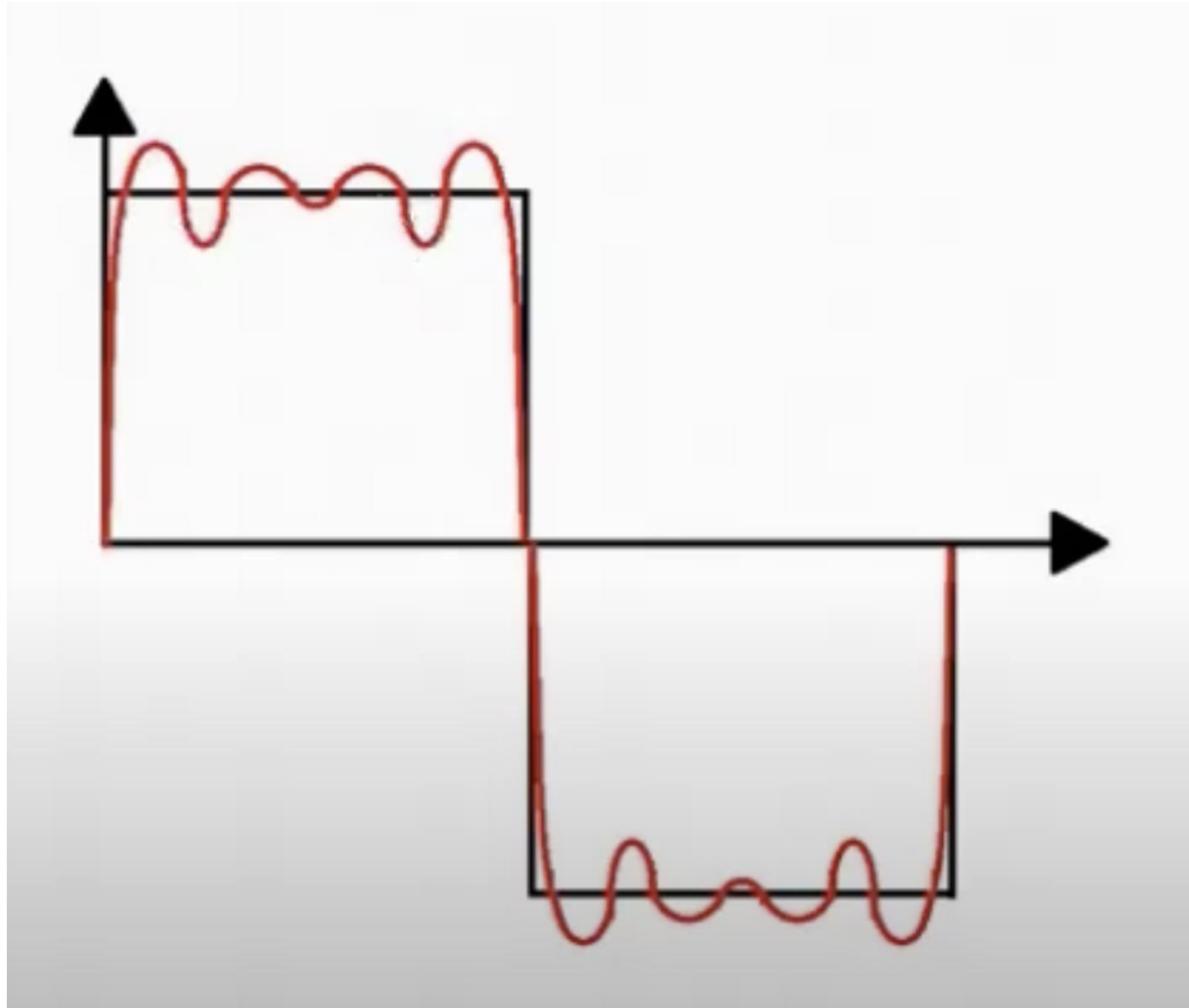


Representing using Simple Sinusoidal Signal

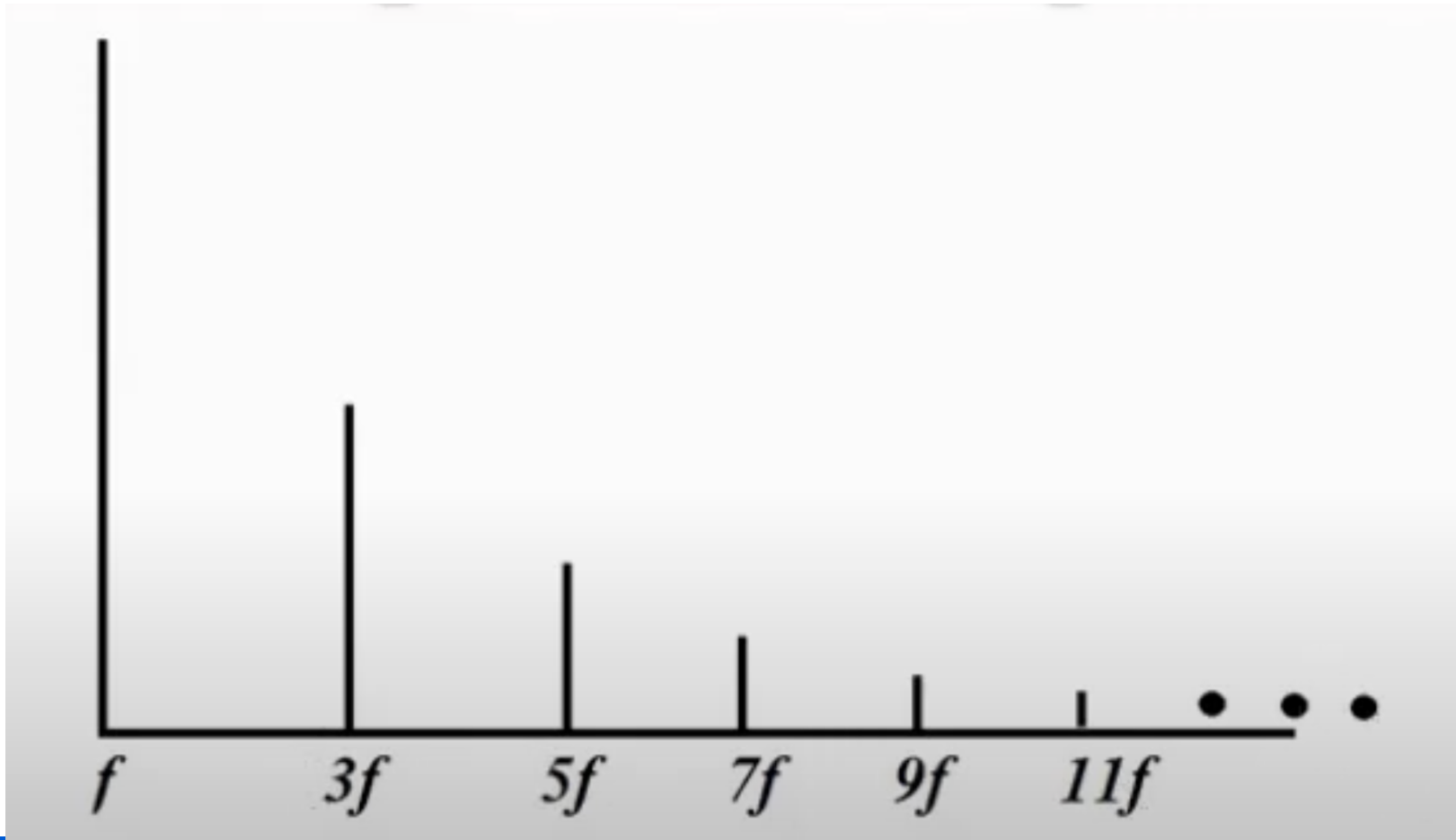


Representing using Composite Sinusoidal Signal

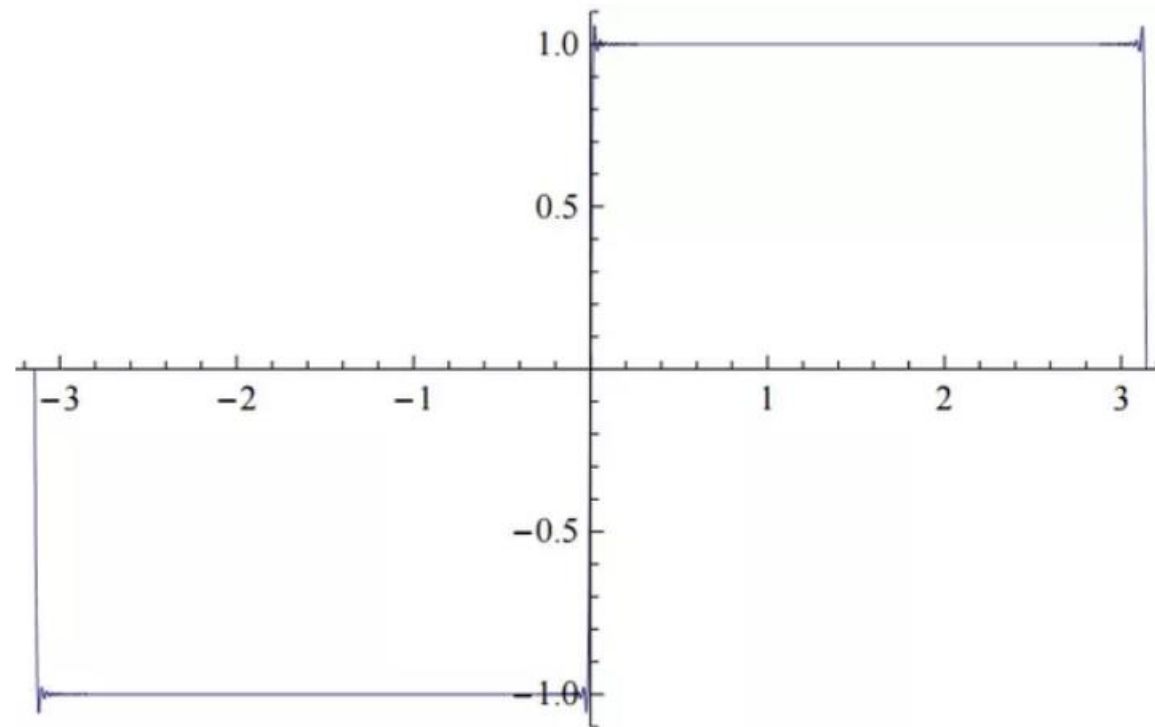




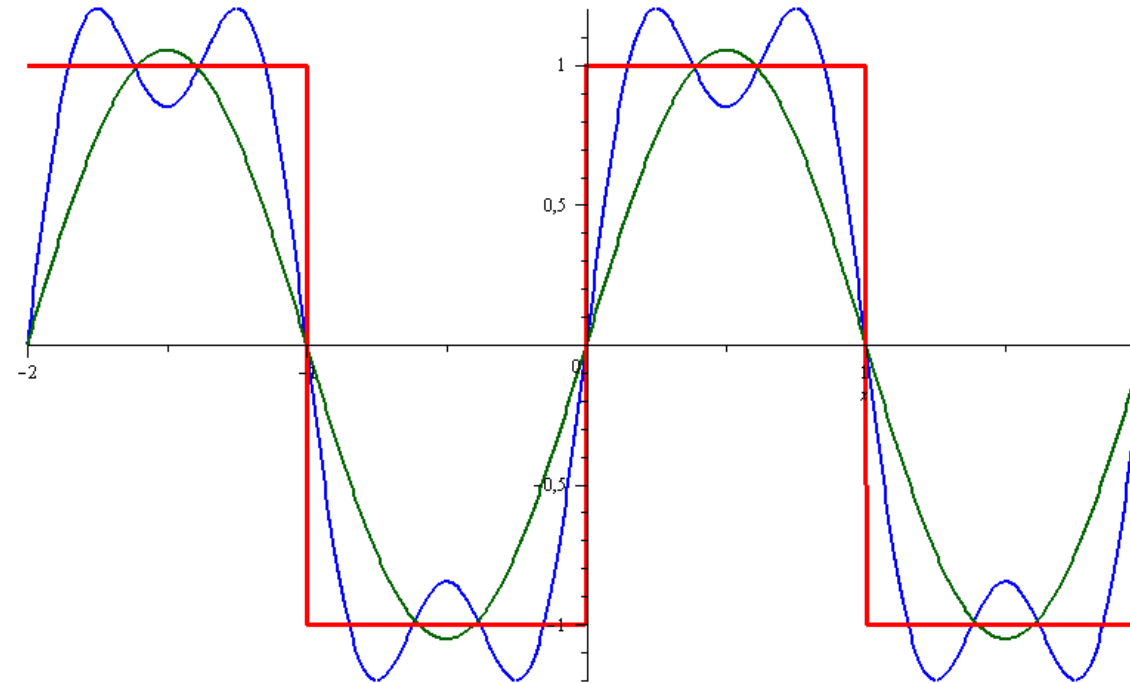
Frequency Domain Representation of a Digital Periodic Sign



Digital Signal as Composite Analog Signal



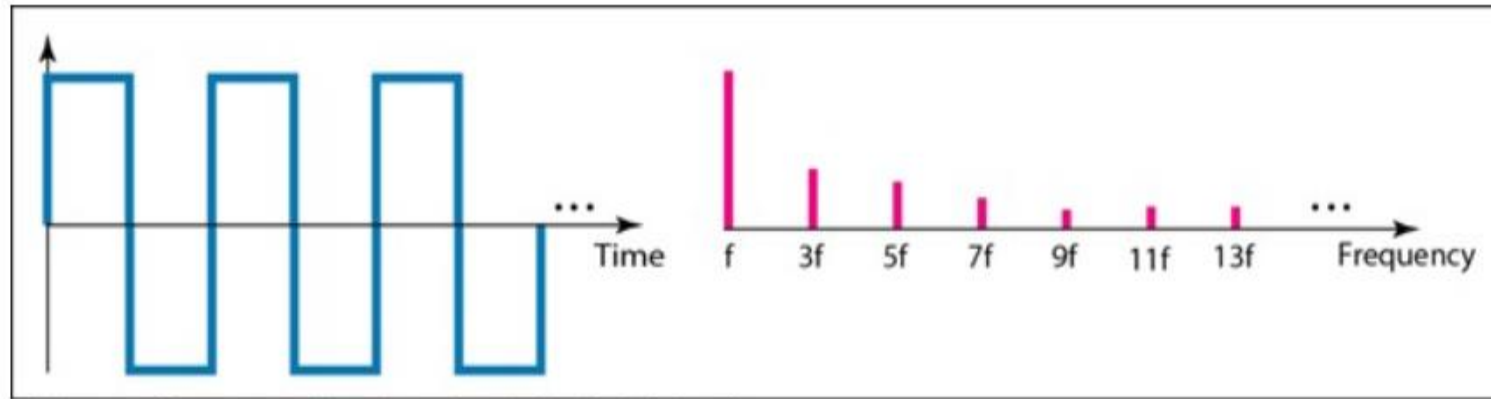
Digital Signal as Composite Analog Signal



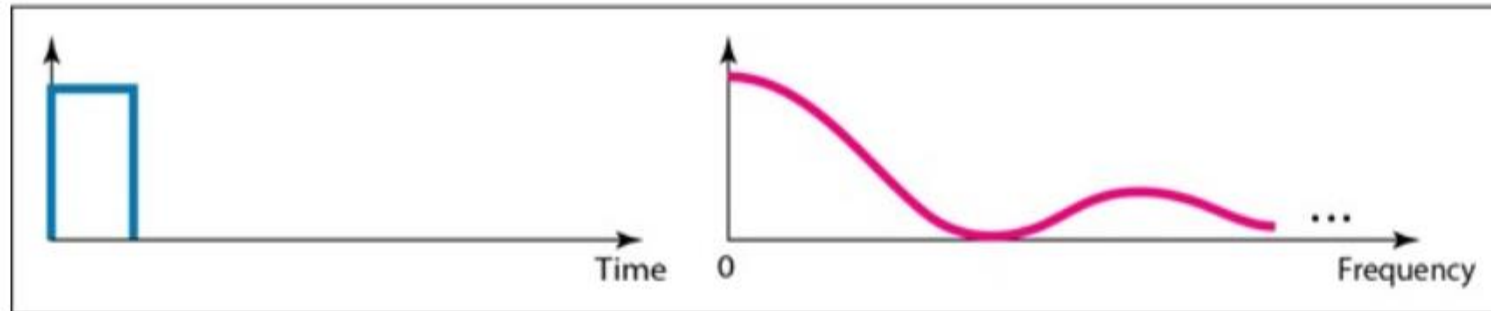
A digital signal is a composite analog signal with an infinite bandwidth.

Bandwidth of composite signal is the difference between the highest and lowest frequencies contained in the signal

Digital Signal as Composite Analog Signal



a. Time and frequency domains of periodic digital signal



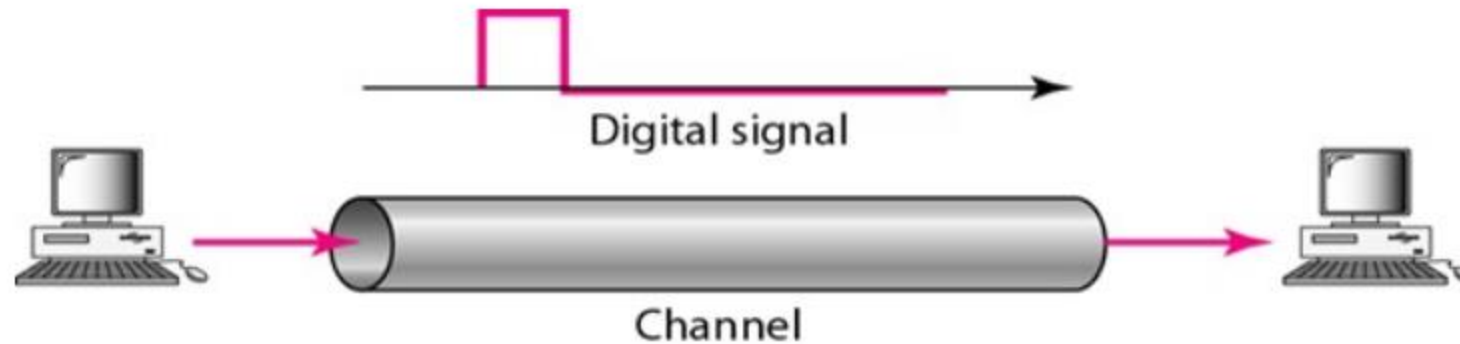
b. Time and frequency domains of nonperiodic digital signal

Transmission of Digital Signals

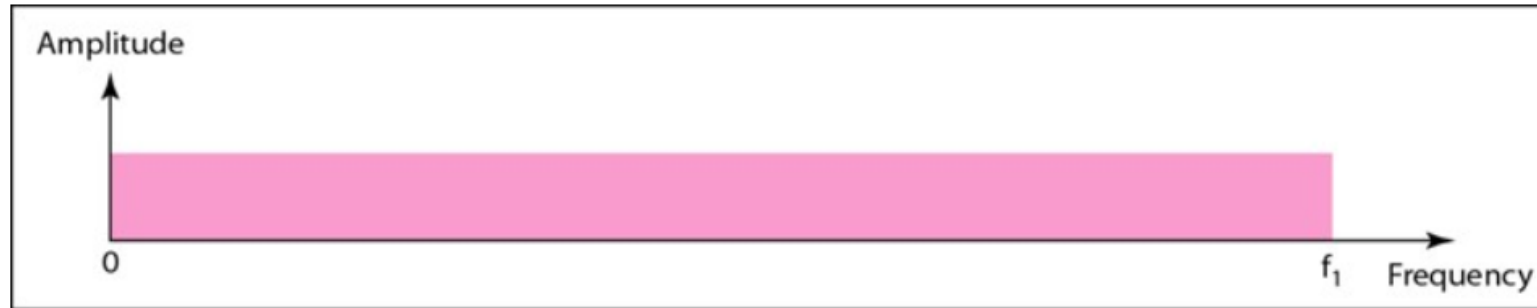
- How can we send a digital signal from point A to point B ?
- Two approaches
 - Baseband
 - Broadband

Baseband Transmission

- Sending a digital signal over a channel without changing the digital signal to analog.
- Baseband transmission requires a **low-pass channel**, a channel with a bandwidth that starts from zero.



Baseband Transmission



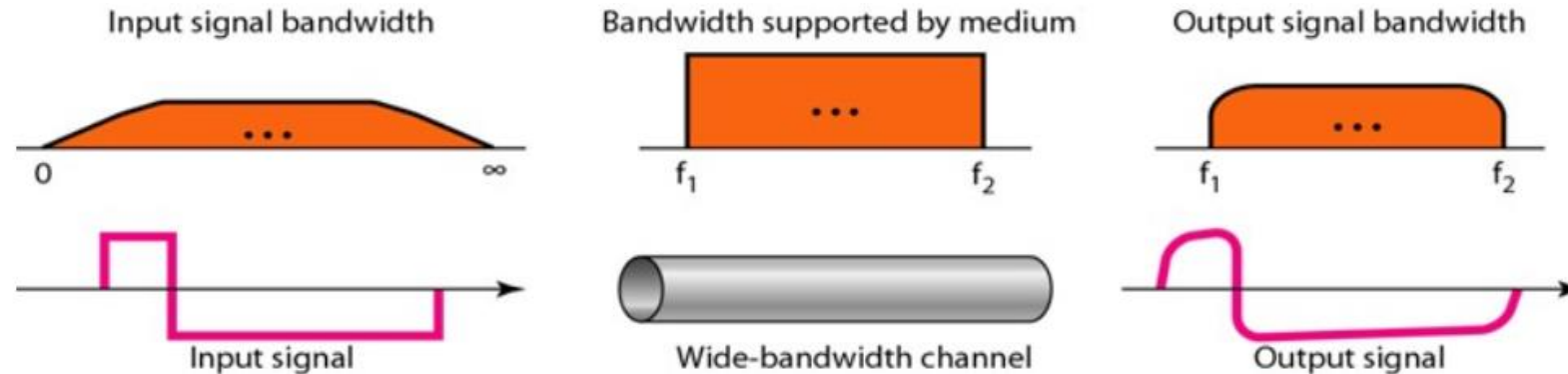
a. Low-pass channel, wide bandwidth



b. Low-pass channel, narrow bandwidth

Baseband Transmission

- *Case 1: Low-Pass Channel with Wide Bandwidth*
- To preserve the exact form of a nonperiodic digital signal, entire spectrum of frequencies must be sent.
- Possible if we have a dedicated medium with an infinite bandwidth between the sender and receiver.



Baseband Transmission

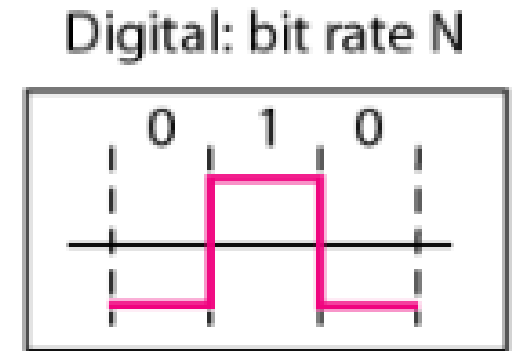
- *Case 2: Low-Pass Channel with Limited Bandwidth*
- In a low-pass channel with limited bandwidth, we approximate the digital signal with an analog signal.
- The level of approximation depends on the bandwidth available.

Baseband Transmission

- Assume that we have a digital signal of bit rate N .
- If we want to send analog signals to roughly simulate this signal, we need to consider the worst case, a maximum number of changes in the digital signal.
- Consider signals like 01010101 or 10101010

Example – Signal 01010101

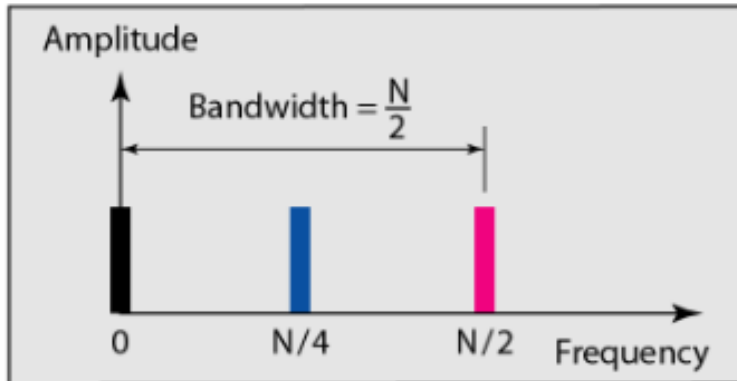
- Bit Rate $N = 8$ bits
 - Each bit stays for $1/N$ time: $1/8 = 0.125$ seconds
- Maximum Frequency $f = N/2$
 - pattern like 01010101 means the signal flips every bit — it goes from $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \dots$
 - That's 1 full cycle every 2 bits.
- Cycle Time
 - one bit stays for $1/8$ second
 - 2 bits stay for $1/4$ second (one complete cycle)
 - So, in 1 second $\rightarrow 1/(1/4) = 4$ cycles per second (or) 4 Hz.



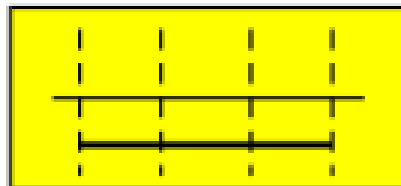
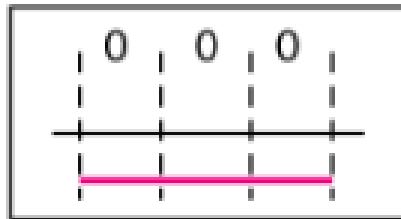
Example – Signal 01010101

- if $N = 8$ bps
- Frequency $f = 4$ Hertz
- $f = N/2$
 - One cycle always requires 2 bits. Bitrate/2 gives max frequency

Baseband Transmission

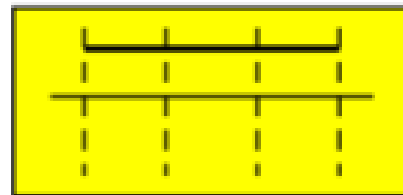
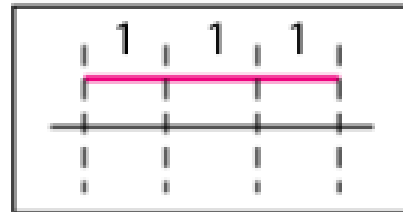


Digital: bit rate N



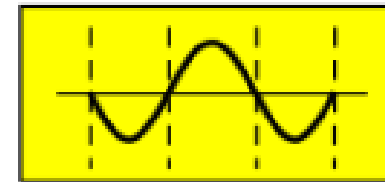
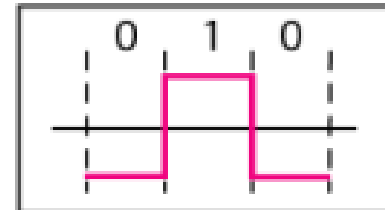
Analog: $f = 0$, $p = 180$

Digital: bit rate N



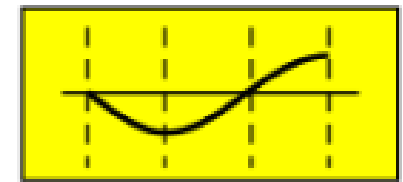
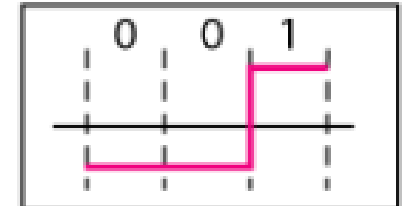
Analog: $f = 0$, $p = 0$

Digital: bit rate N



Analog: $f = N/2$, $p = 180$

Digital: bit rate N



Analog: $f = N/4$, $p = 180$

This rough approximation is referred to as using the first harmonic ($N/2$) frequency.

Baseband Transmission

- Bandwidth requirements
 - range of signals in Hz

<i>Bit Rate</i>	<i>Harmonic 1</i>	<i>Harmonics 1, 3</i>	<i>Harmonics 1, 3, 5</i>
$n = 1$ kbps	$B = 500$ Hz	$B = 1.5$ kHz	$B = 2.5$ kHz
$n = 10$ kbps	$B = 5$ kHz	$B = 15$ kHz	$B = 25$ kHz
$n = 100$ kbps	$B = 50$ kHz	$B = 150$ kHz	$B = 250$ kHz

$$B = n/2$$

$$B = 3n/2$$

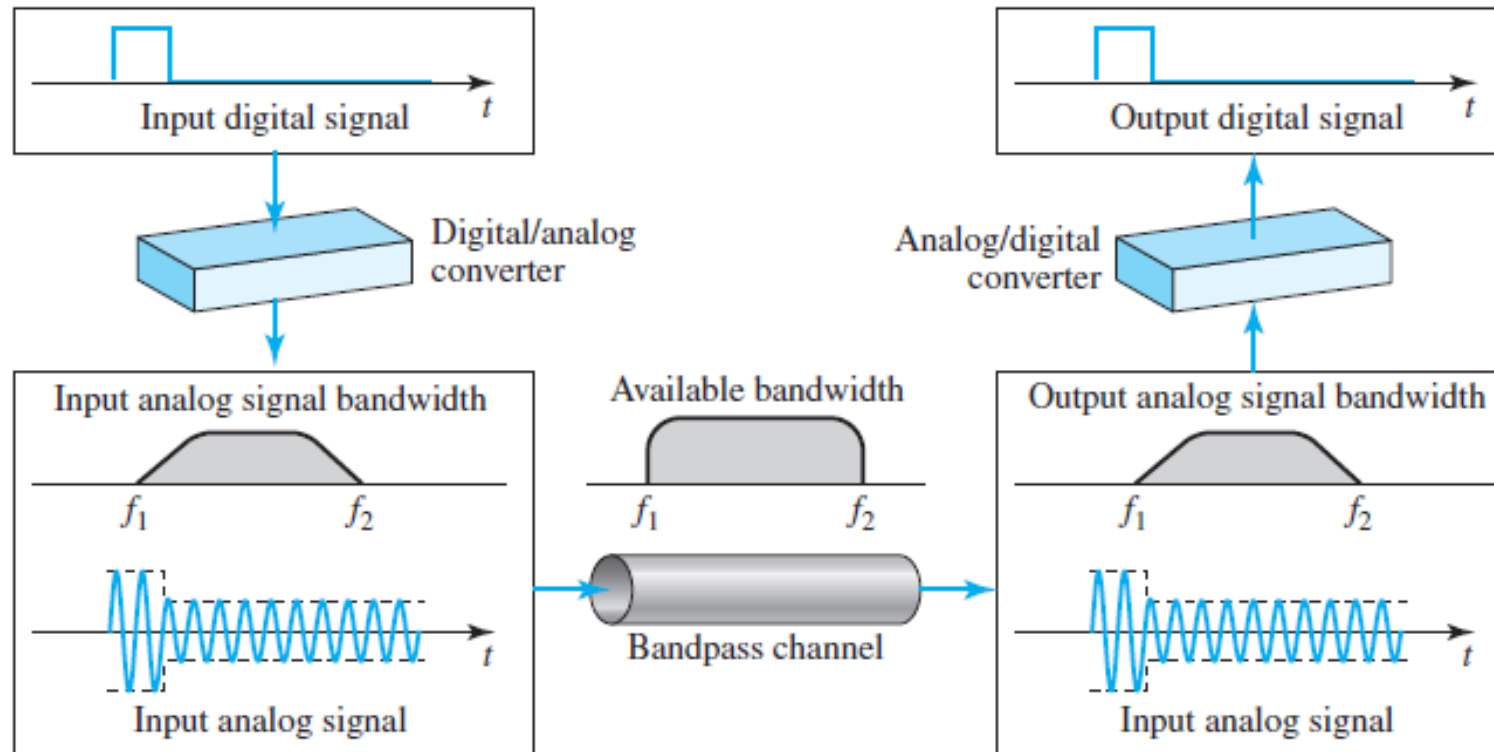
$$B = 5n/2$$

Each added harmonic makes the signal closer to the ideal square wave, but also increases the bandwidth required.

Broadband Transmission

- **Broadband transmission** or modulation means changing the digital signal to an analog signal for transmission.
- Modulation allows us to use a **bandpass channel**—a channel with a bandwidth that does not start from zero

Broadband Transmission



Performance

- Bandwidth

1. Bandwidth in Hz

- The range of frequencies contained in a composite signal or the range of frequencies a channel can pass

2. Bandwidth in Bits Per Second

- Refer to the number of bits per second that a channel, a link, or even a network can transmit

Performance

- Throughput
 - A measure of how fast we can actually send data through a network
 - The bandwidth is a potential measurement of a link; the throughput is an actual measurement of how fast we can send data

Performance

- Latency
 - Defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

$$\text{Latency} = \text{propagation time} + \text{transmission time} + \text{queuing time} + \text{processing delay}$$

Performance

- *Propagation Time*

- Measures the time required for a bit (first bit) to travel from the source to the destination.

$$\text{Propagation time} = \text{Distance} / (\text{Propagation Speed})$$

Performance

- *Transmission Time*
 - Time required to **push all the bits** of the message into the channel
 - The **transmission time** of a message depends on the size of the message and the bandwidth of the channel.

$$\text{Transmission time} = (\text{Message size}) / \text{Bandwidth}$$

Performance

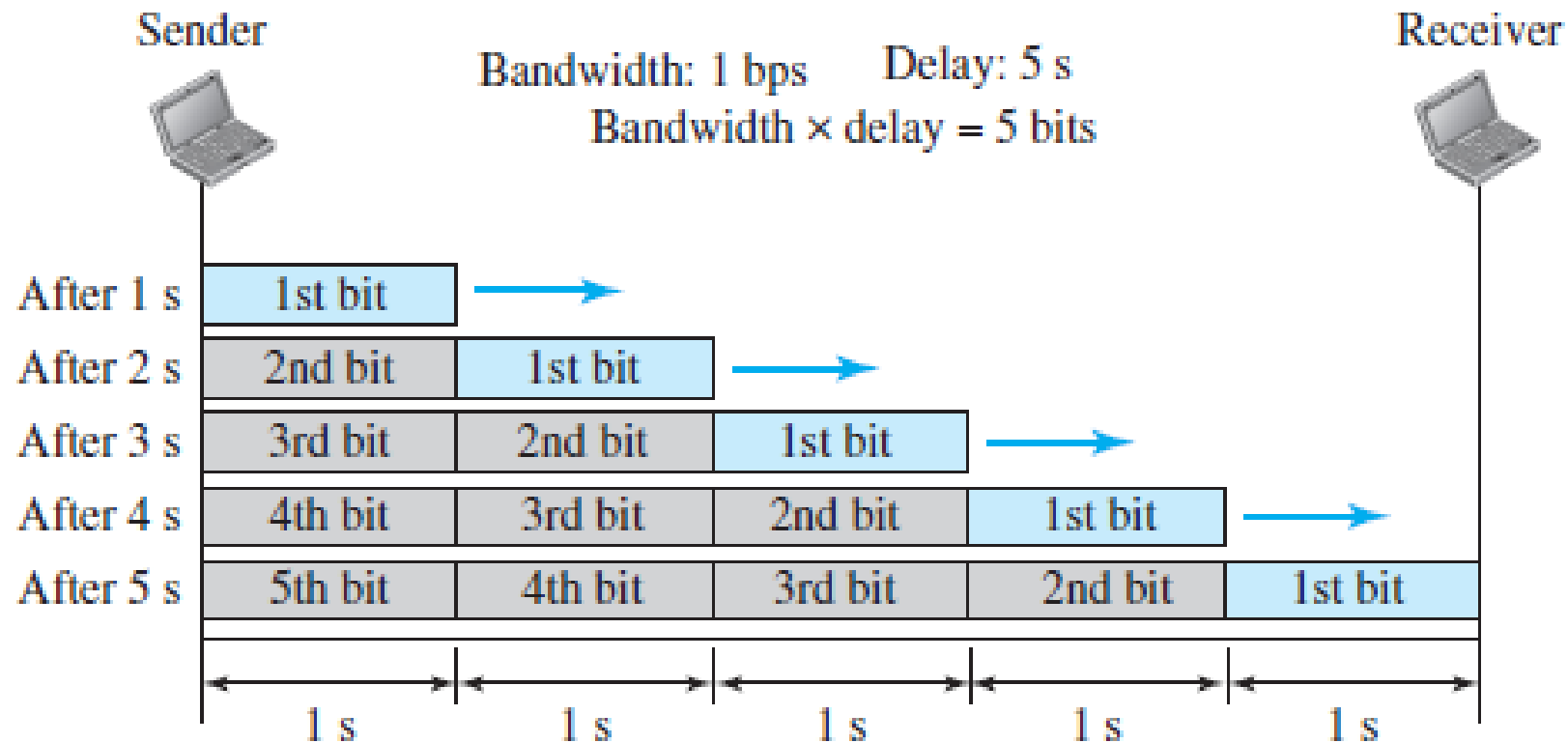
- *Queuing Time*
 - The time needed for each intermediate or end device to hold the message before it can be processed.

Processing Delay

- The time it takes for the routers to process the packet header.

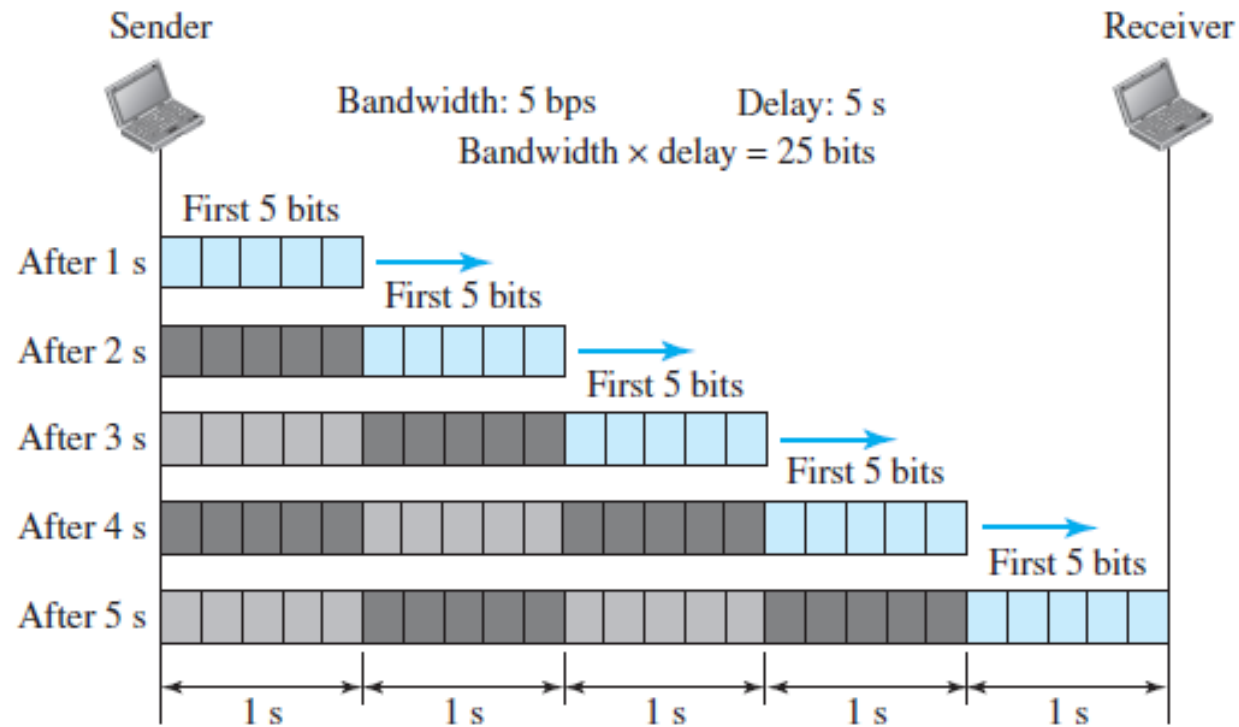
Performance

- Bandwidth-Delay Product



Performance

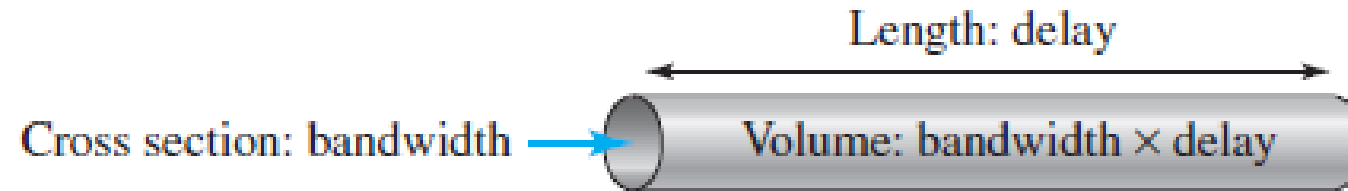
- Bandwidth-Delay Product



- The product of bandwidth and delay is the number of bits that can fill the link

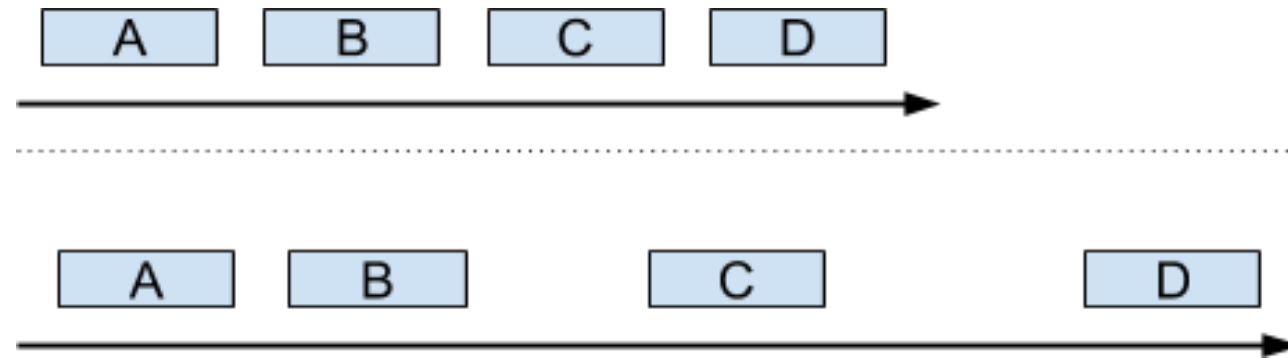
Performance

- Bandwidth-Delay Product

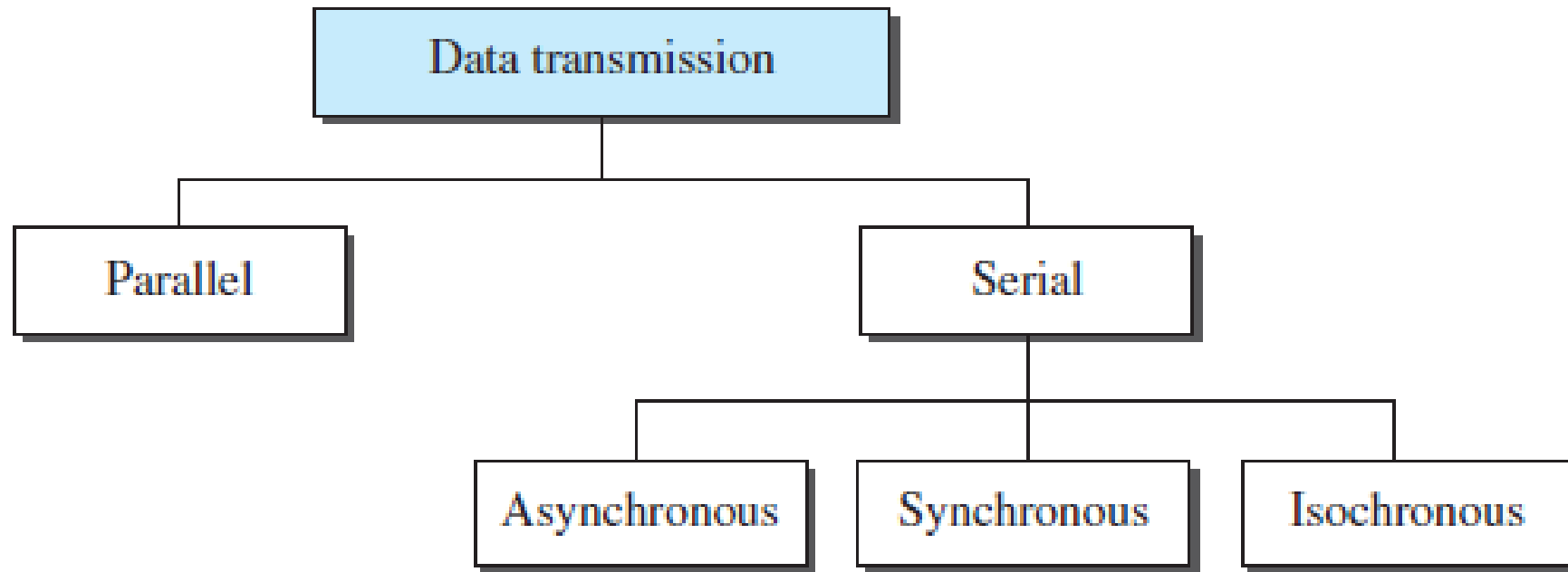


Performance

- Jitter
 - The problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data)

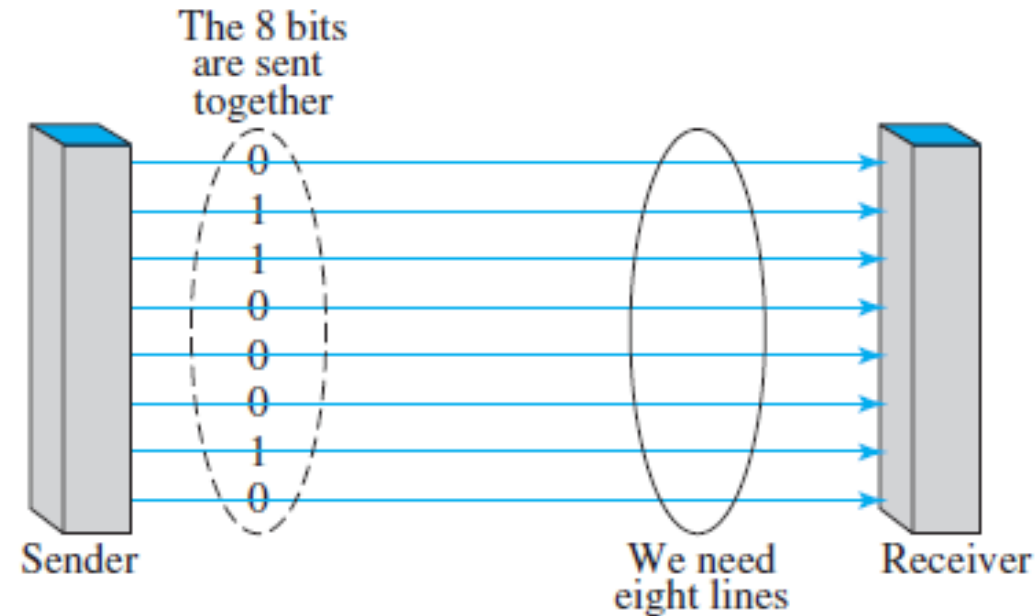


Transmission Modes



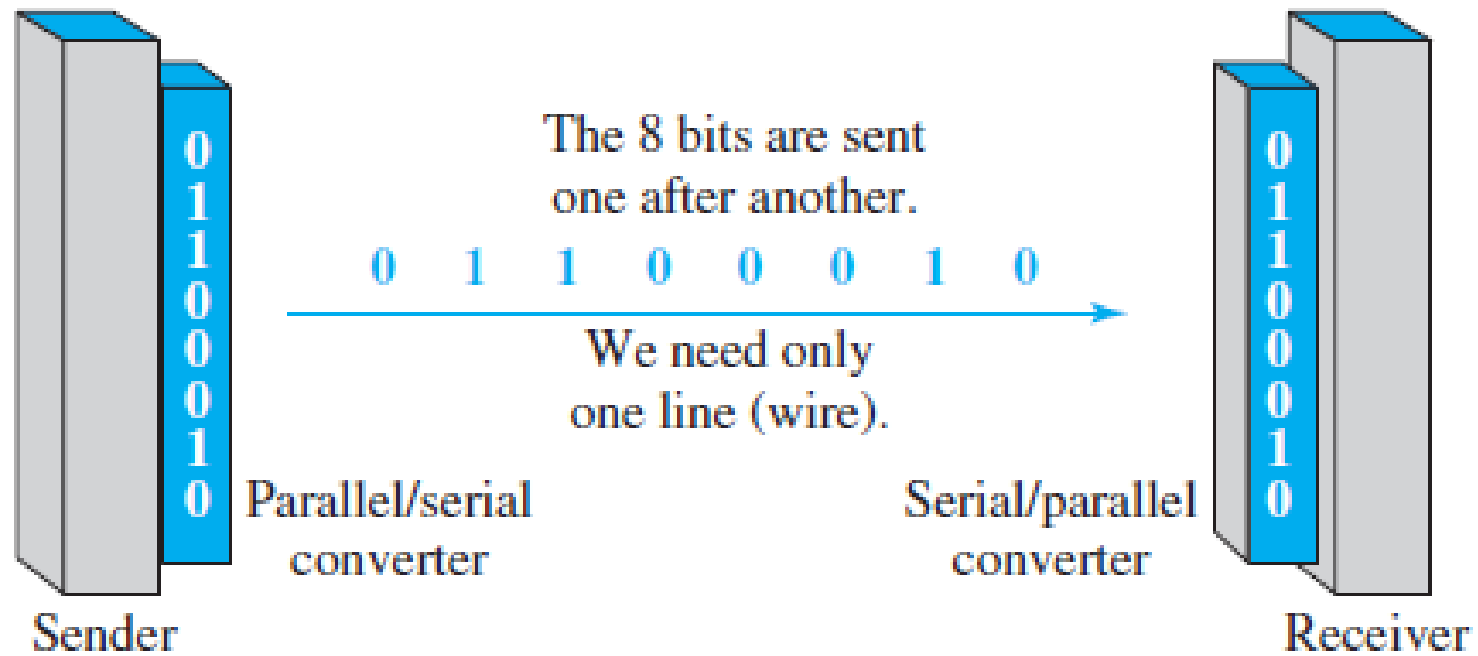
Parallel Transmission

- Binary data may be organized into groups of n bits each.
 - By grouping, we can send data n bits at a time instead of 1.



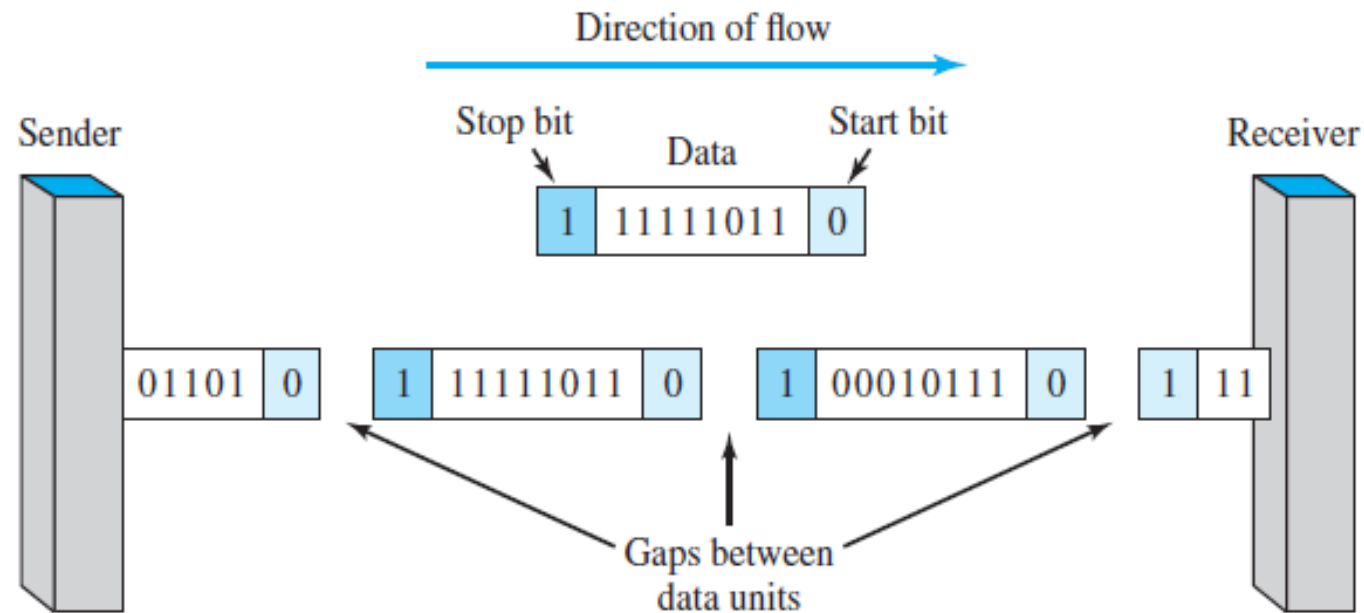
- Advantage: Speed, Disadvantage: Cost

Serial Transmission



Serial Transmission

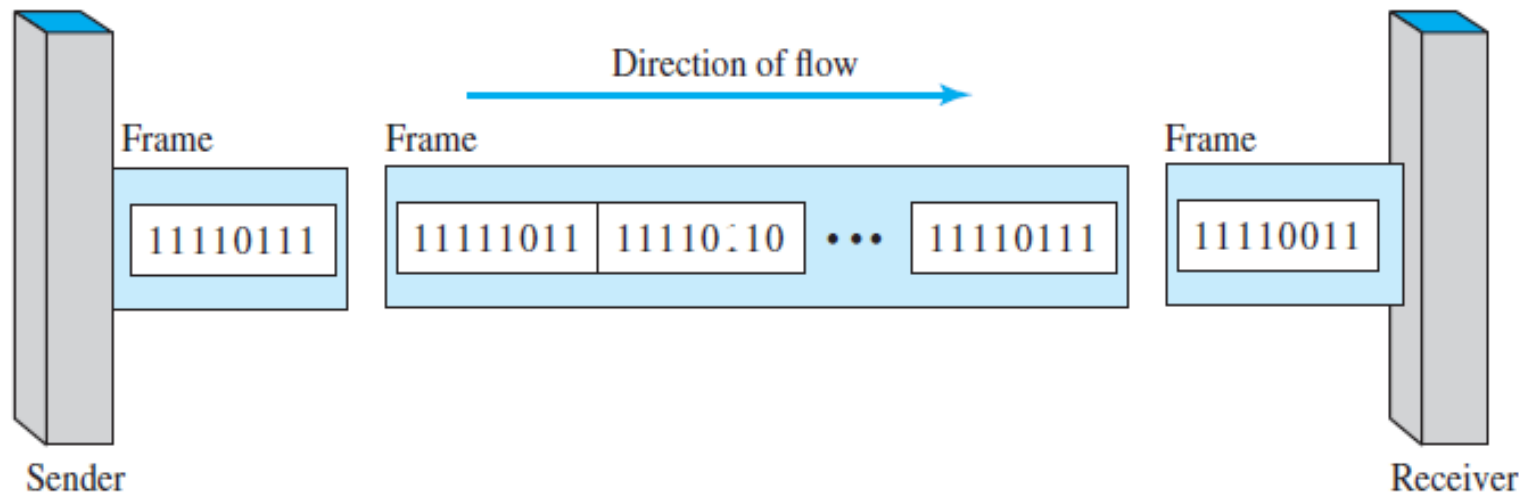
- **Asynchronous** means “asynchronous at the byte level,” but the bits are still synchronized; their durations are the same.



- Advantage: Cheap and effective

Serial Transmission

- In **synchronous transmission**, we send bits one after another without start or stop bits or gaps.
 - It is the responsibility of the receiver to group the bits.



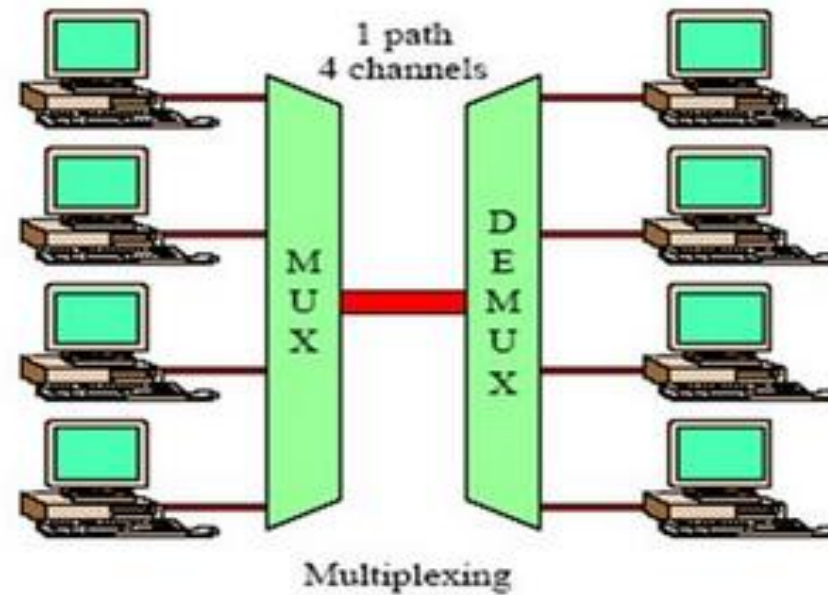
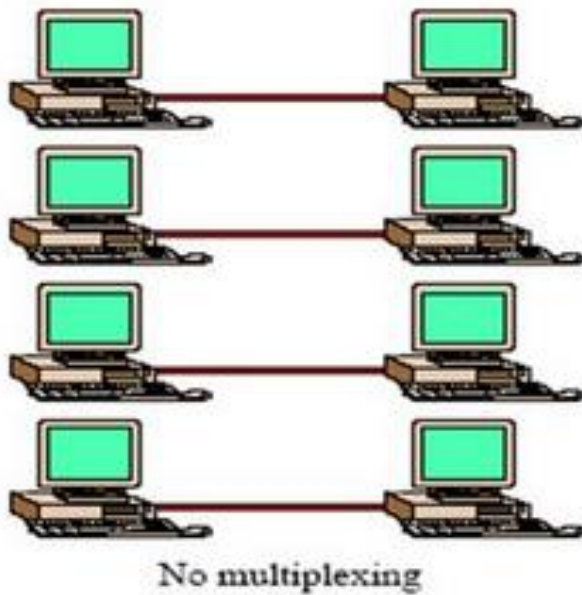
- Advantage: Speed

Serial Transmission

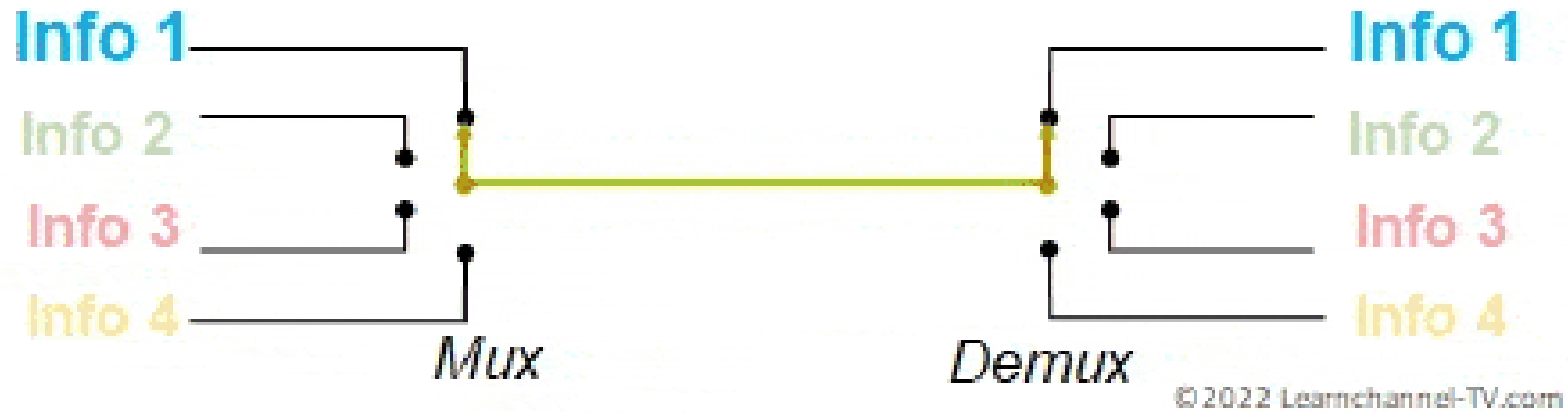
- When both sender and receiver cannot have unacceptable delays,
- The **isochronous transmission** guarantees that the data arrive at a fixed rate.

Multiplexing

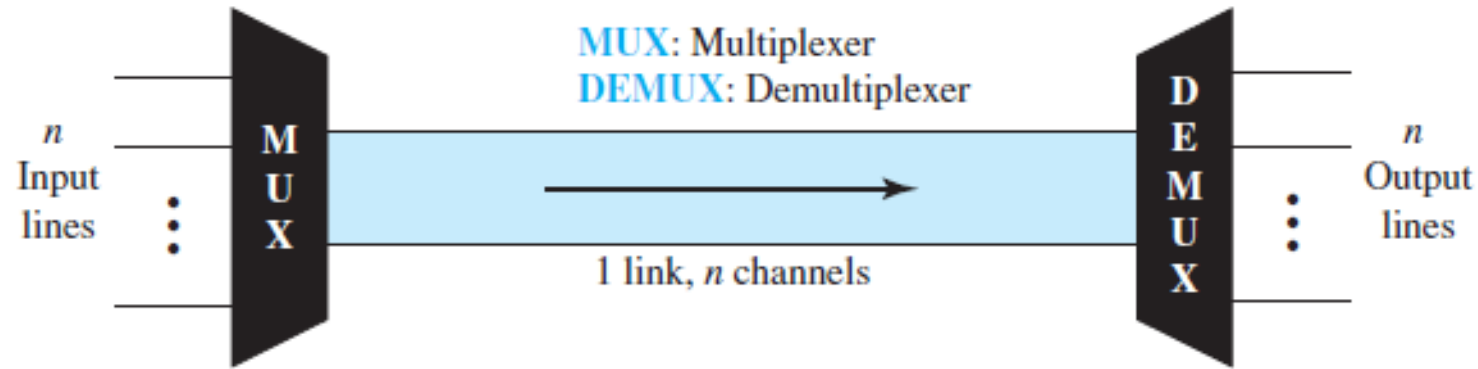
- Set of techniques that allow the simultaneous transmission of multiple signals across a single data link.



Multiplexing

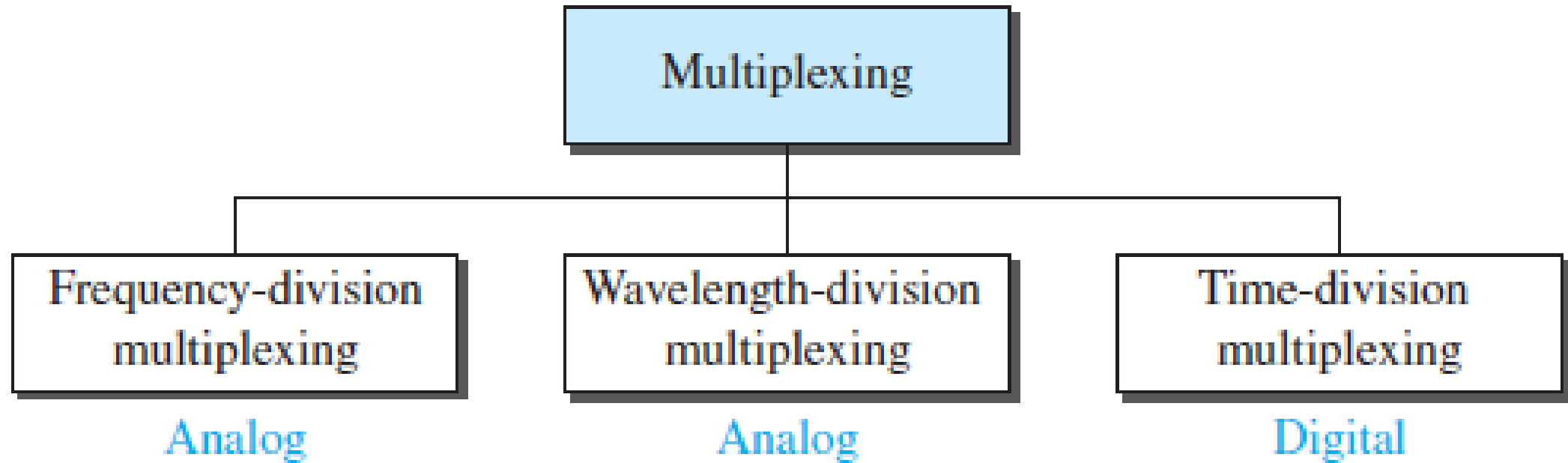


Multiplexing



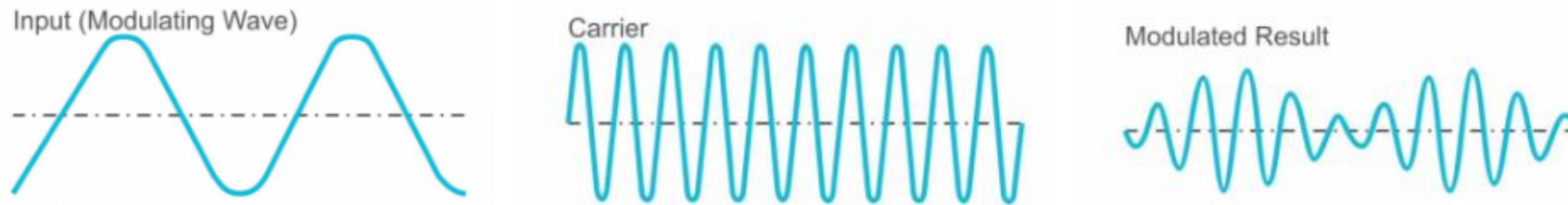
- **Multiplexer (MUX)** combines lines into a single stream (many-to-one).
- **Demultiplexer (DEMUX)** separates the stream back into its component transmissions (one-to-many).
- **Link** refers to the physical path.
- **Channel** refers to the portion of a link that carries a transmission between a given pair of lines.

Multiplexing

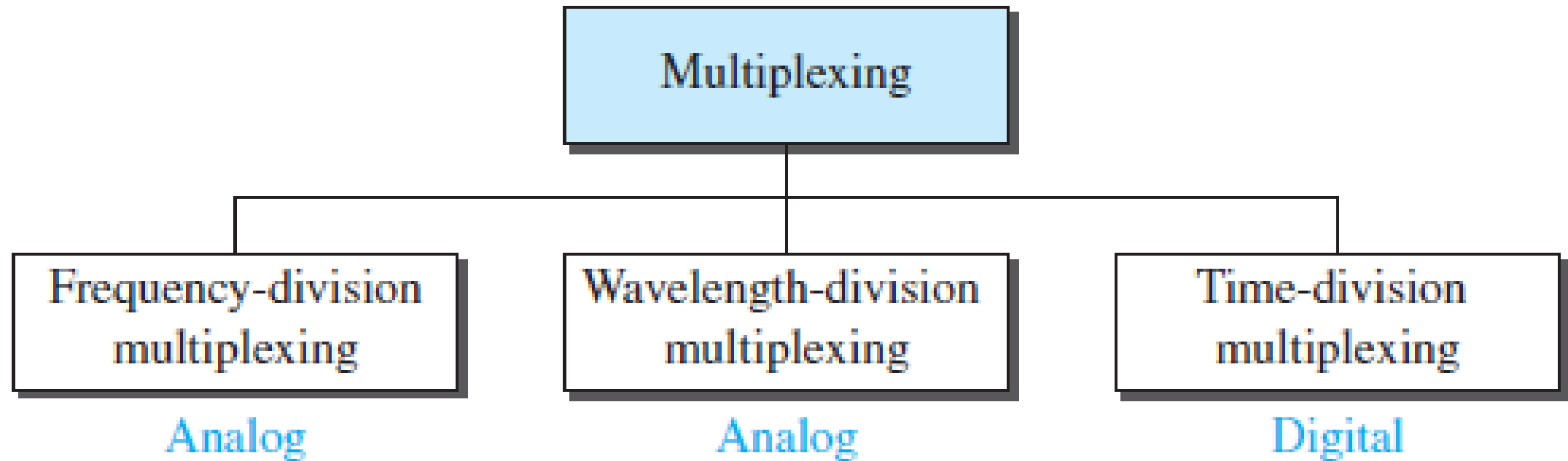


Terms to Know

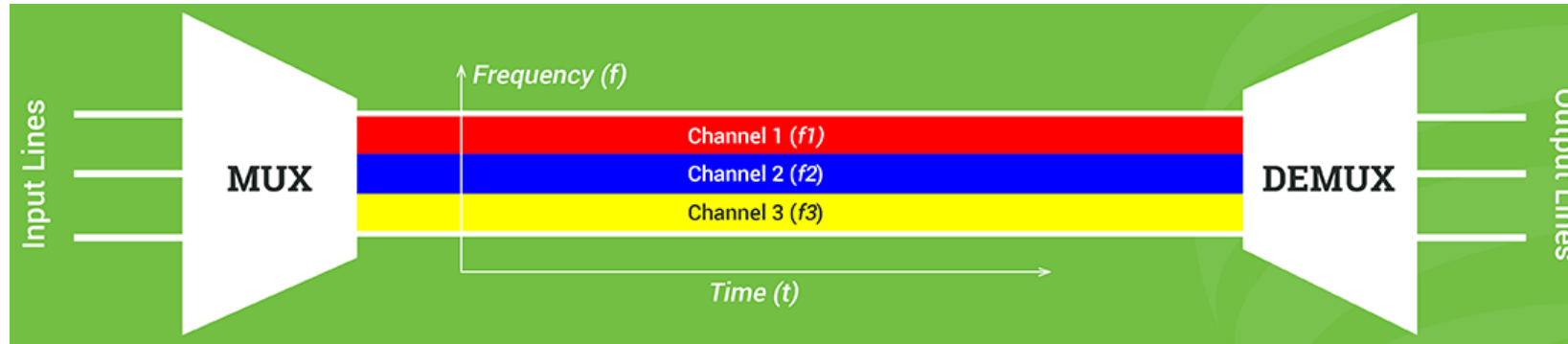
- A **carrier signal** is a signal with constant frequency.
 - They carry little or no information by themselves.
 - They have sufficient power and frequency to be transmitted easily without interference.
- By imposing the input signal on to a carrier signal, the input signal can be transmitted along with it. This is known as **modulation**.



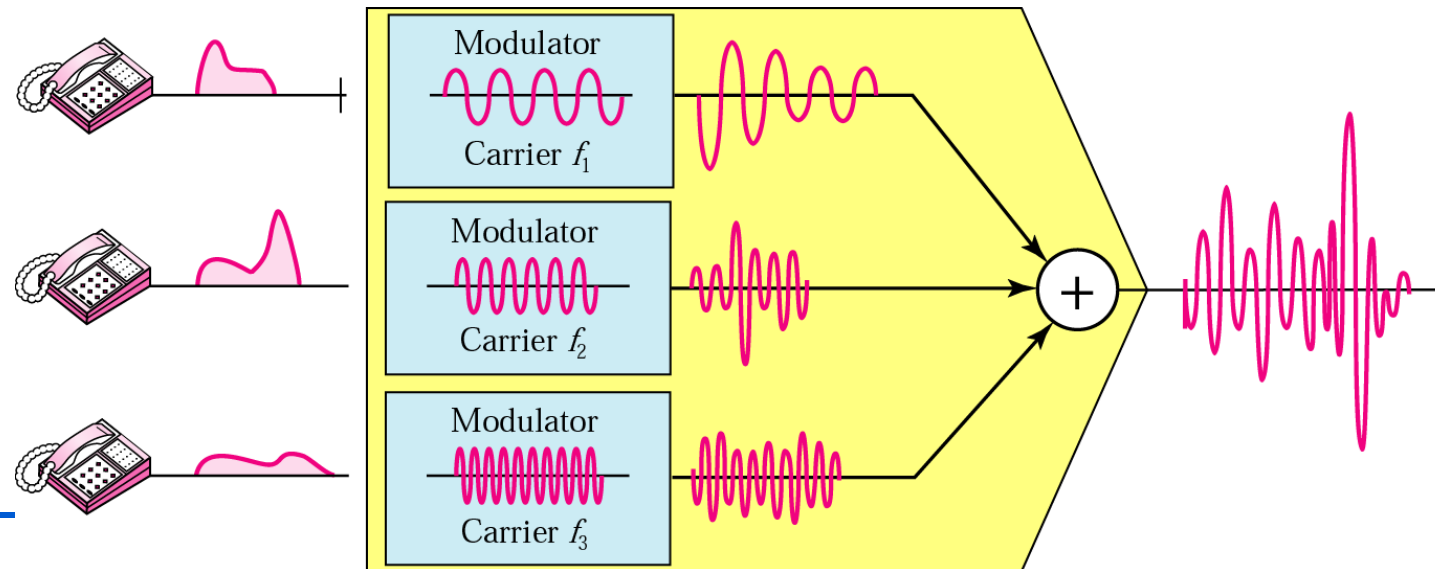
Multiplexing



FDM - Multiplexing Process



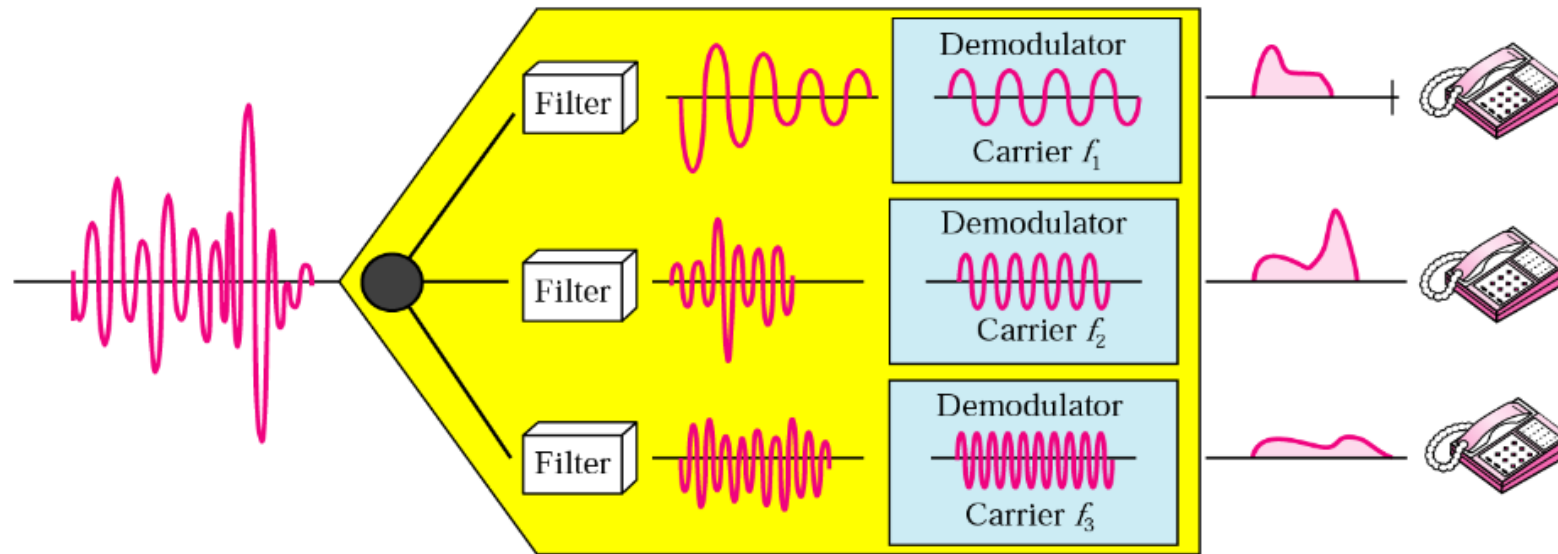
Channels can be separated by strips of unused bandwidth—**guard bands**—to prevent signals from overlapping



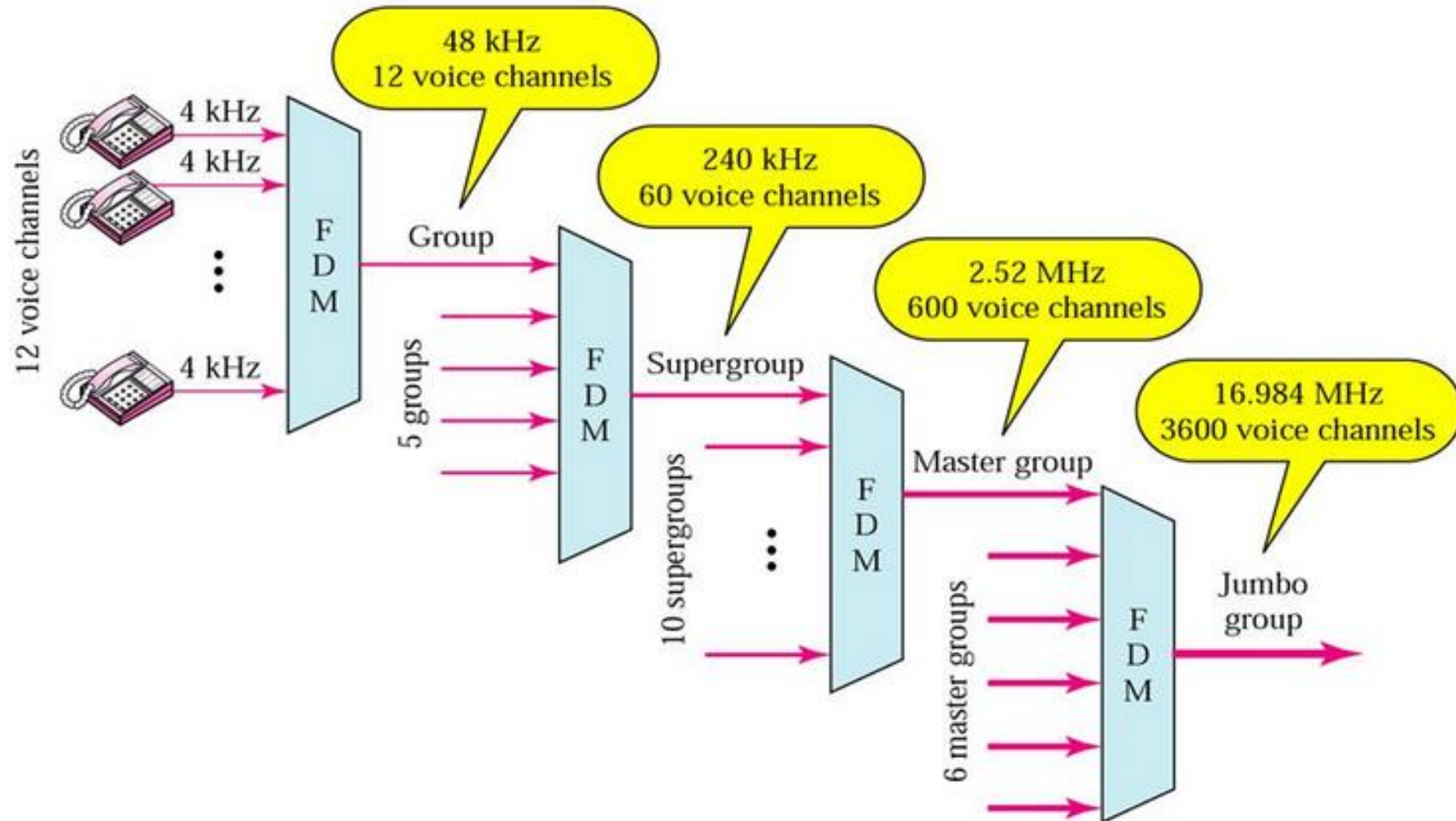
Frequency-Division Multiplexing (FDM)

- Analog technique that can be applied when the bandwidth of a link is greater than the combined bandwidths of the signals.
- In FDM, signals generated by each sending device modulate different carrier frequencies.
- These modulated signals are then combined into a single composite signal that can be transported by the link.

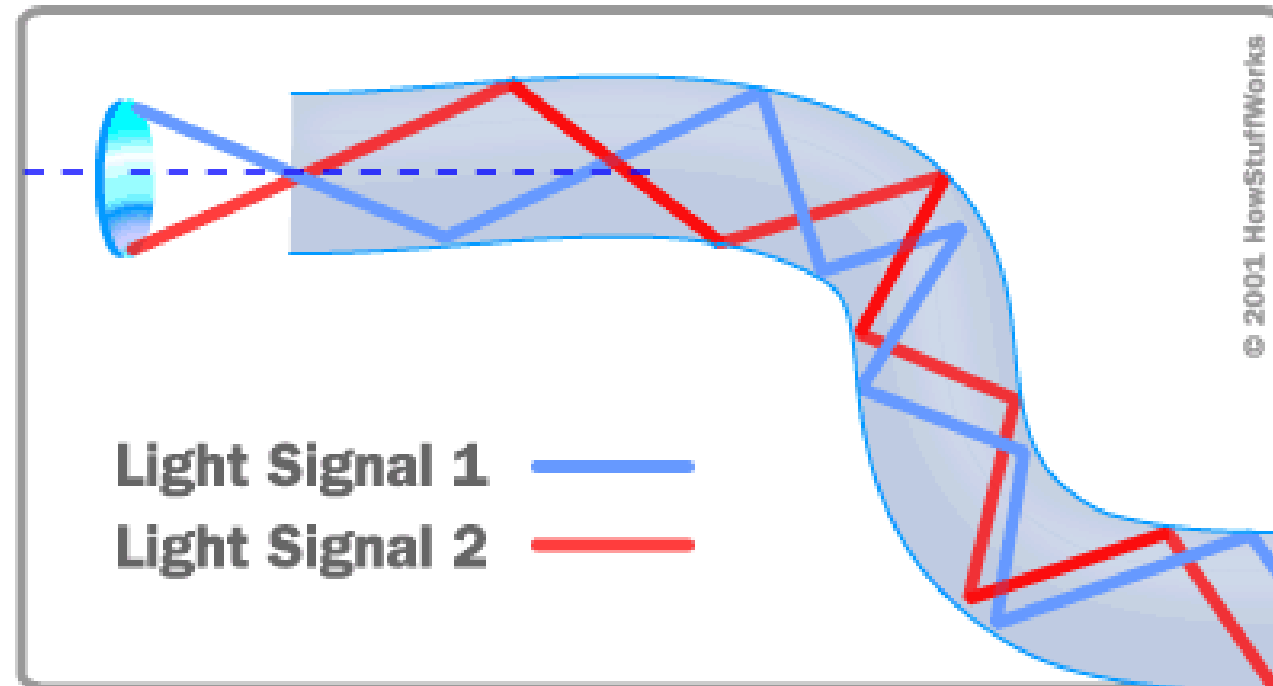
FDM - Demultiplexing Process



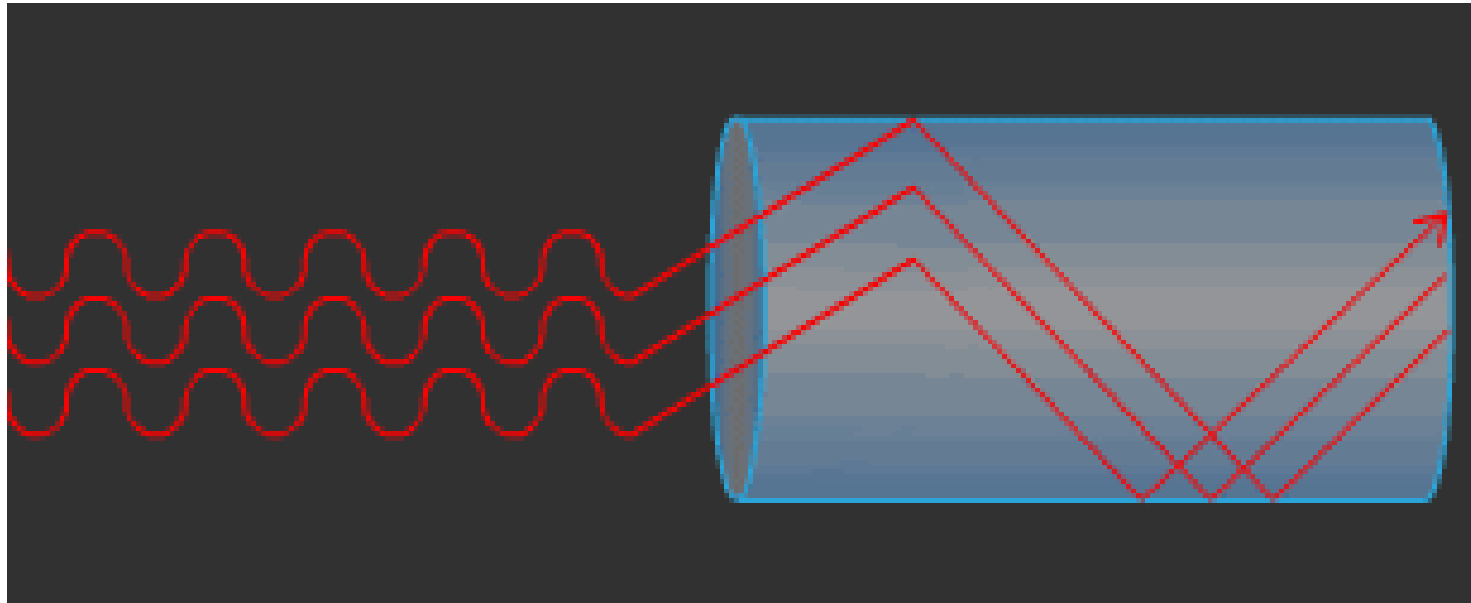
Analog Carrier System



Fiber-Optic Cable



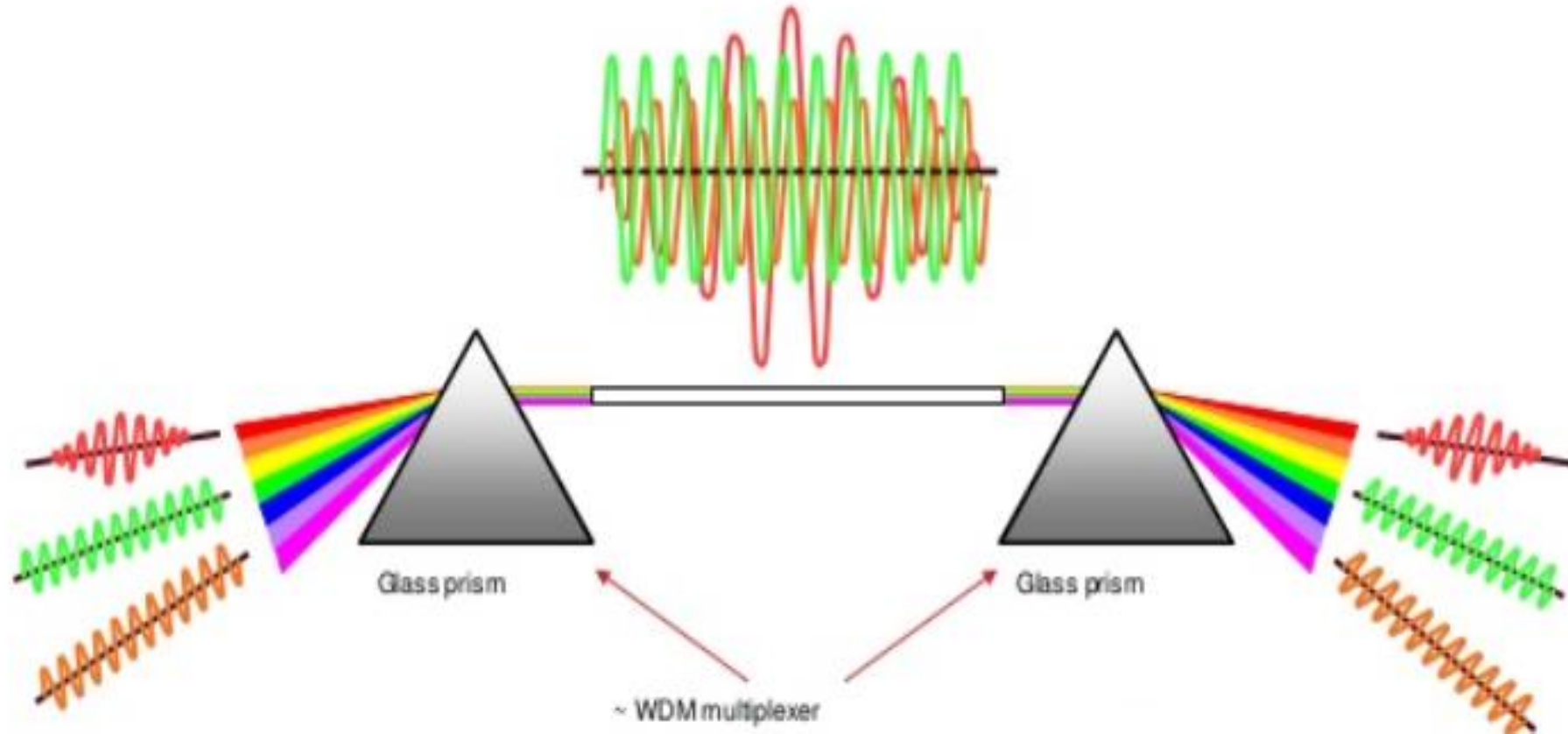
Fiber-Optic Cable



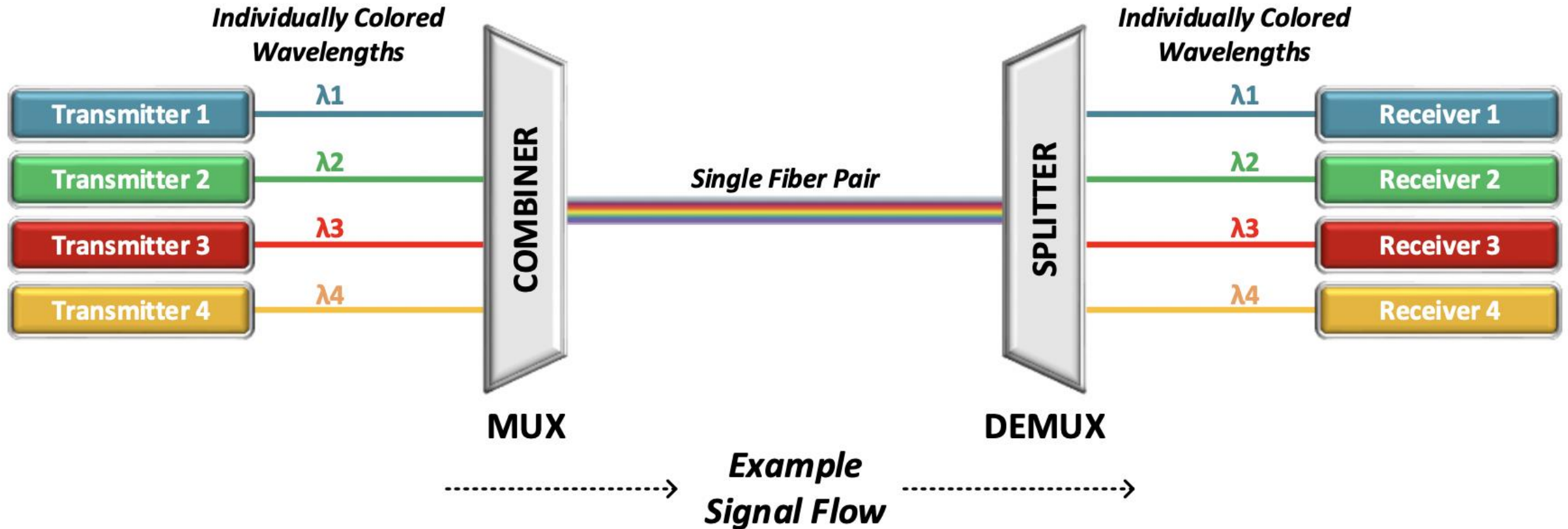
Wavelength-Division Multiplexing (WDM)

- Designed to use the high-data-rate capability of fiber-optic cable.
 - Using a fiber-optic cable for a single line wastes the available bandwidth.
- Multiplexing allows us to combine several lines into one.
 - The multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels.
- The idea is the same like **FDM** but the difference is that the frequencies are very high in **WDM**.

WDM

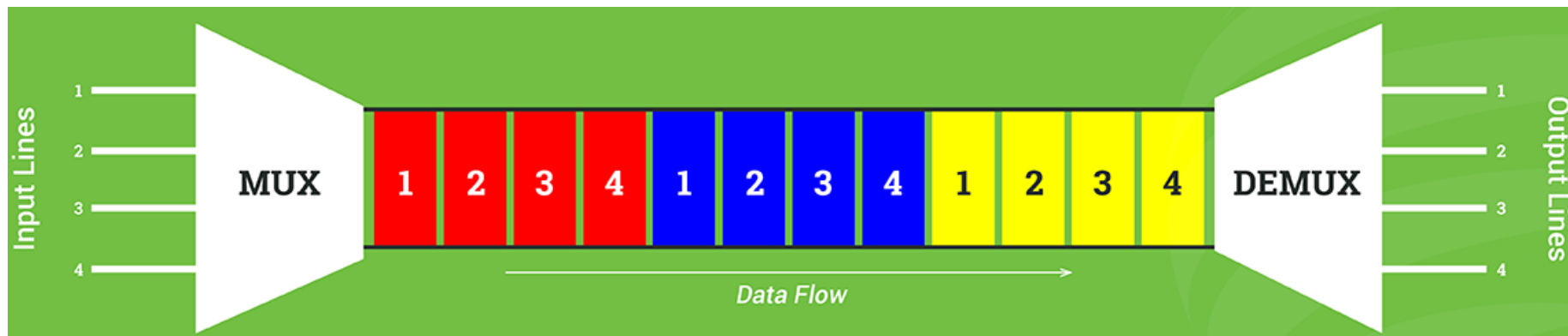


WDM



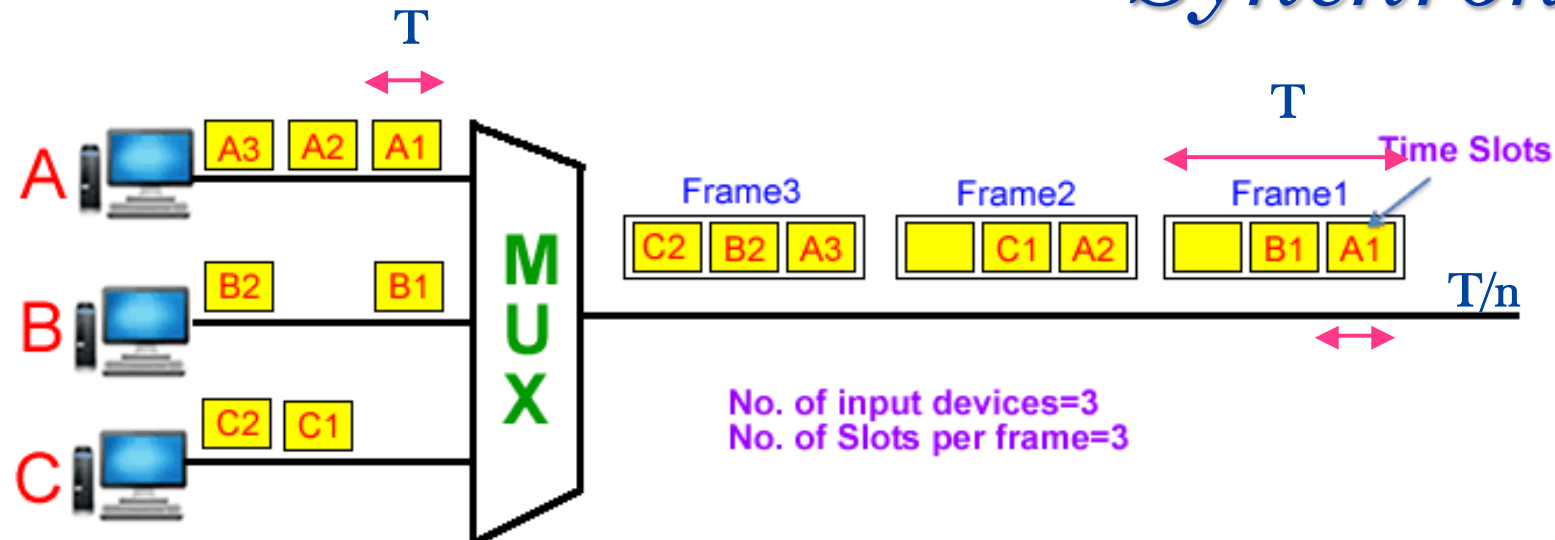
Time-Division Multiplexing (TDM)

- A digital process that allows several connections to share the high bandwidth of a link.
- Instead of sharing a portion of the bandwidth as in FDM, time is shared.
- Each connection occupies a portion of time in the link.



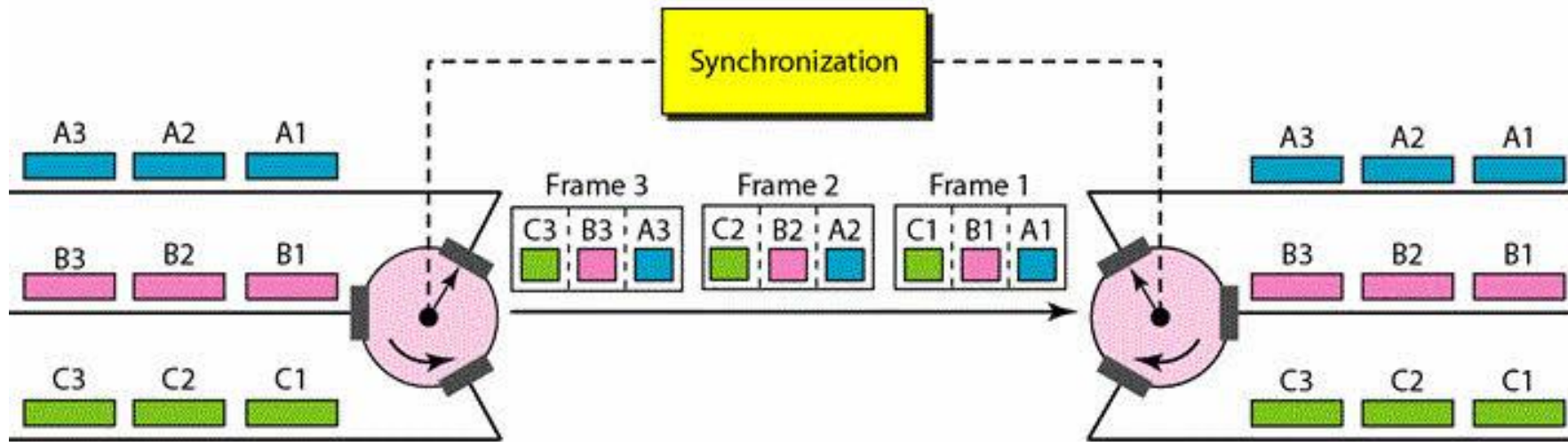
- Two different schemes:
 - Synchronous TDM
 - Statistical TDM

Synchronous TDM



- If we have n connections, a frame is divided into n time slots and one slot is allocated for each unit, one for each input line.
- If the duration of the input unit is T , the duration of each slot is T/n and the duration of each frame is T .
- In synchronous TDM, the data rate of the link is n times faster, and the unit duration is n times shorter.

Interleaving in TDM

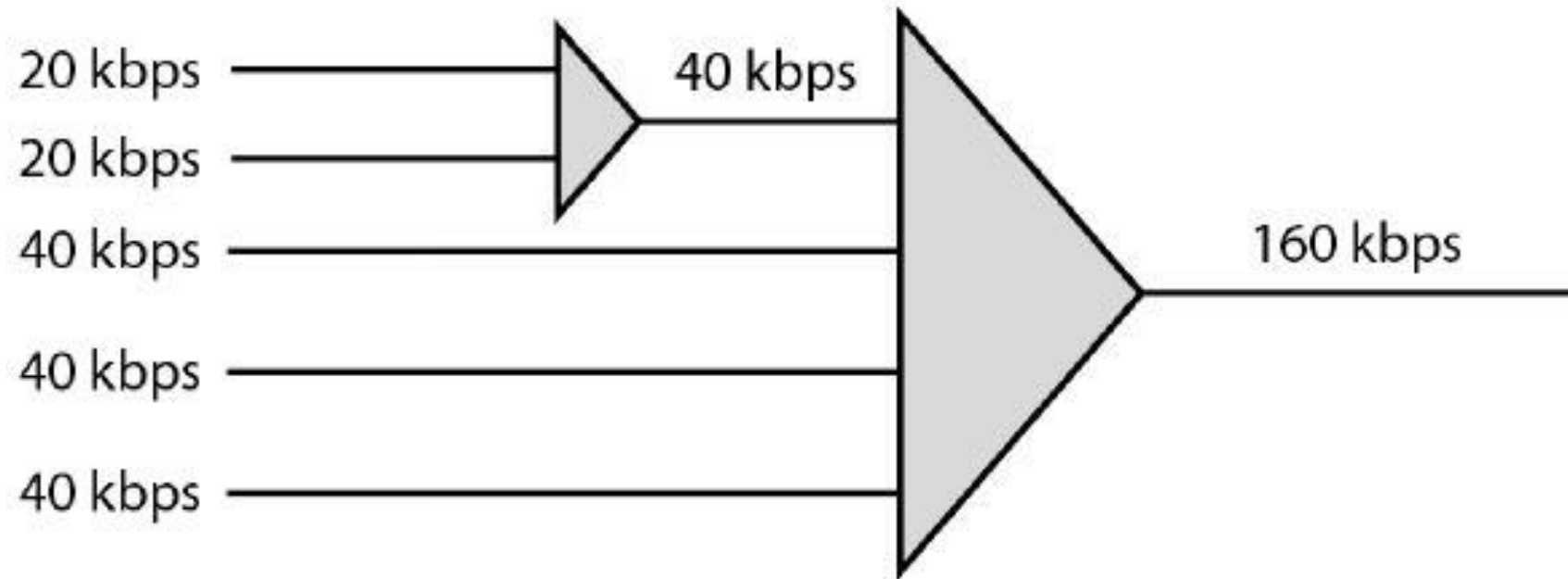


- The switches are synchronized and rotate at the same speed, but in opposite directions.
- On the multiplexing side, as the switch opens in front of a connection, that connection has the opportunity to send a unit onto the path - this is called **interleaving**.

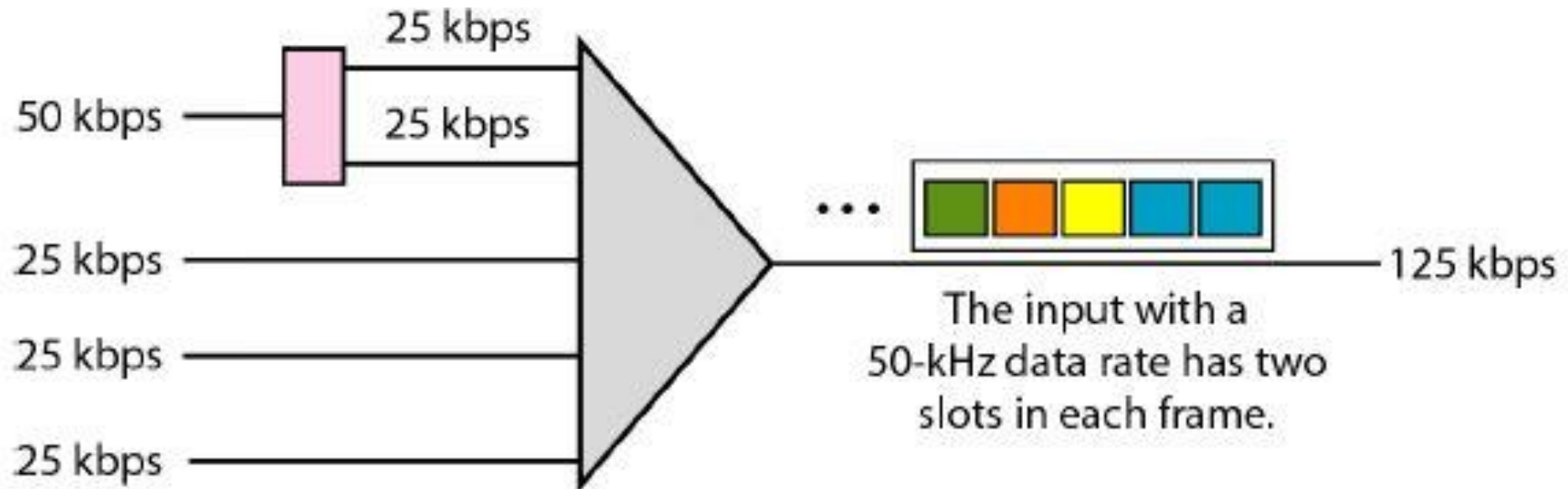
Data Rate Management - TDM

- One problem with TDM is how to handle a disparity in the input data rates.
- Assume three devices sending 100 bps, 40 bps, 10 bps. Which one needs more bandwidth?
- If data rates are not the same, **three strategies**, or a combination of them, can be used.

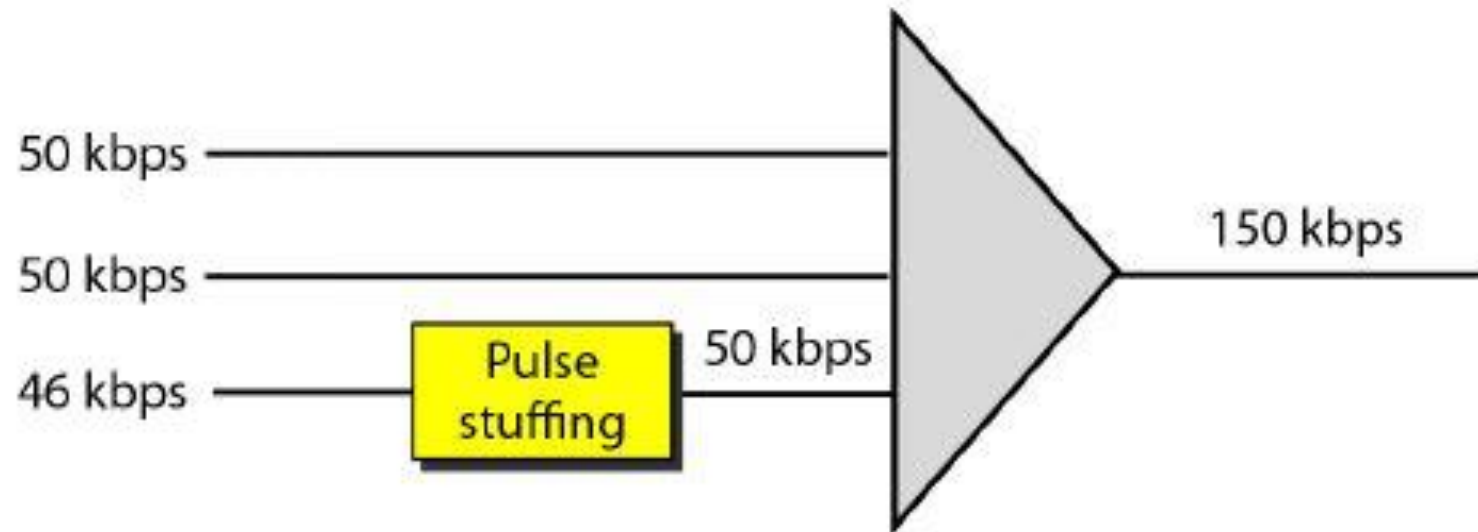
Multilevel Multiplexing



Multiple-slot Allocation



Pulse Stuffing/Bit Padding/Bit Stuffing

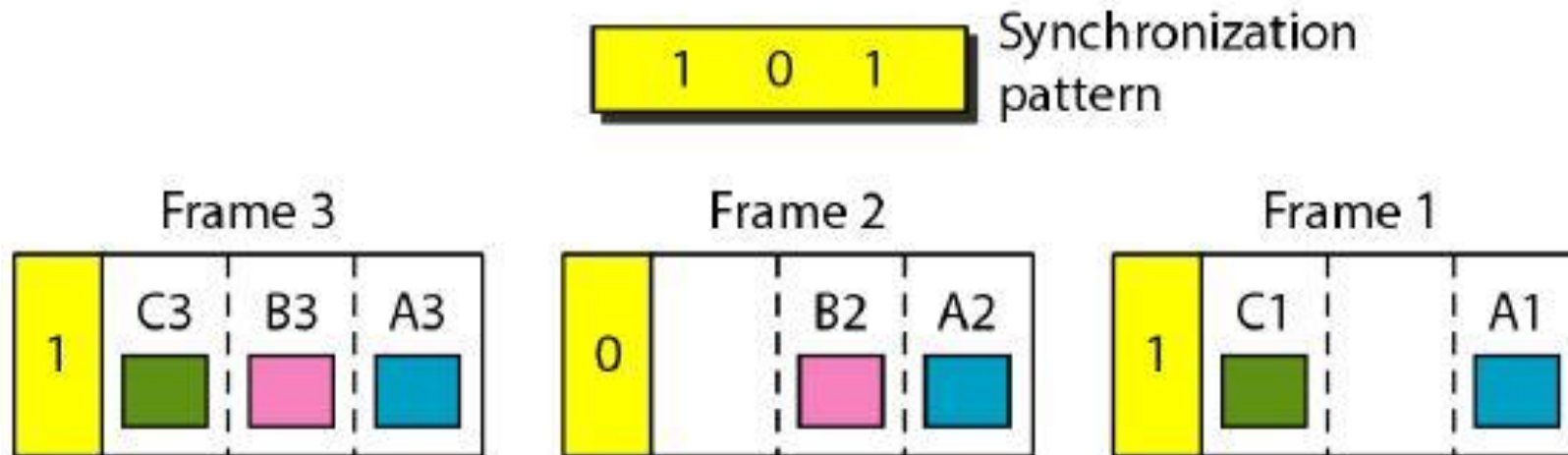


Three Strategies - TDM

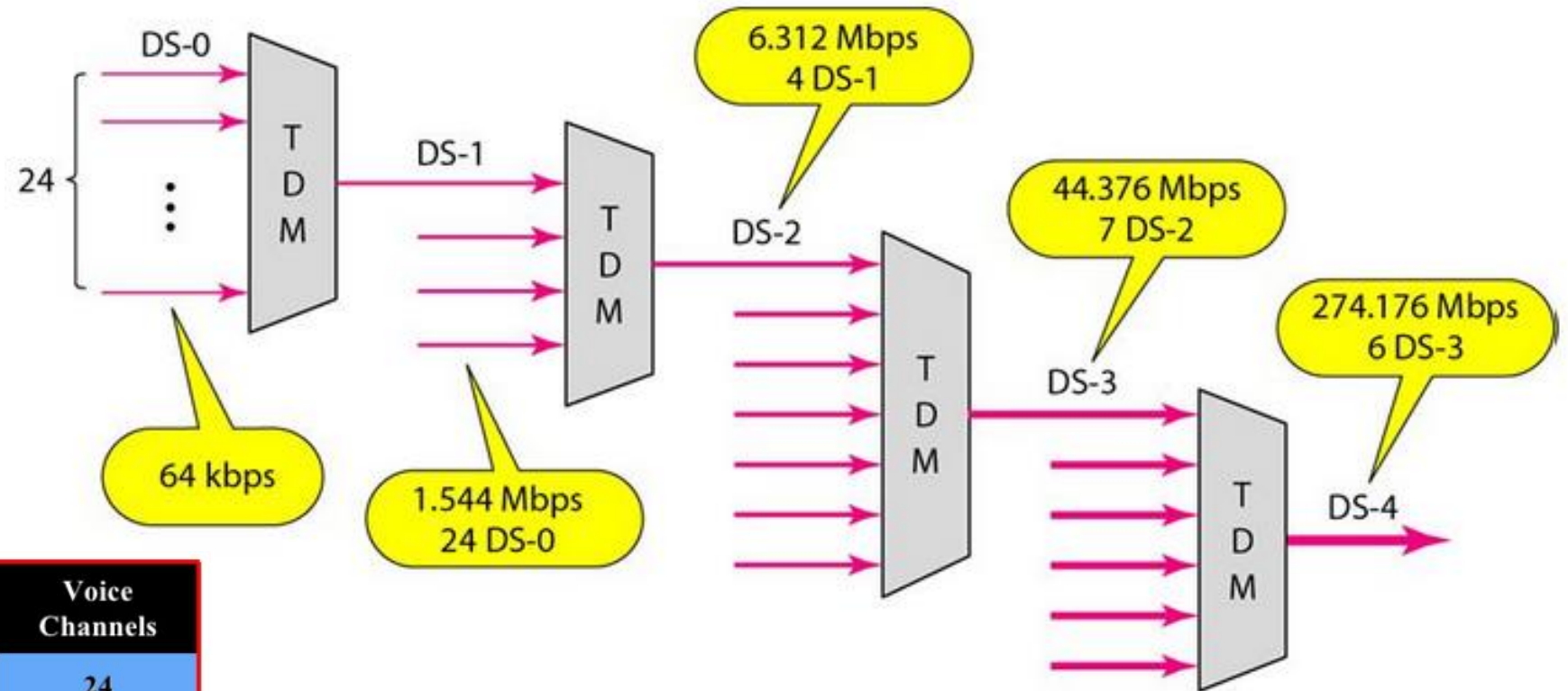
- Multilevel multiplexing
- Multiple-slot allocation
- Pulse stuffing

Frame Synchronization - TDM

- If the multiplexer and the demultiplexer are not synchronized, a bit belonging to one channel may be received by the wrong channel.
- Solution: **Framing Bits**



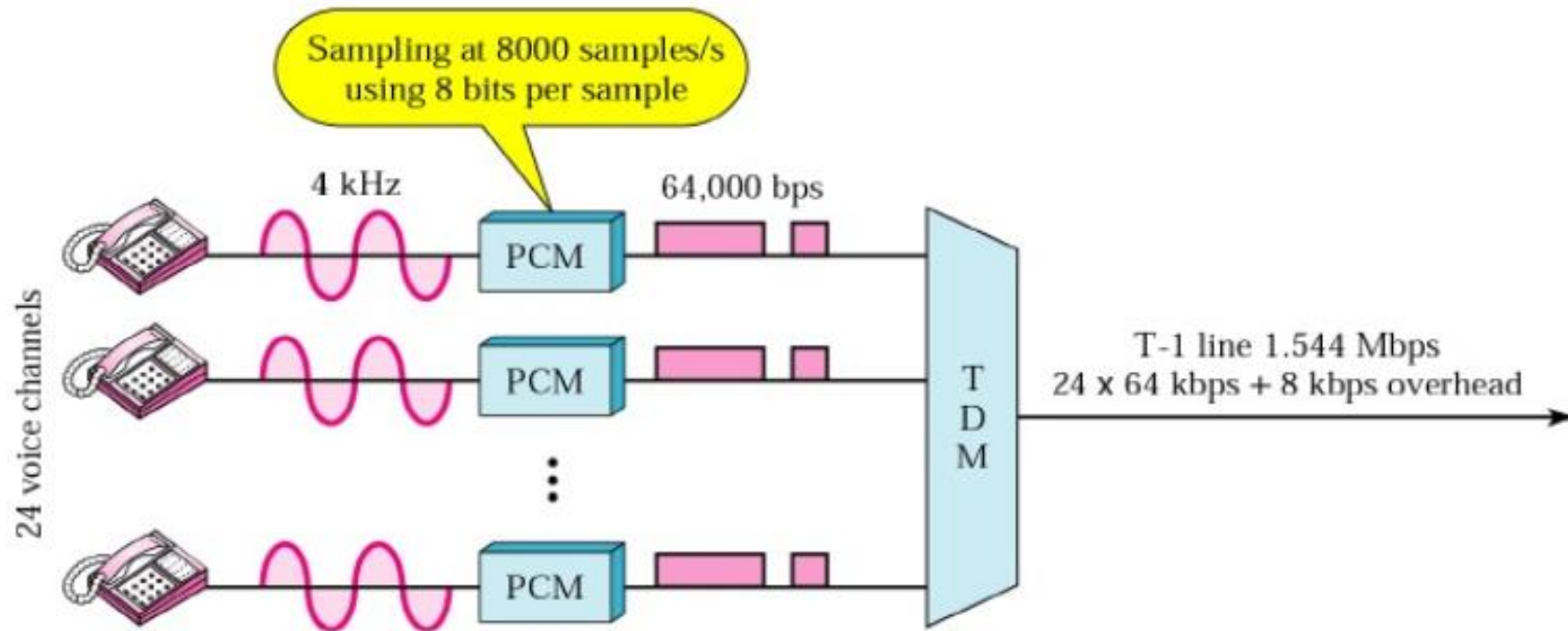
Digital Signal Service - Digital Hierarchy



Service	Line	Rate (Mbps)	Voice Channels
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

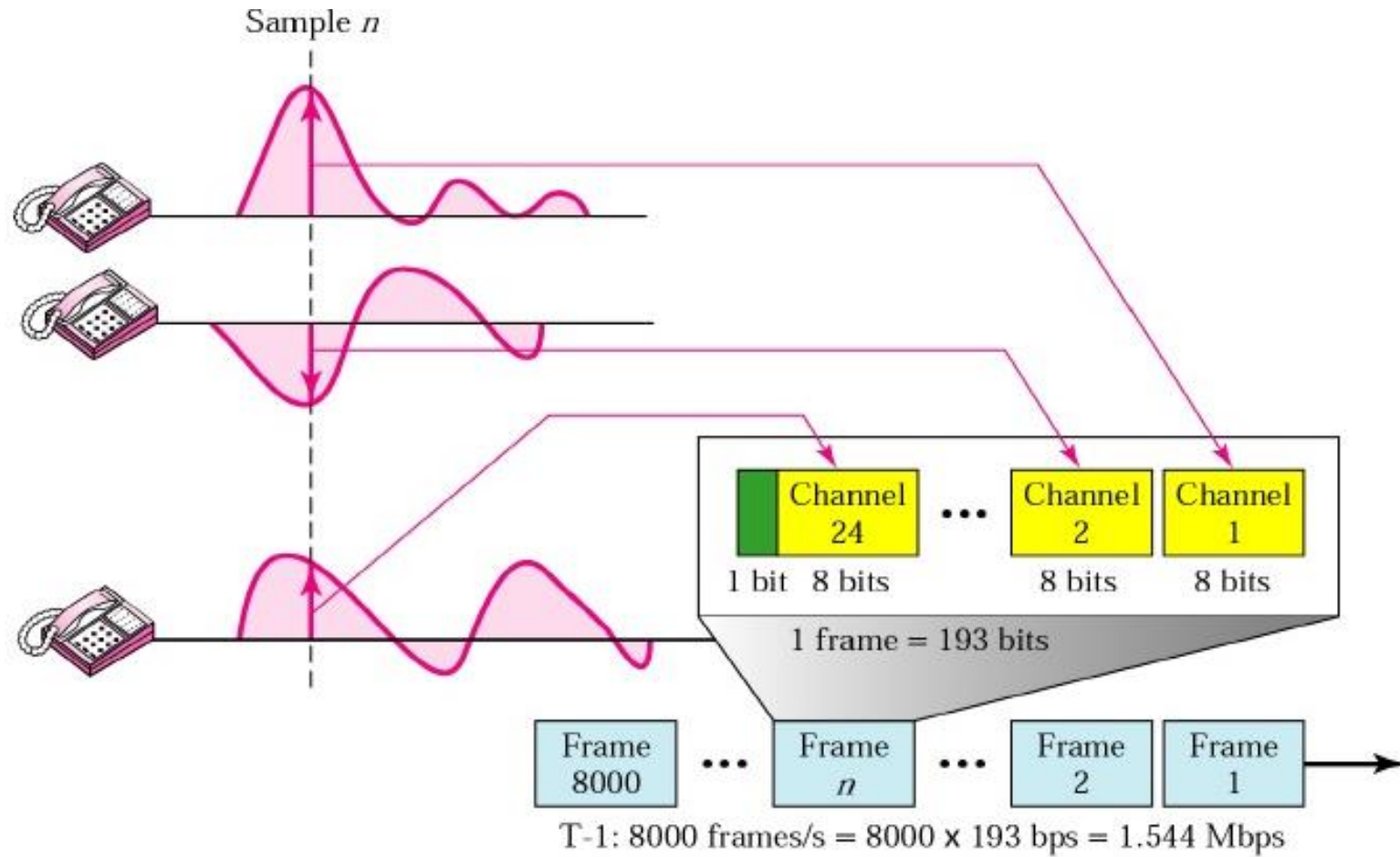
T Lines for Analog Transmission

$4 \text{ kHz} \times 2 = 8 \text{ kHz}$ (sampled twice the maximum frequency) = 8000 times per sec = 8000 samples per sec

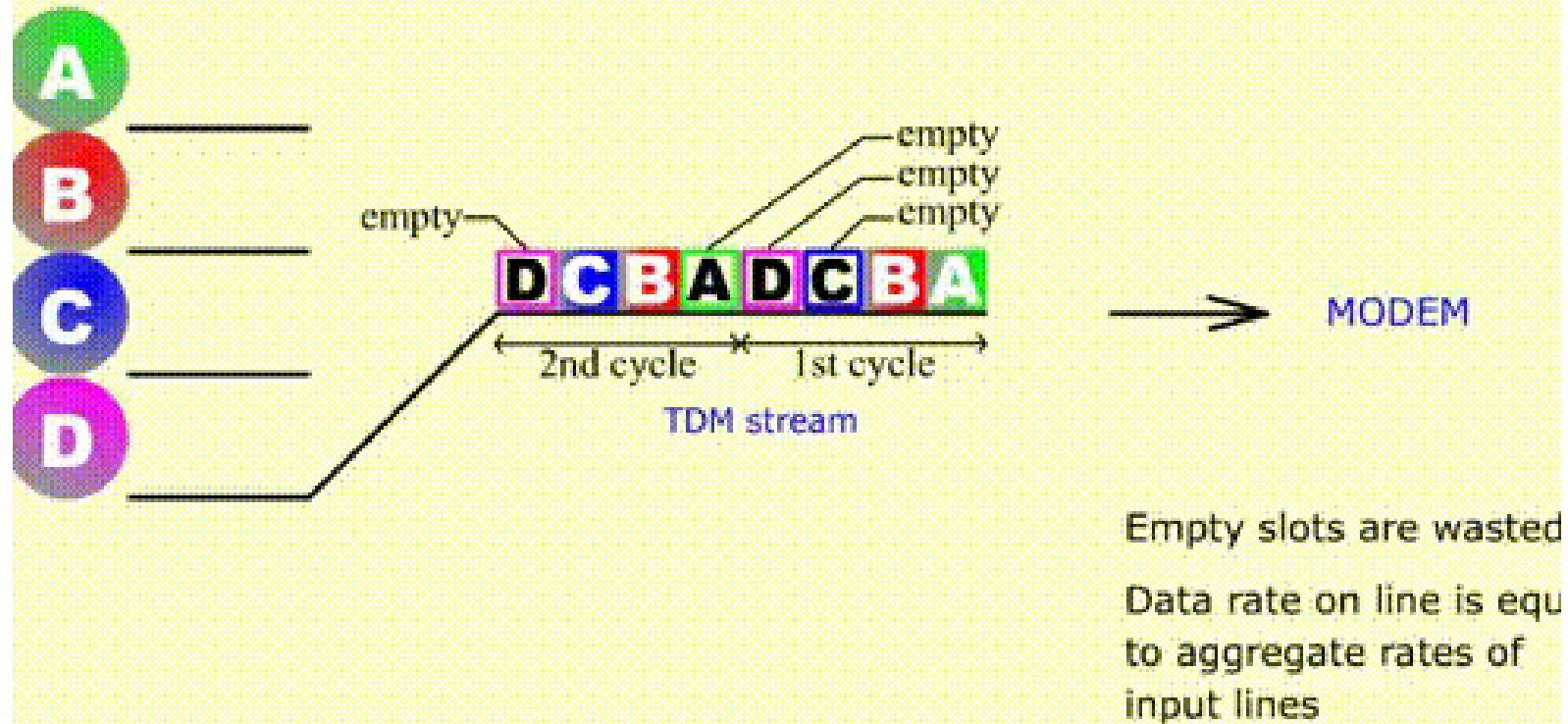


Pulse-code modulation (**PCM**) is a method used to digitally represent sampled analog signals. Sampling is the reduction of a continuous-time signal to a discrete-time signal

T-1 Frame Structure



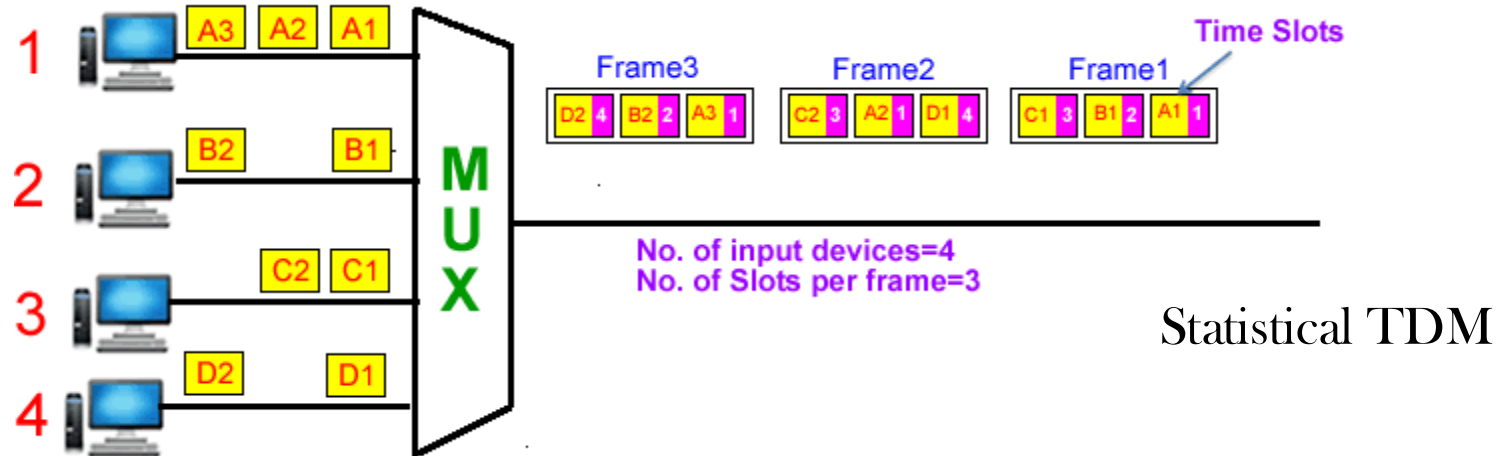
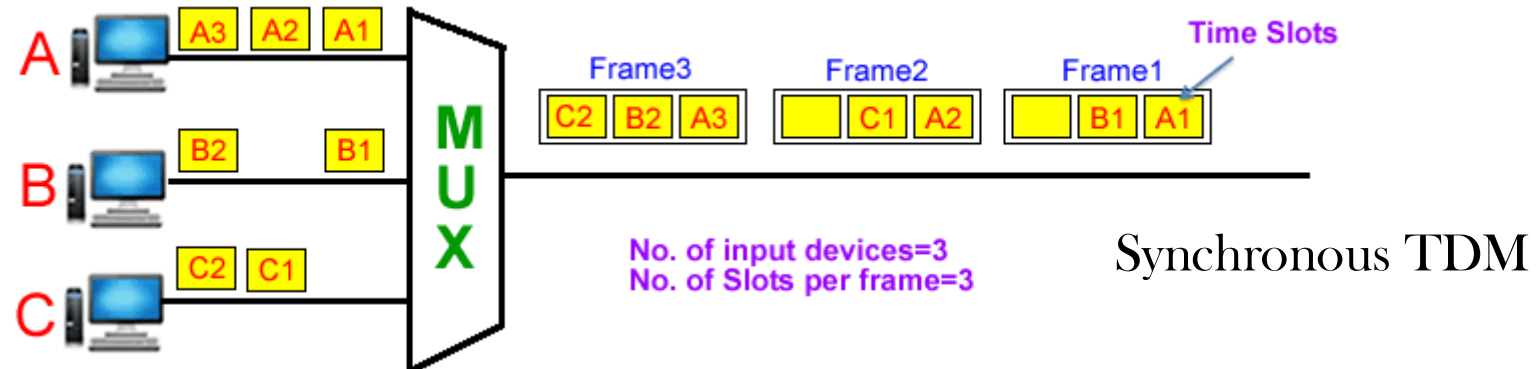
Time Division Multiplexing



Statistical Time-Division Multiplexing

- Slots are dynamically allocated to improve bandwidth efficiency
- The number of slots in each frame is less than the number of input lines.
- The multiplexer checks each input line in round-robin fashion
- It allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line

Statistical Time-Division Multiplexing



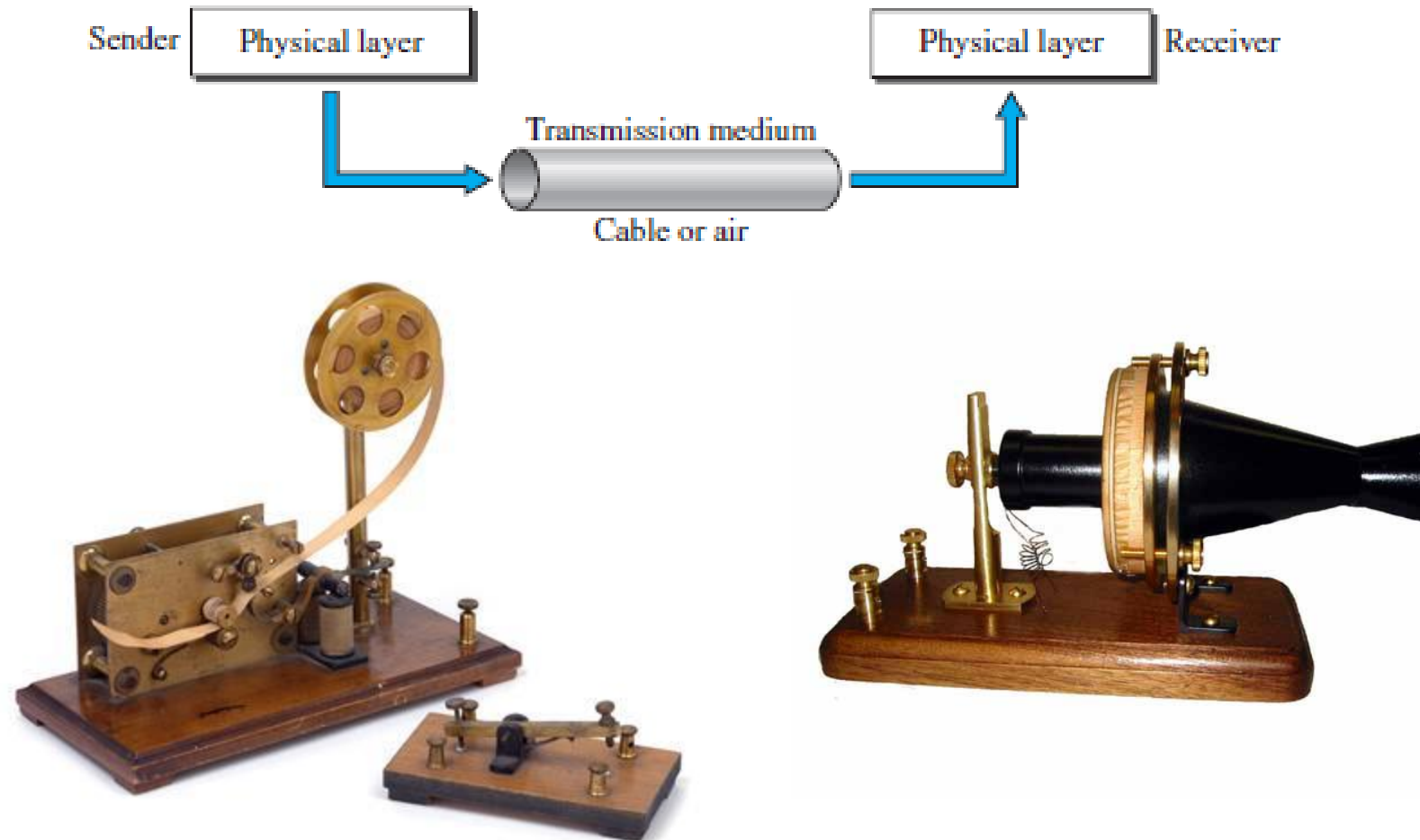
Addressing - Statistical TDM

- A slot needs to carry data as well as the address of the destination.
 - In synchronous TDM, there is no need for addressing
- The addressing in its simplest form can be n bits to define N different output lines with $n = \log_2 N$.
 - For example, for eight different output lines, we need a 3-bit address

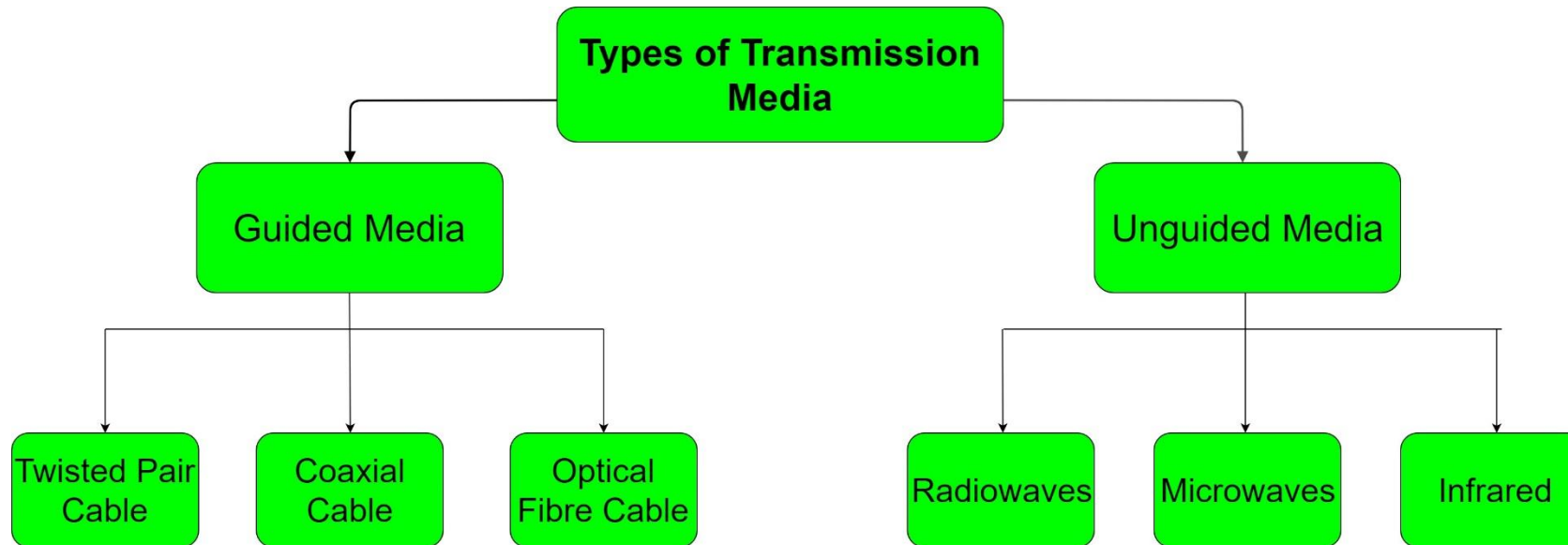
Statistical TDM

- *Slot Size*
 - Since a slot carries both data and an address, the ratio of the data size to address size must be reasonable to make transmission efficient.
- *No Synchronization Bit (because they use address)*
- *Bandwidth*
 - The capacity of the link is normally less than the sum of the capacities of each channel.
 - The designers define the capacity of the link based on the statistics of the load for each channel.

Transmission Media



Transmission Media



Guided Media

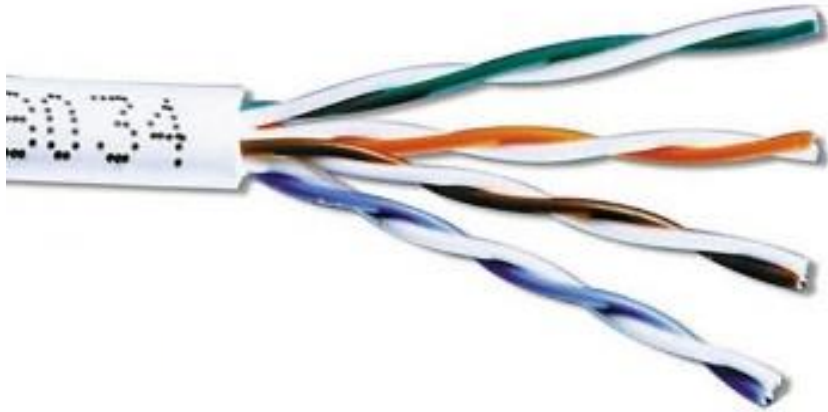
- Those that provide a channel from one device to another.
- Includes
 - Twisted-pair cable
 - Coaxial cable
 - Fiber-optic cable
- A signal traveling along any of these media is directed and contained by the physical limits of the medium.

Twisted Pair Cable

- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.



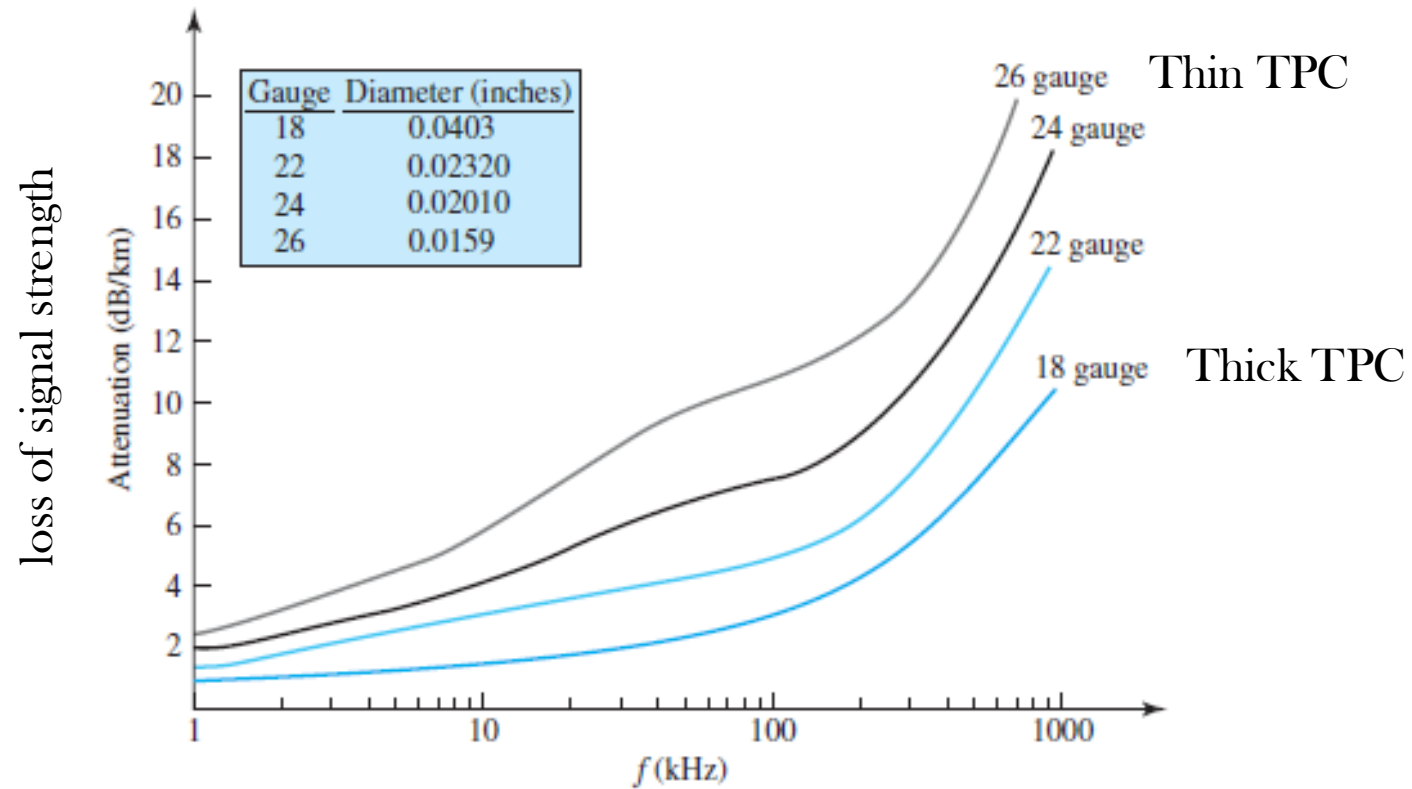
Unshielded Versus Shielded Twisted-Pair Cable



Twisted Pair Cable Connectors – RJ45



Twisted Pair Cable Performance

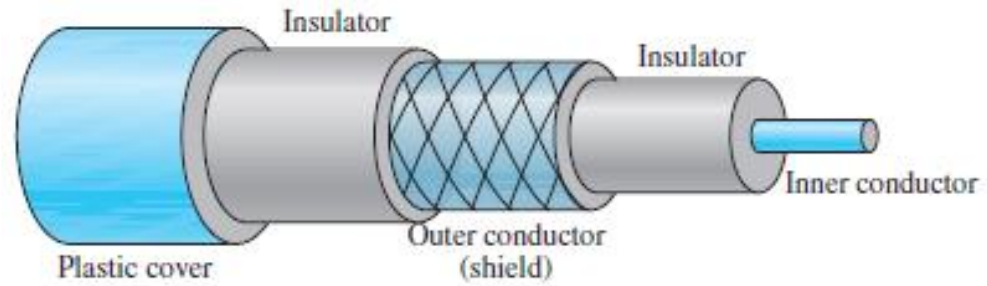


gauge is a measure of the thickness of the wire.

Twisted Pair Cable Applications

- In telephone lines to provide voice and data channels.
- DSL lines that provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.

Coaxial Cable

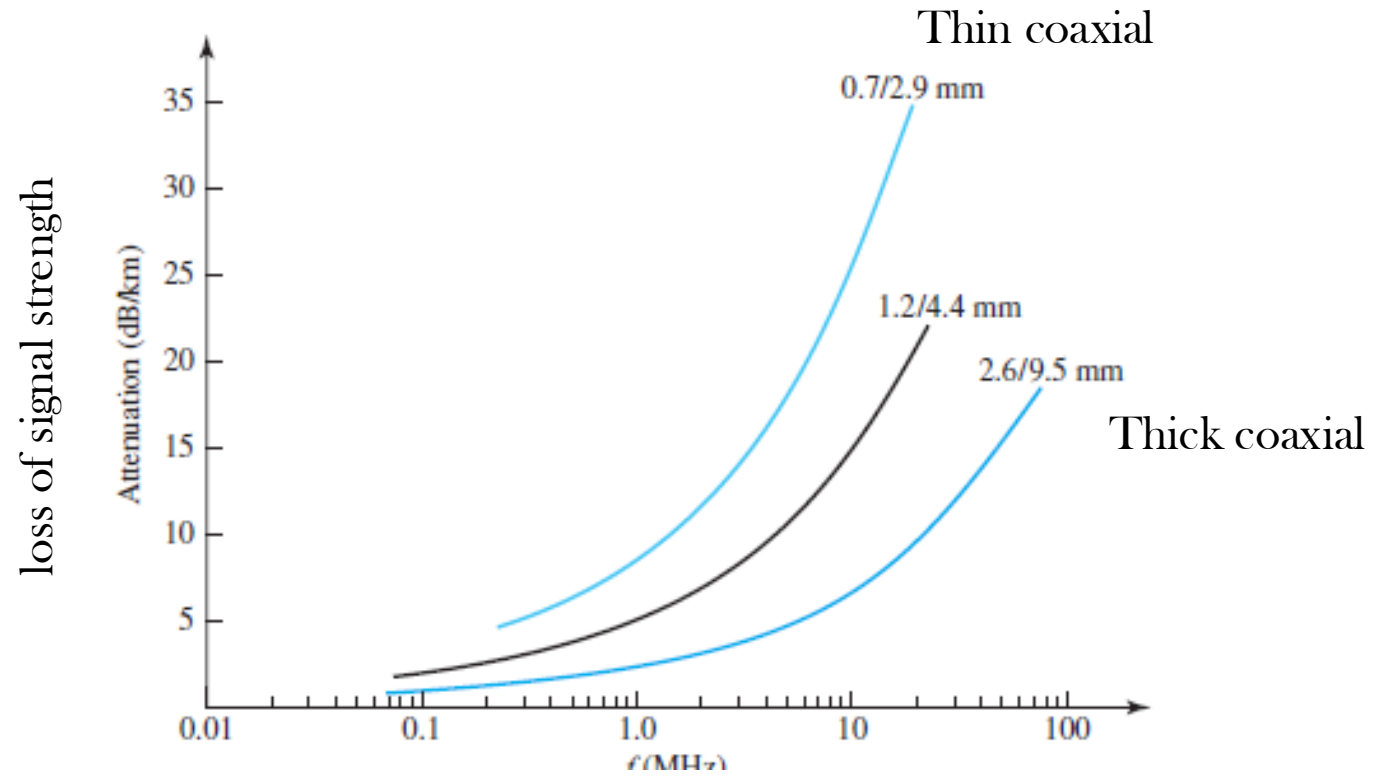


Coaxial Cable Connectors

- Bayonet Neill-Concelman (BNC) connector



Coaxial Cable Performance

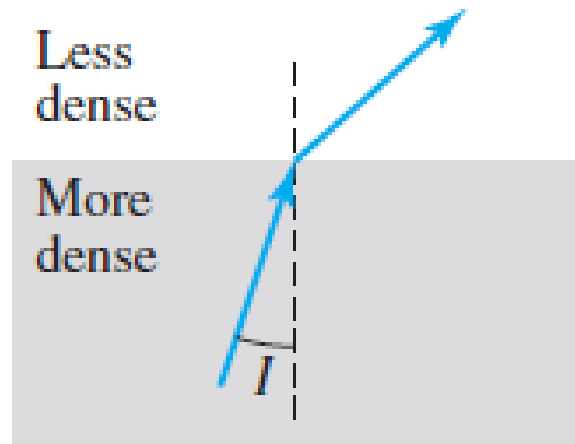


Coaxial Cable Applications

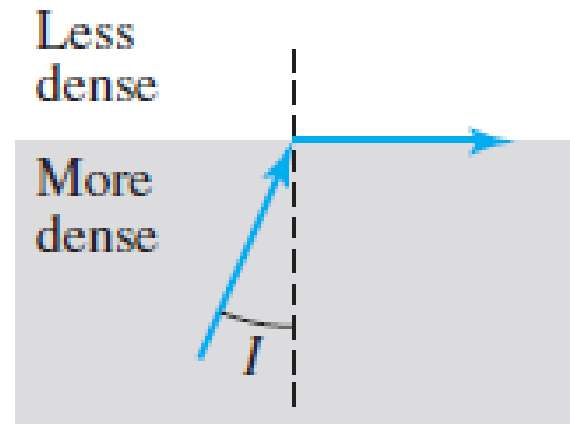
- Widely used in analog telephone networks where a single coaxial network could carry 10,000 voice signals.
- Later it was used in digital telephone networks where a single coaxial cable could carry digital data up to 600 Mbps.

Fiber-Optic Cable

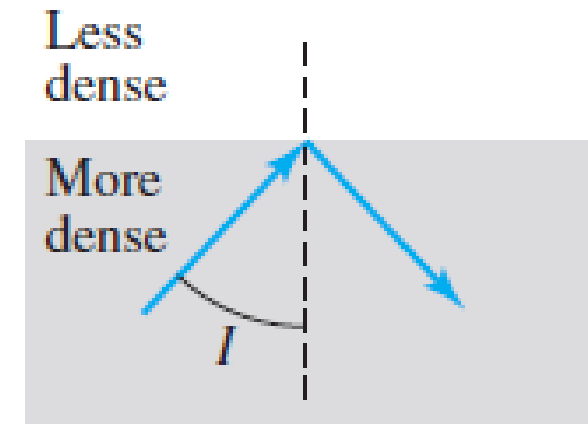
- Made of glass or plastic and transmits signals in the form of light.



$I < \text{critical angle,}$
refraction



$I = \text{critical angle,}$
refraction

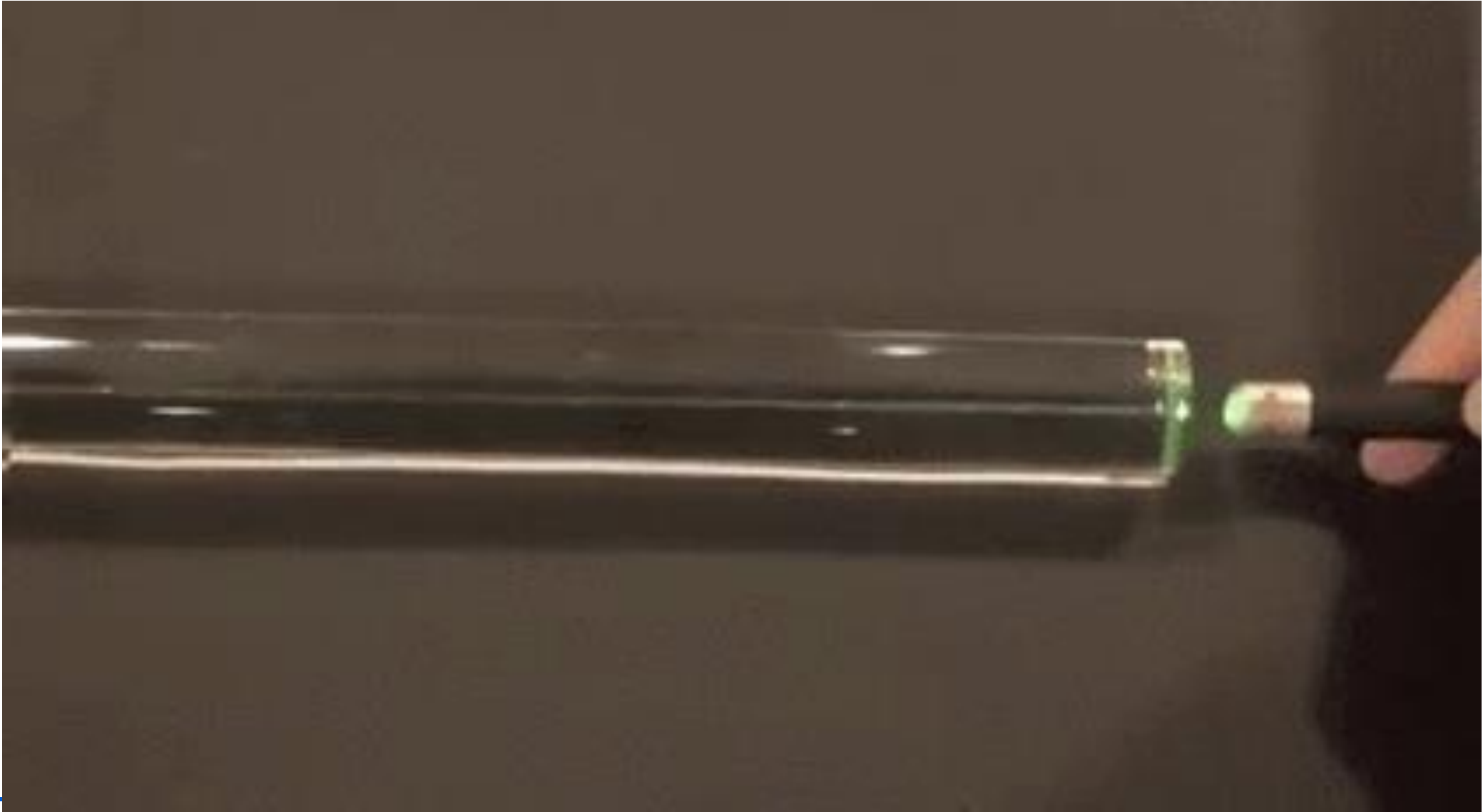


$I > \text{critical angle,}$
reflection

I - angle of incidence



Image Source: <https://www.thefoa.org/tech/ref/basic/fiber.html>

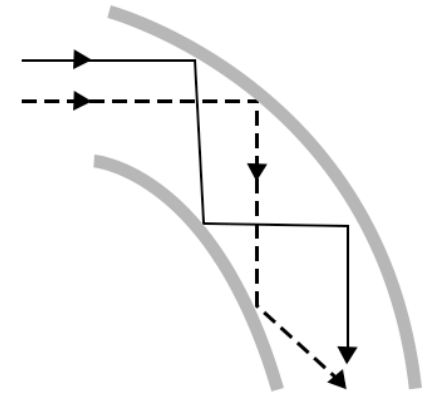
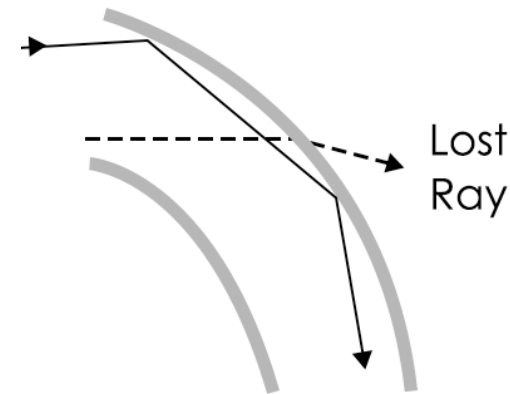
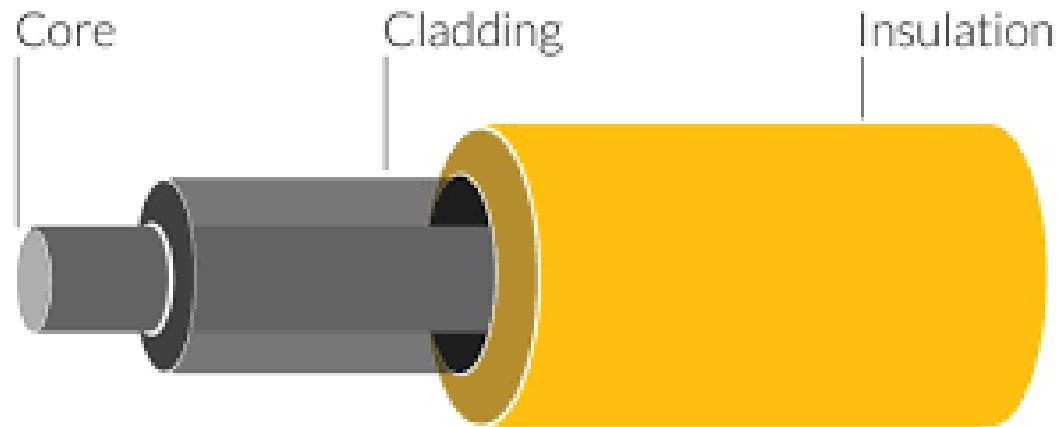


Unit 2

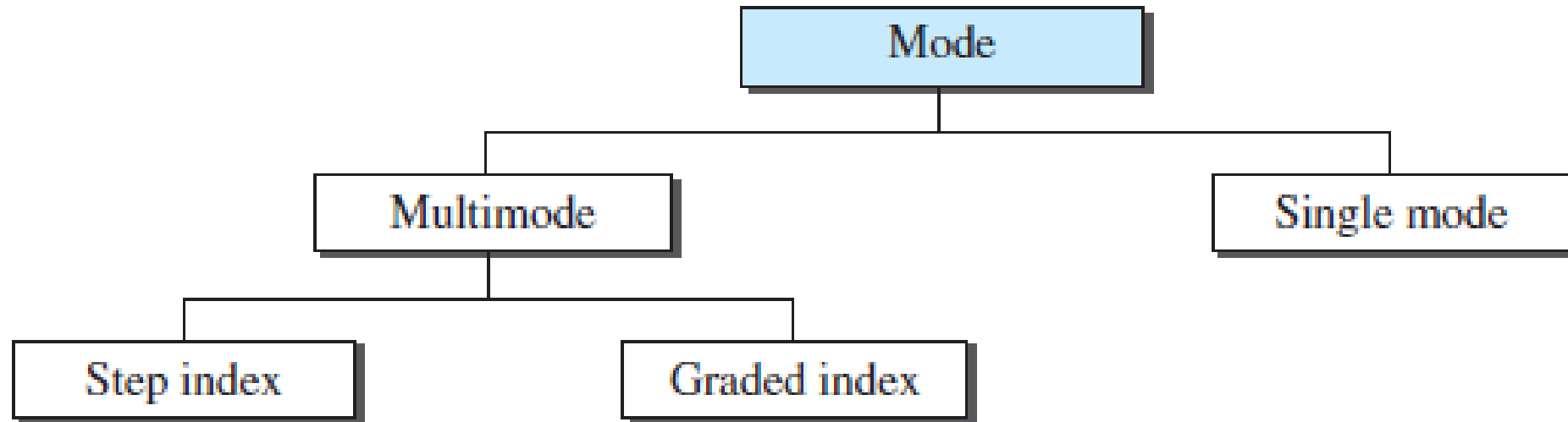
Image Source: <https://giflounge.tumblr.com/post/166178491910/path-of-light-traveling-through-a-fiber-optic>

Fiber-Optic Cable

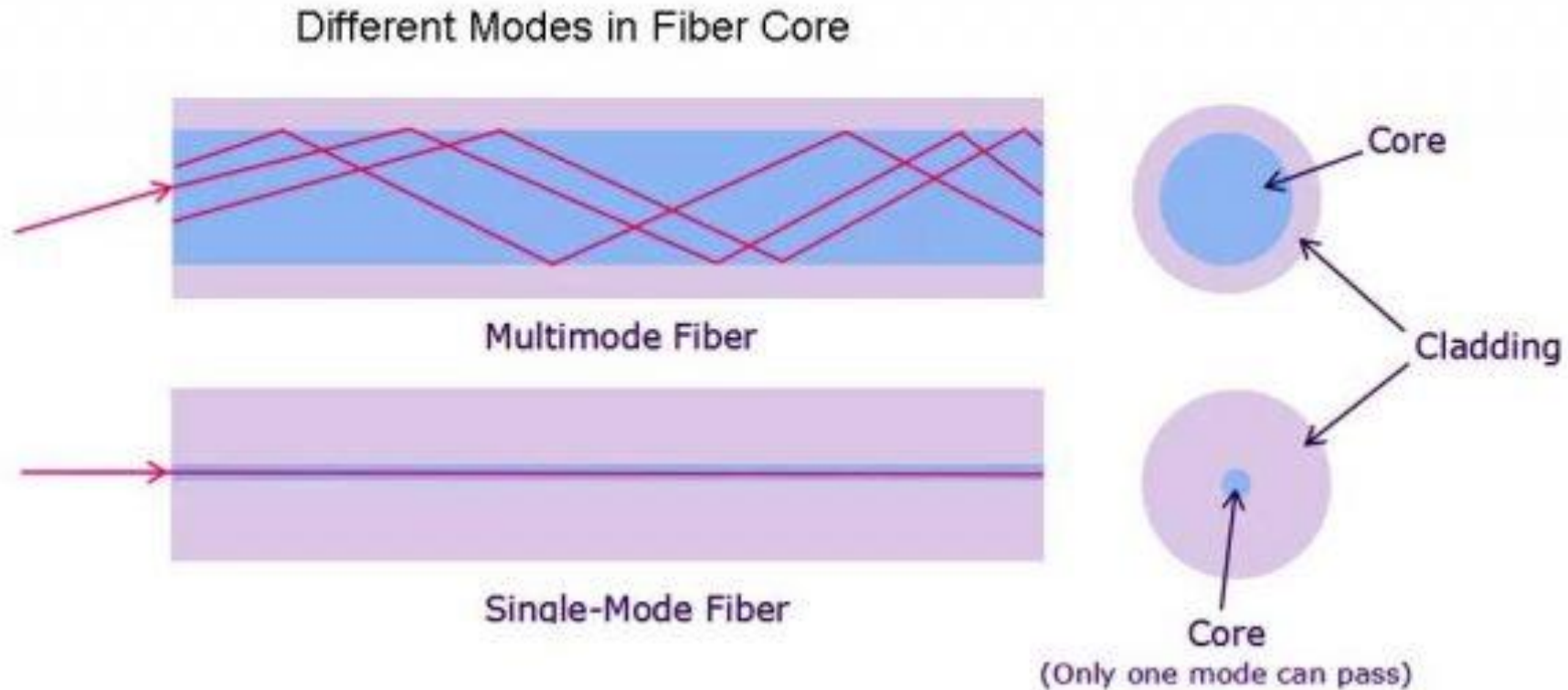
- Optical fibers use reflection to guide light through a channel.
- A glass or plastic **core** is surrounded by a **cladding** of less dense glass or plastic.



Fiber-Optic Cable Propagation Modes



Single Mode & Multimode



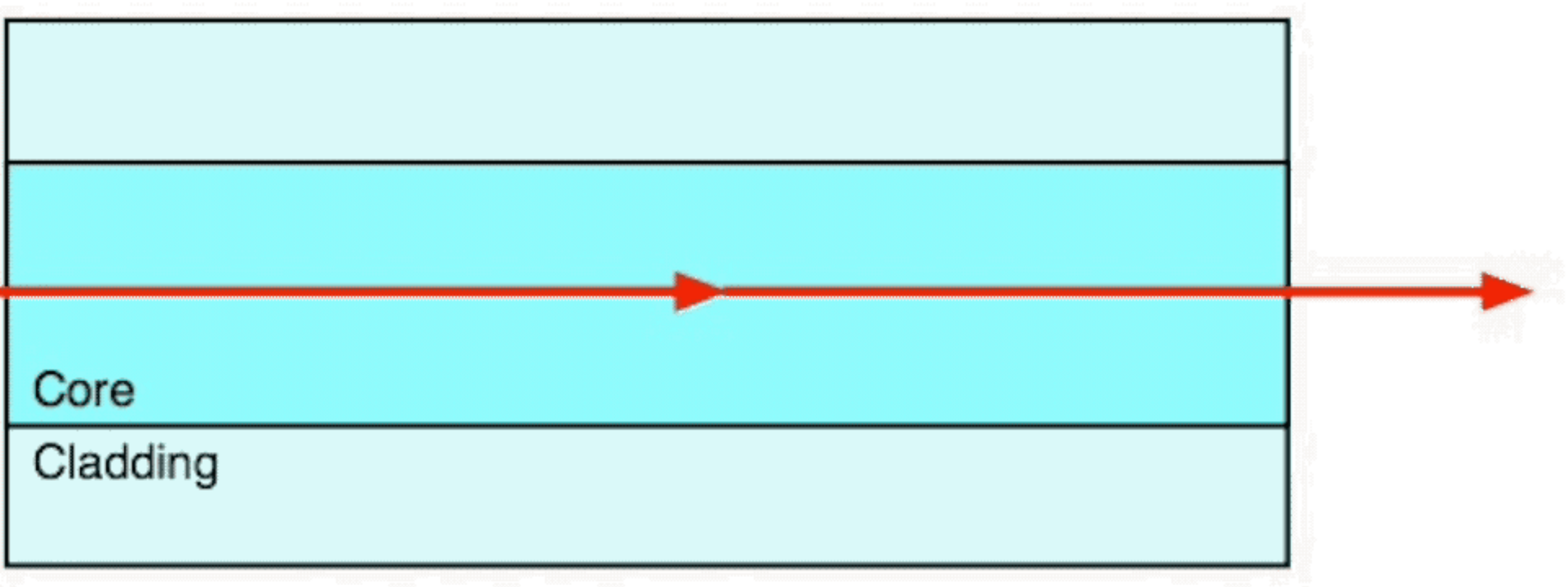
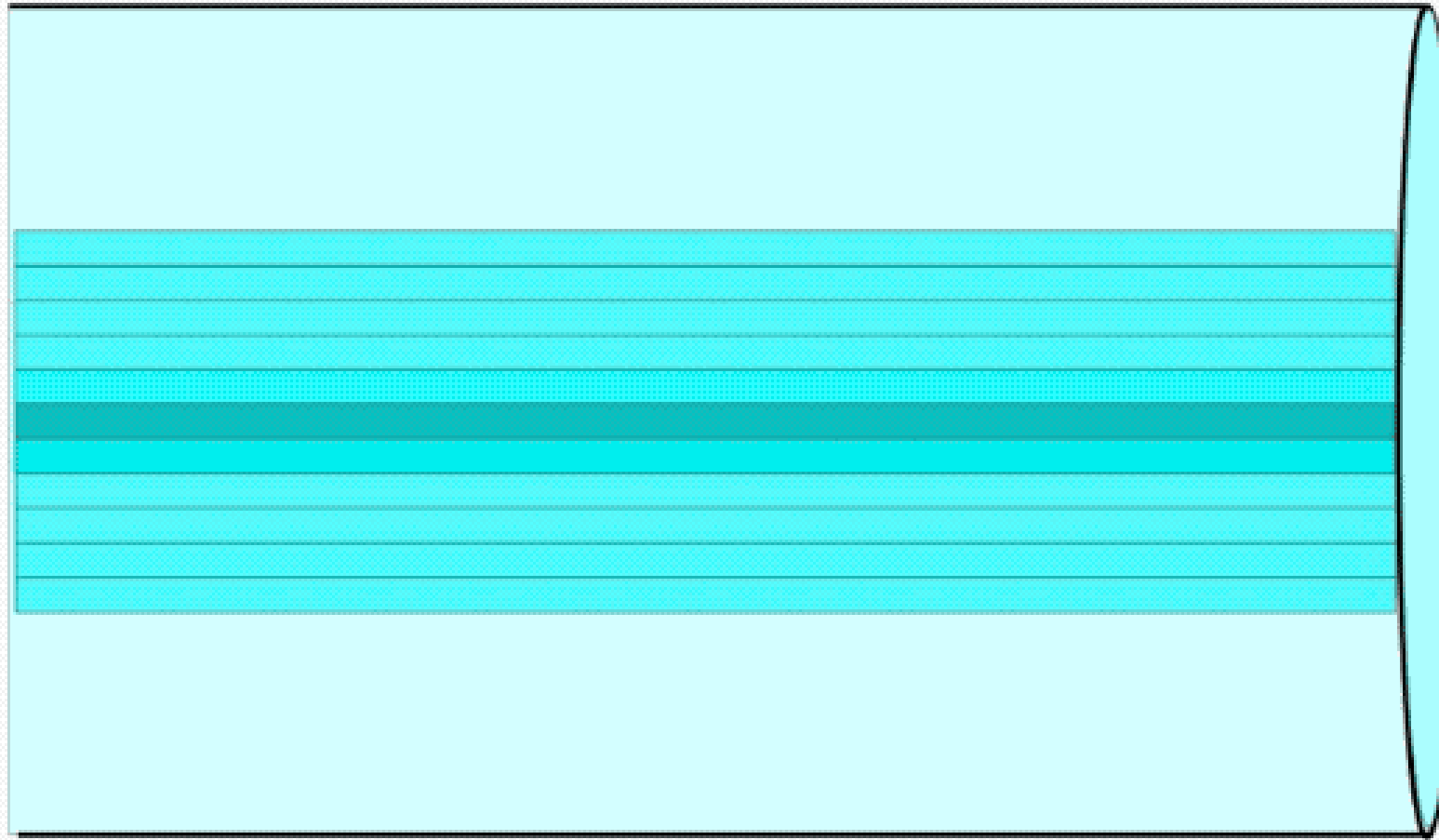
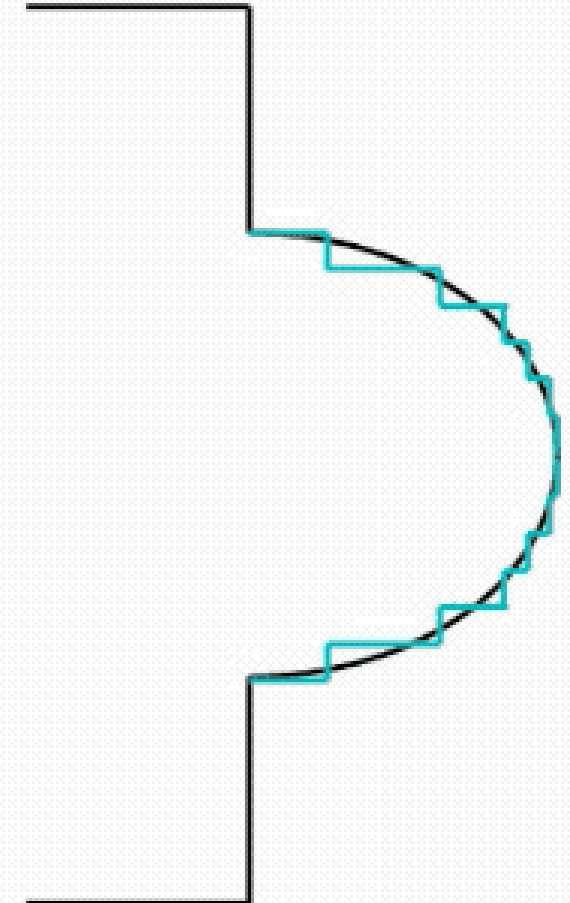


Image Source: <https://www.thefoa.org/tech/ref/basic/fiber.html>

Wide core with uniform density. Light reflects in zig-zag paths.



Path Of Light In Core



Index Profile
Of Fiber

Core index decreases continuously from the center out toward the cladding.

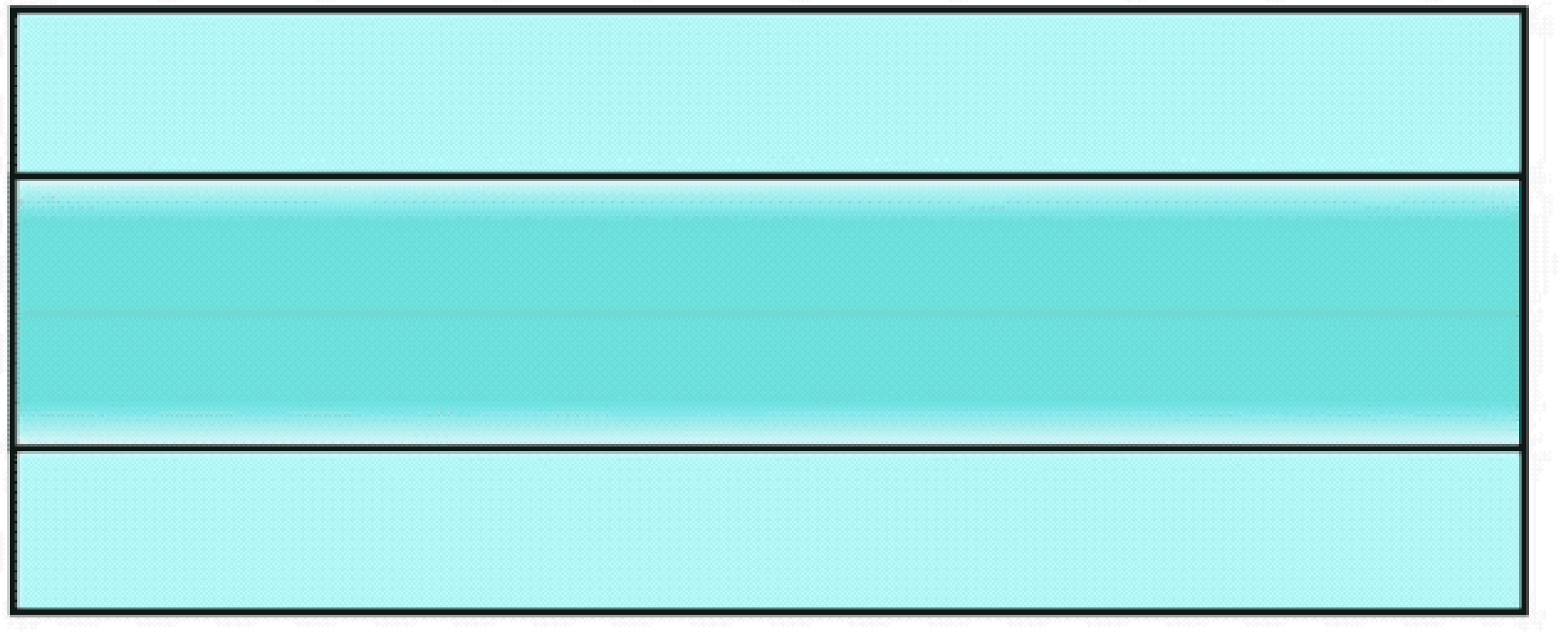
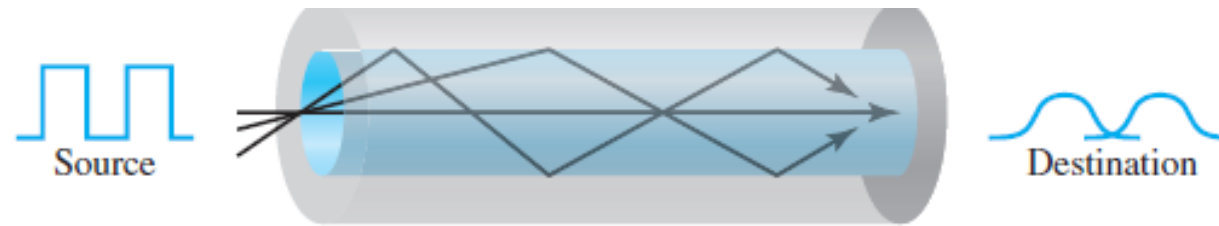
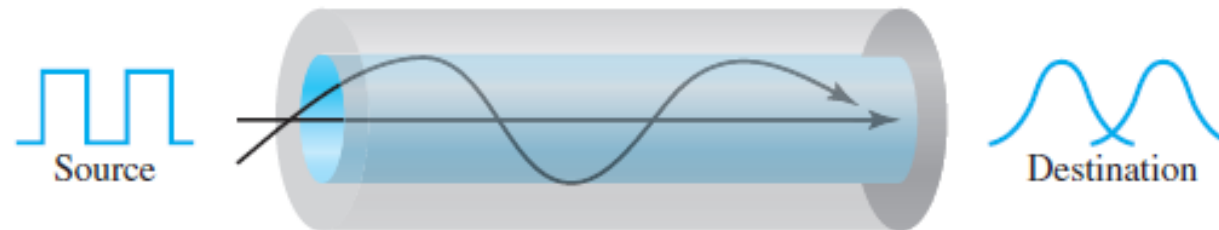


Image Source: <https://www.thefoa.org/tech/ref/basic/fiber.html>

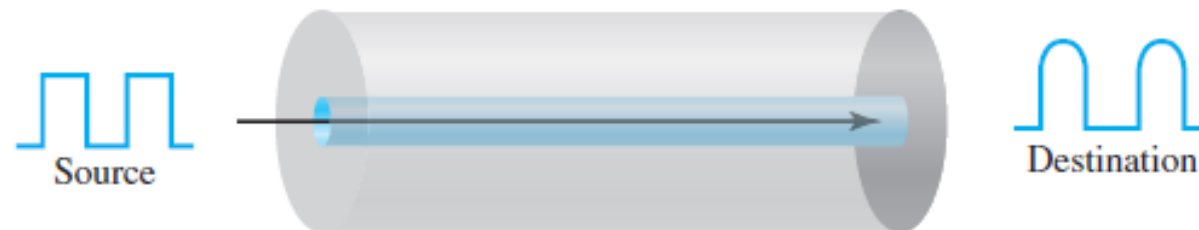
Fiber-Optic Cable Propagation Modes



a. Multimode, step index

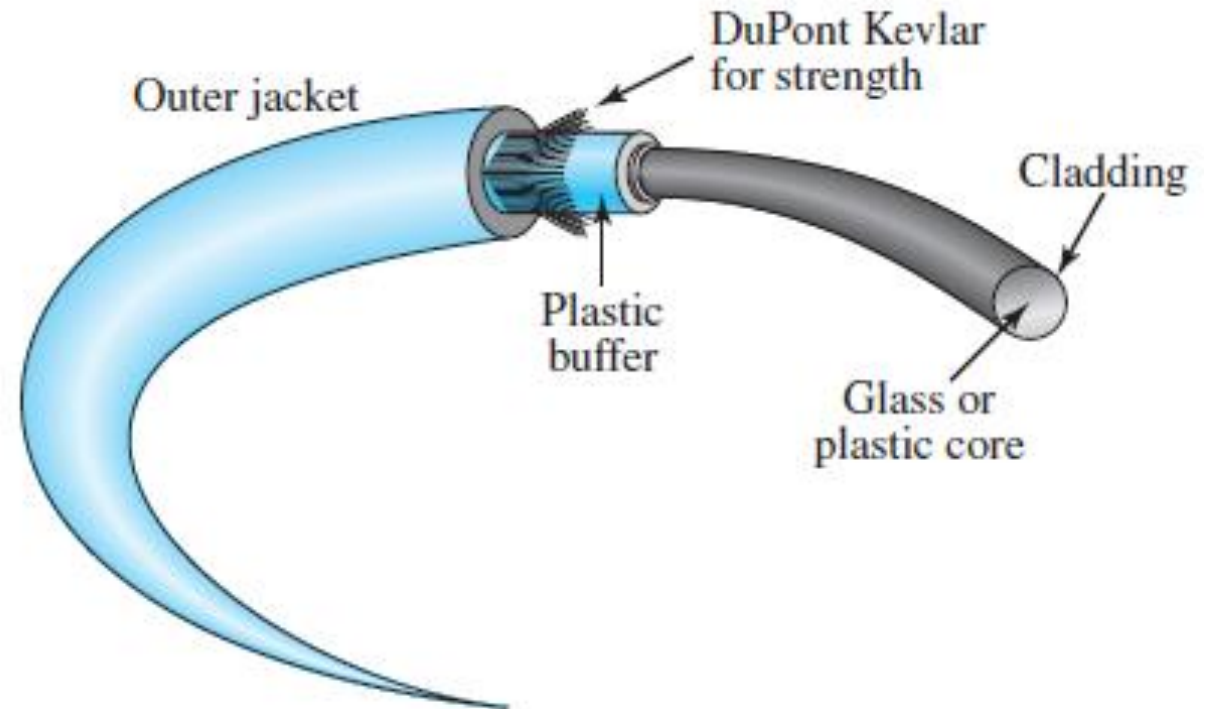
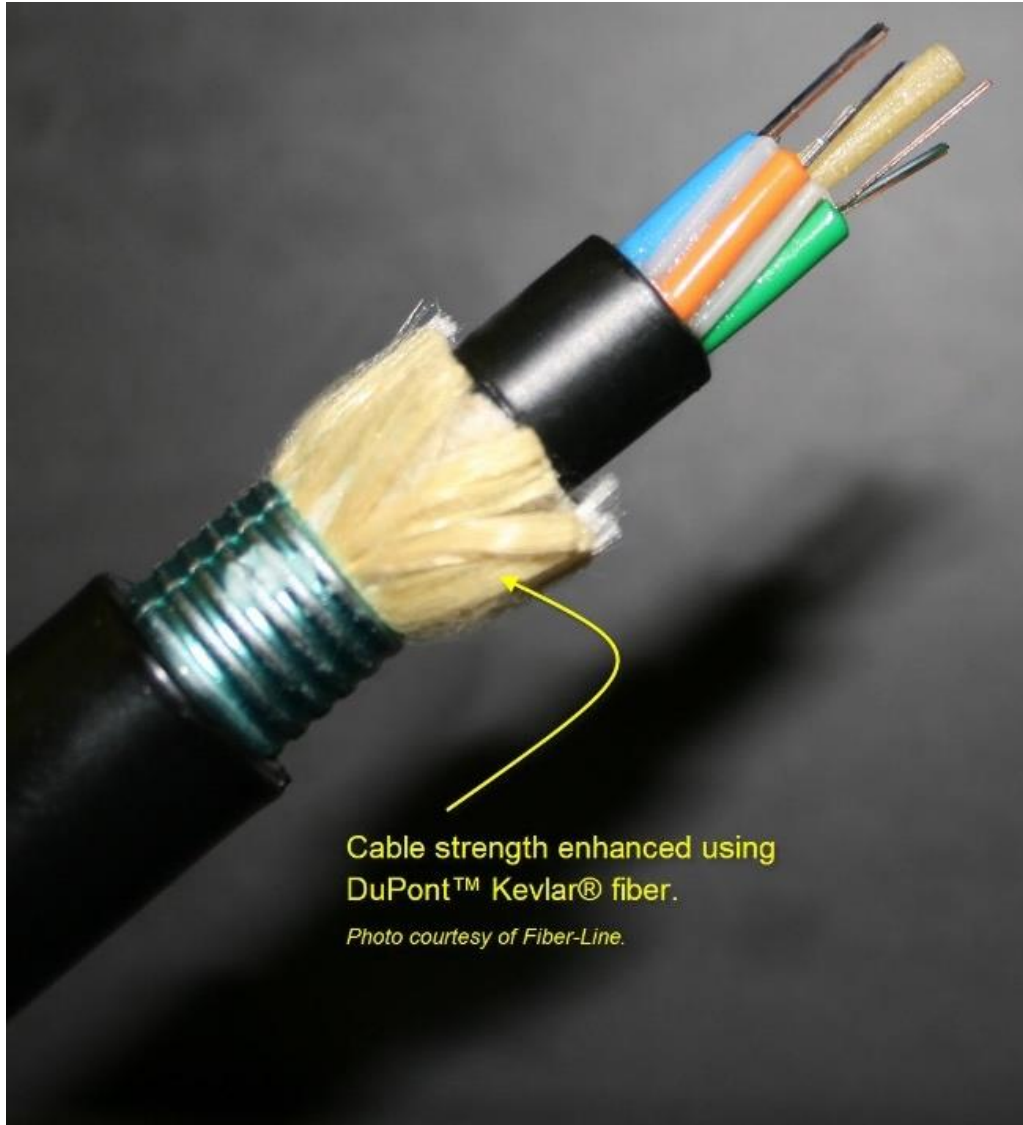


b. Multimode, graded index



c. Single mode

Cable Composition



Fiber-Optic Cable Connectors



ST
type connector

Straight Tip



SC
type connector

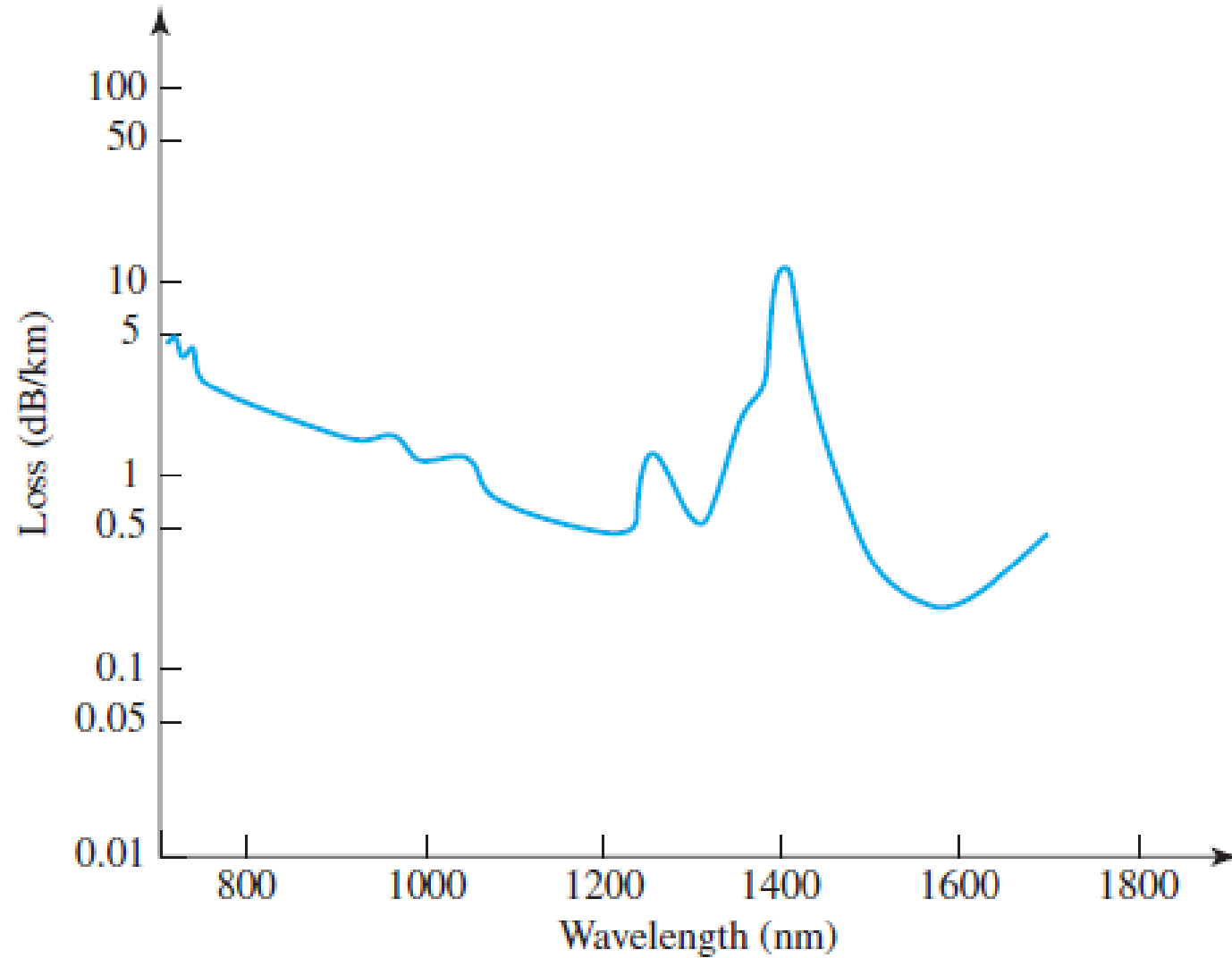
Subscriber Channel



MTRJ
type connector

Mechanical Transfer Registered Jack

Fiber Optic Performance



Fiber Optic Applications

- Backbone Networks
- SONET (Synchronous Optical Network)
- Cable TV companies use a combination of fiber optic and coaxial

Fiber Optic Advantages

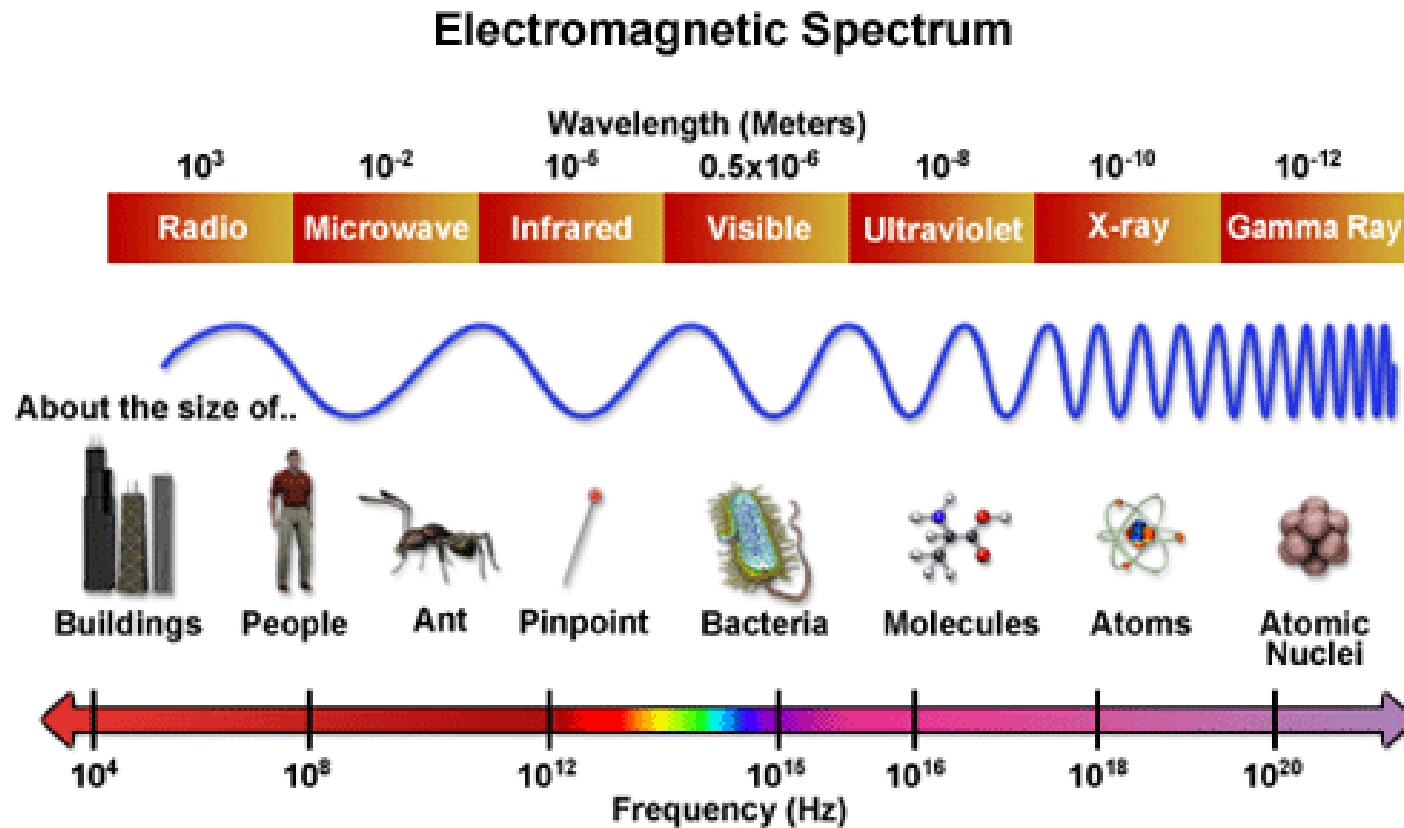
- Higher bandwidth
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials
- Light weight
- Greater immunity to tapping

Fiber Optic Disadvantages

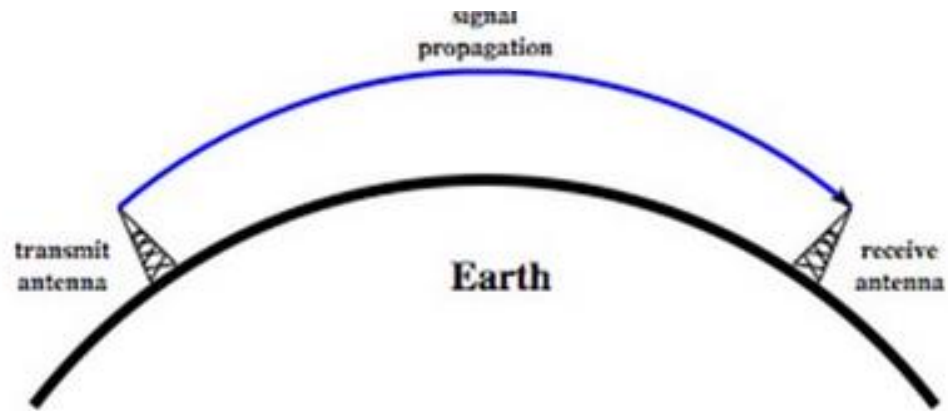
- Installation and maintenance
- Unidirectional light propagation
- Cost

Unguided Media: Wireless

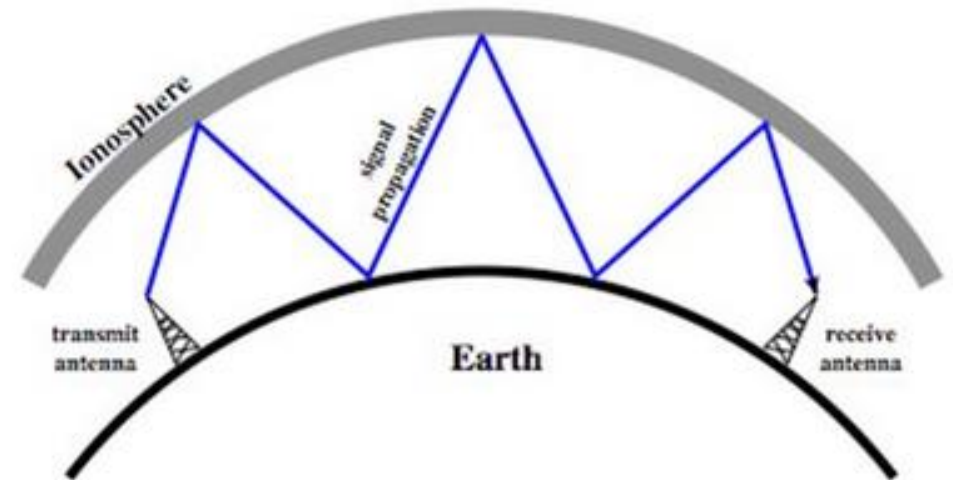
- Transport electromagnetic waves without using a physical conductor.



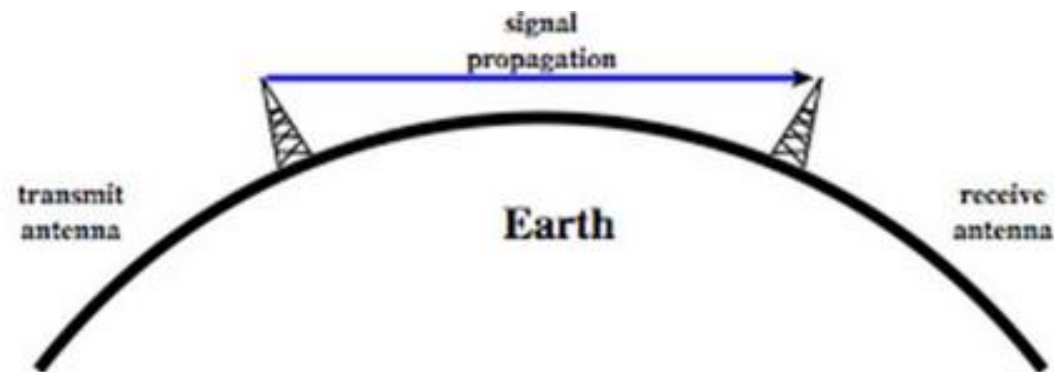
Propagation Methods



(a) Ground-wave propagation (below 2 MHz)



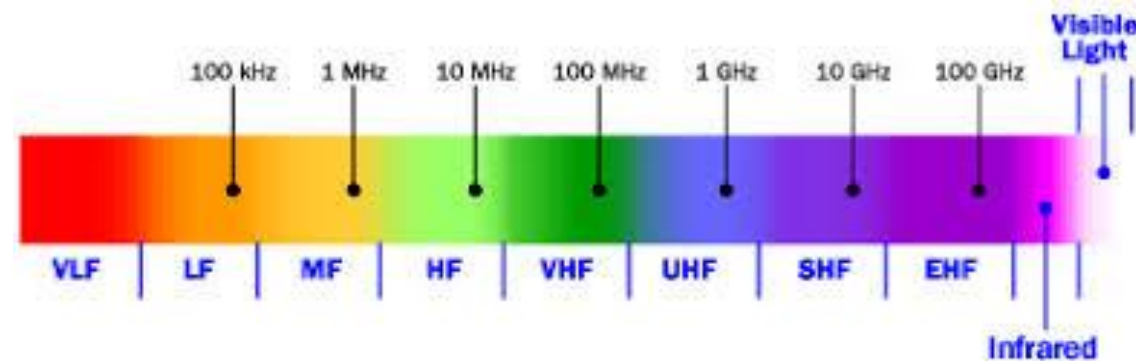
(b) Sky-wave propagation (2 to 30 MHz)



(c) Line-of-sight (LOS) propagation (above 30 MHz)

Bands

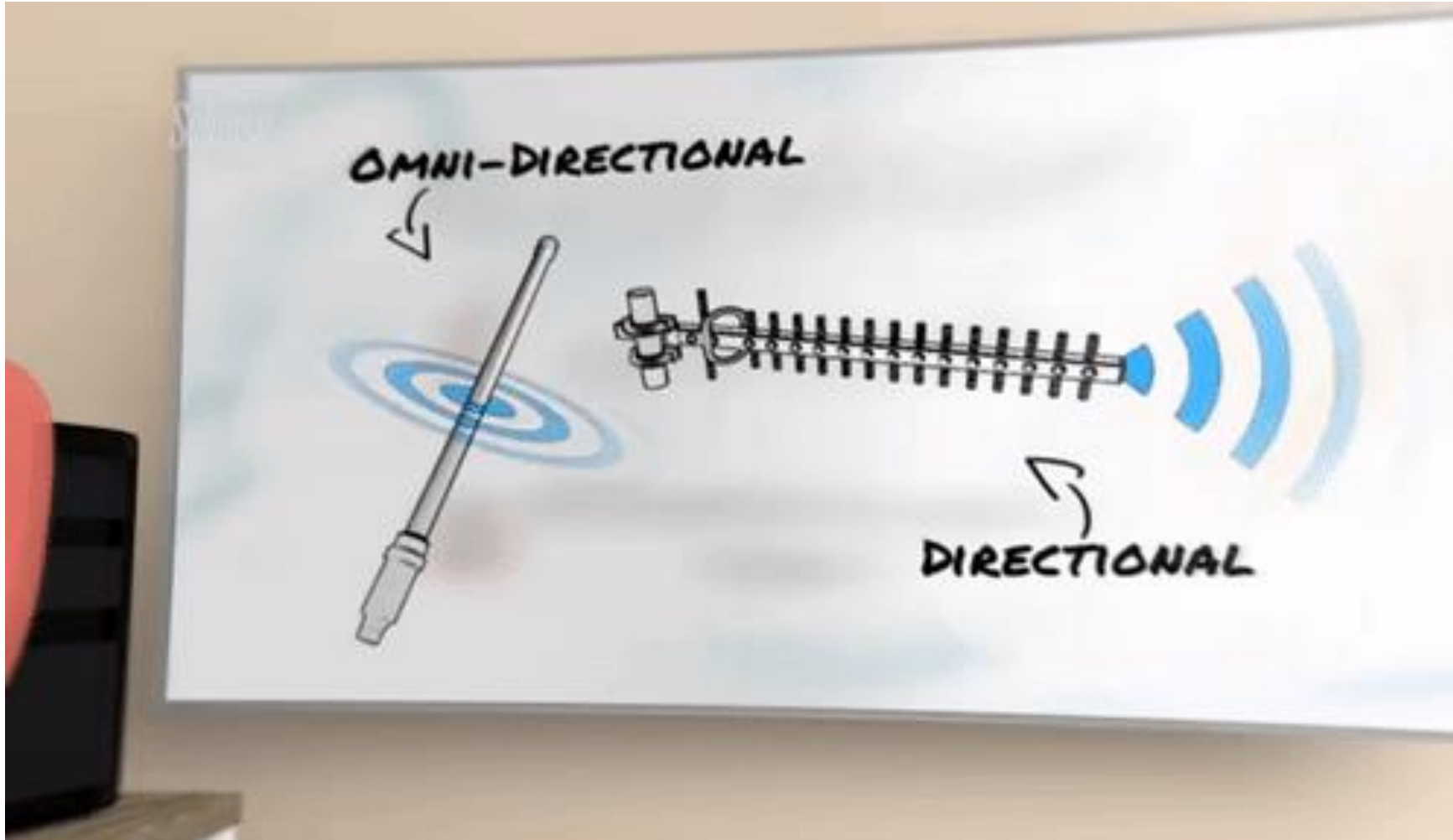
<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
very low frequency (VLF)	3–30 kHz	Ground	Long-range radio navigation
low frequency (LF)	30–300 kHz	Ground	Radio beacons and navigational locators
middle frequency (MF)	300 kHz–3 MHz	Sky	AM radio
high frequency (HF)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft
very high frequency (VHF)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
ultrahigh frequency (UHF)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
superhigh frequency (SHF)	3–30 GHz	Line-of-sight	Satellite
extremely high frequency (EHF)	30–300 GHz	Line-of-sight	Radar, satellite



Radio Waves

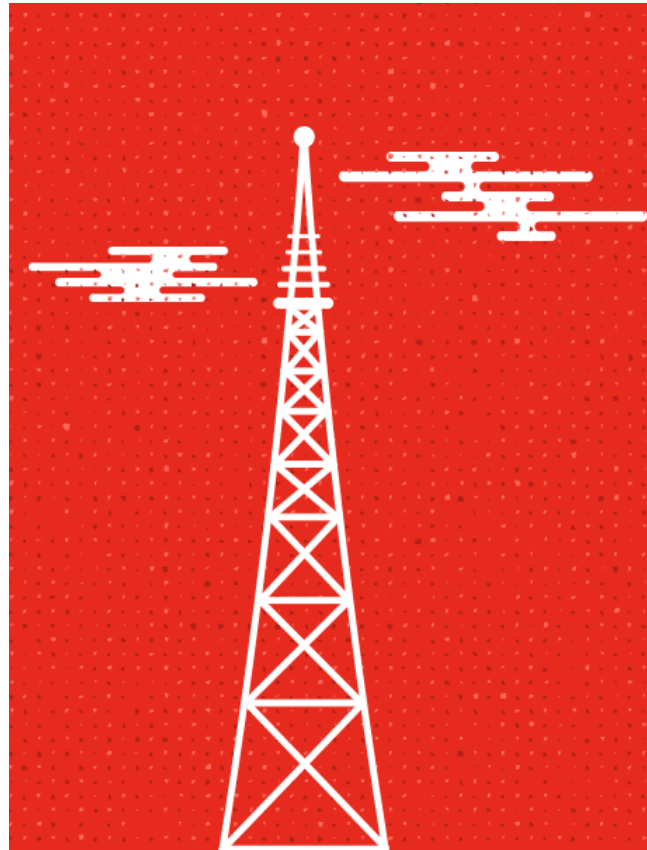
- Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz.
- For the most part, they are omnidirectional.
- Waves that propagate in the sky mode, can travel long distances.
 - This makes for long-distance broadcasting such as AM radio.
- Waves of low and medium frequencies, can penetrate walls.
 - This can be both an advantage and a disadvantage.

Omnidirectional Antenna



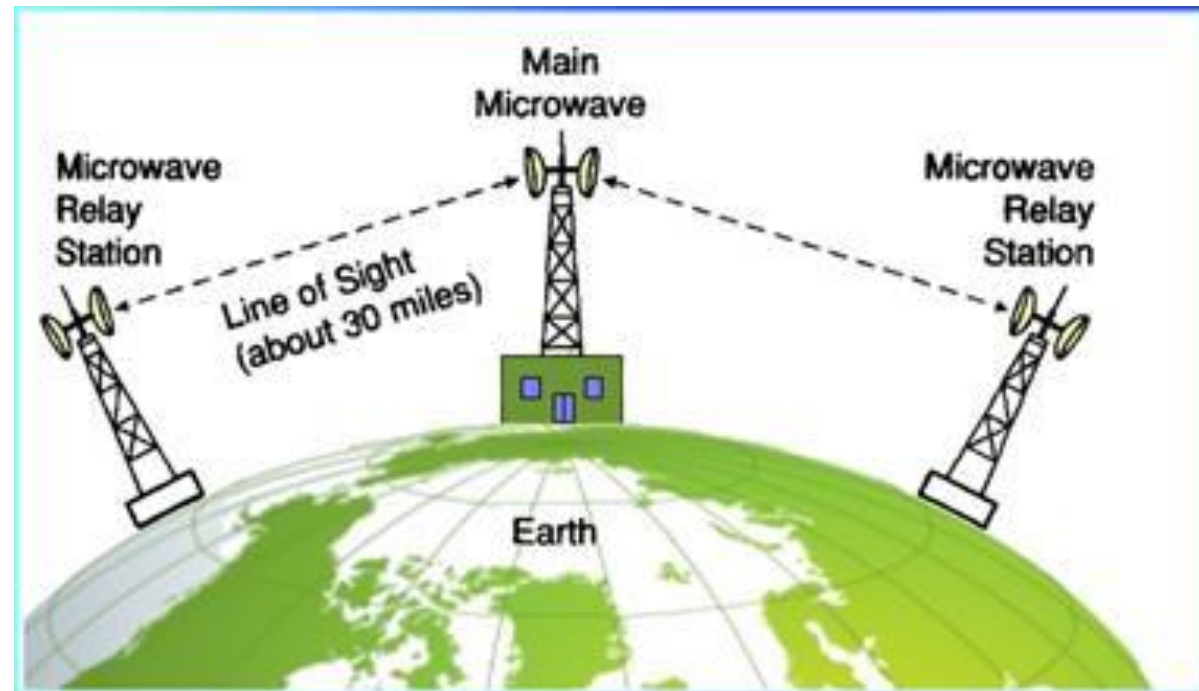
Radio Waves Applications

- The omnidirectional characteristics of radio waves make them useful for multicasting.



Microwaves

- Electromagnetic waves having frequencies between 1 and 300 GHz.
- Unidirectional.
 - When an antenna transmits microwaves, they can be narrowly focused.
 - The sending and receiving antennas need to be aligned.



Microwaves Characteristics

- Microwave propagation is line-of-sight.
 - For long distance, antennas must be tall and repeaters are often needed.
- Very high-frequency microwaves cannot penetrate walls.
 - This characteristic can be a disadvantage if receivers are inside buildings.
- The microwave band is relatively wide, almost 299 GHz.
 - Therefore wider subbands can be assigned, and a high data rate is possible.
- Use of certain portions of the band requires permission from authorities.

Unidirectional Antennas



Parabolic dish antenna



Horn antenna

Microwave Applications

- Microwaves, due to their unidirectional properties, are very useful in unicast (one-to-one) communication.

Infrared

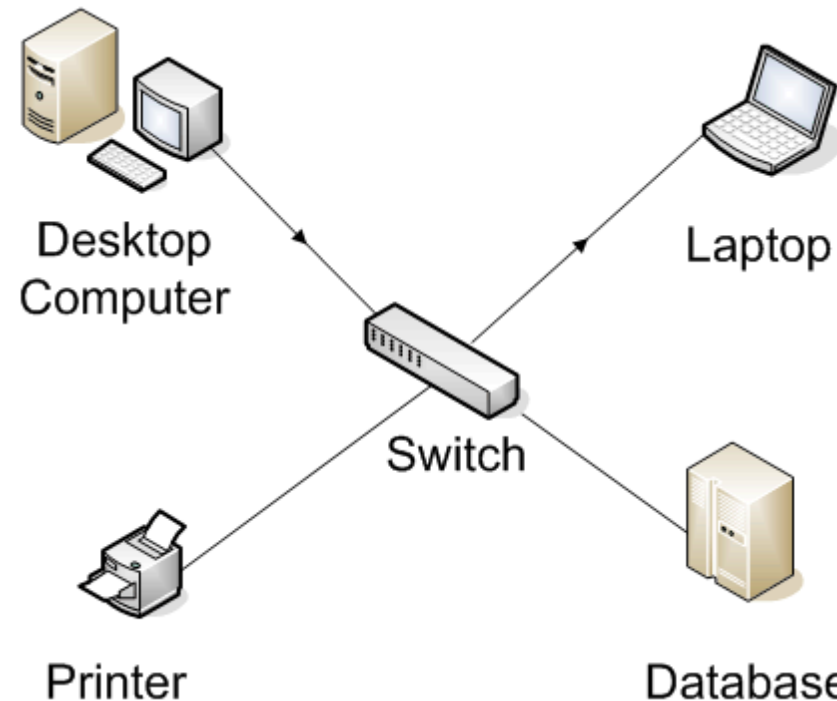
- Waves with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm).
- Can be used for short-range communication.
- Having high frequencies they cannot penetrate walls.

Infrared Applications

- Such a wide bandwidth can be used to transmit digital data with a very high data rate.
- IrDA has established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.
 - *Infrared Data Association* (IrDA), an association for sponsoring the use of infrared waves

Switching

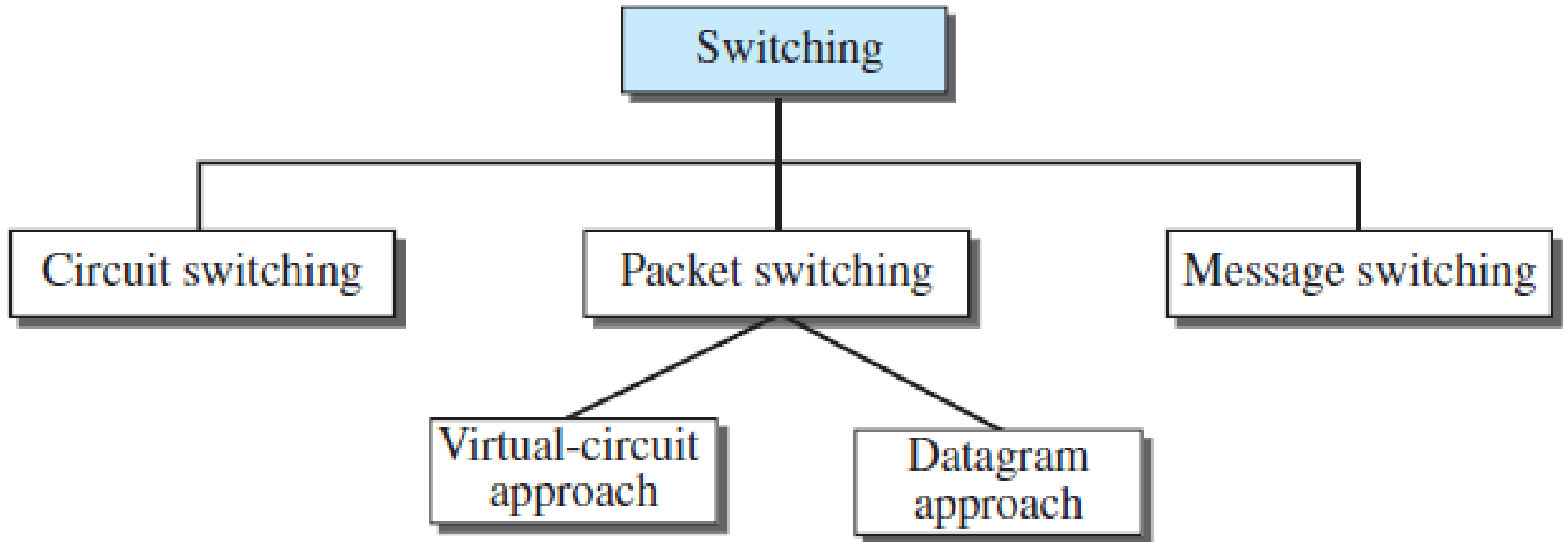
- A switched network consists of a series of interlinked nodes/devices called *switches*.
- Capable of creating temporary connections between two or more devices linked to the switch



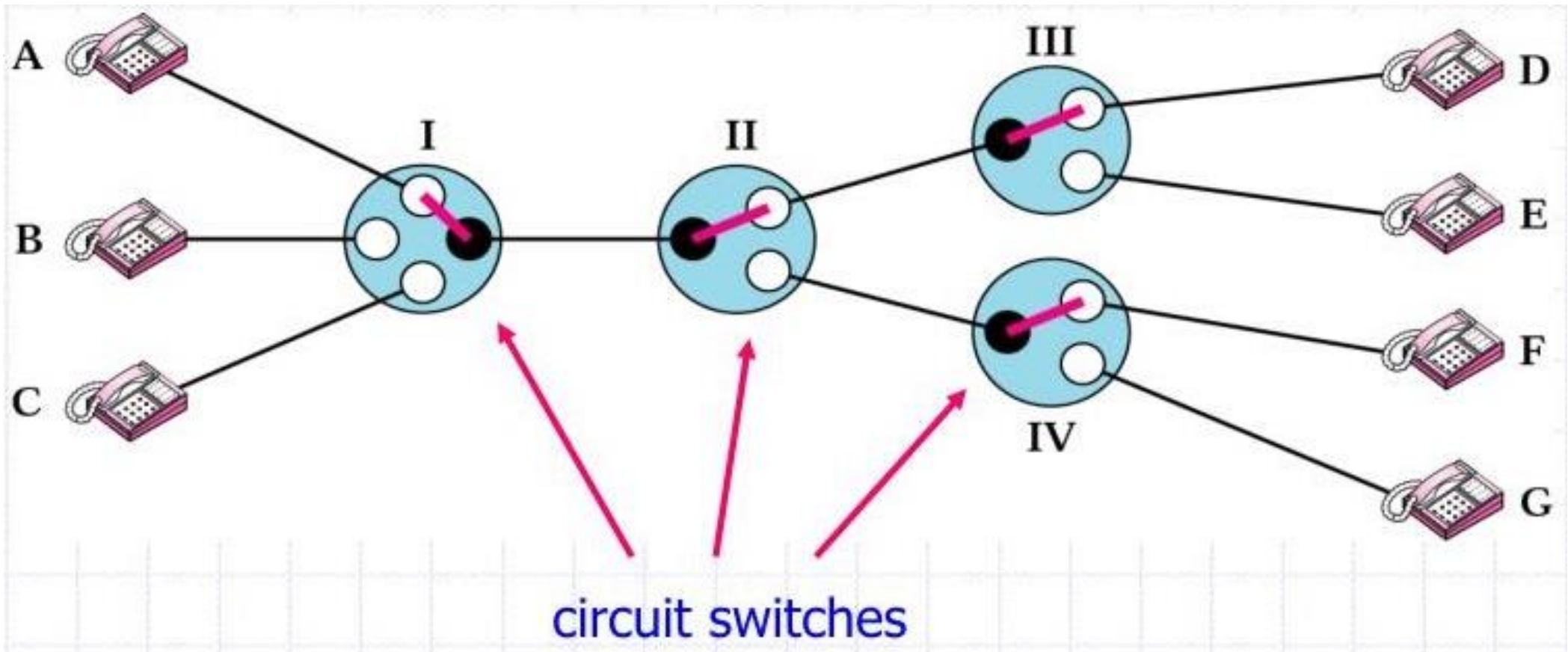
Switch

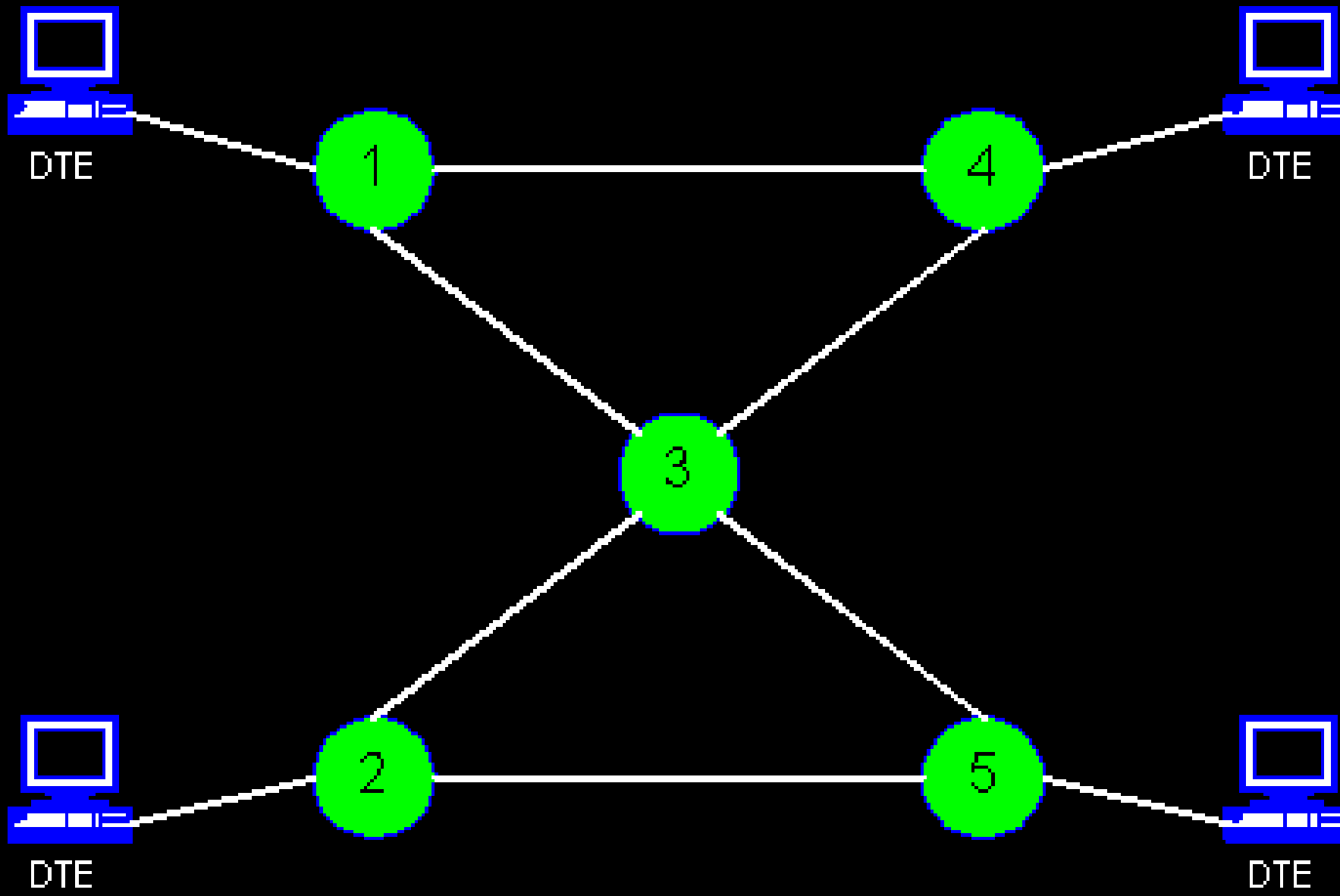


Methods of Switching



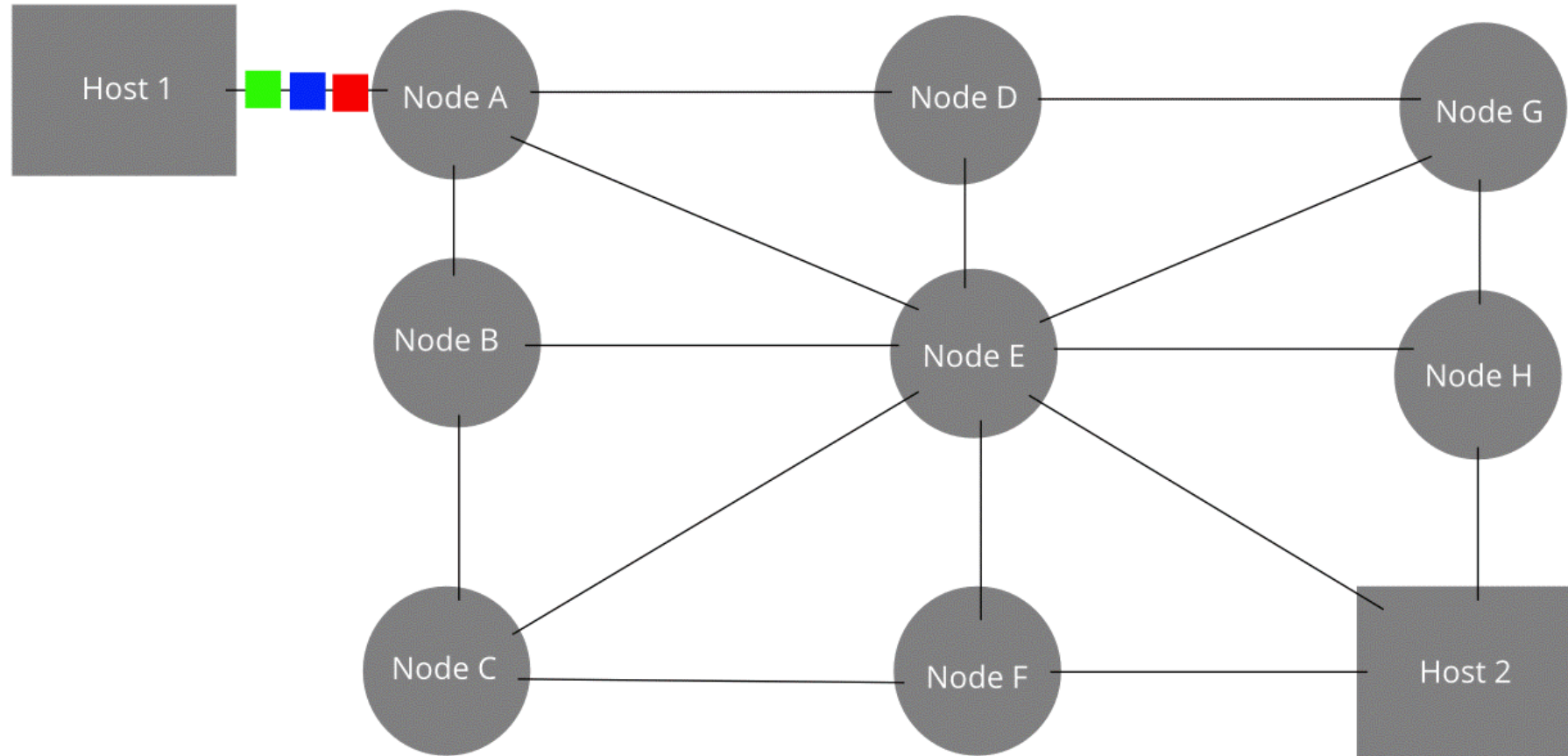
Circuit Switching





Packet Switching

The original message is Green, Blue, Red.

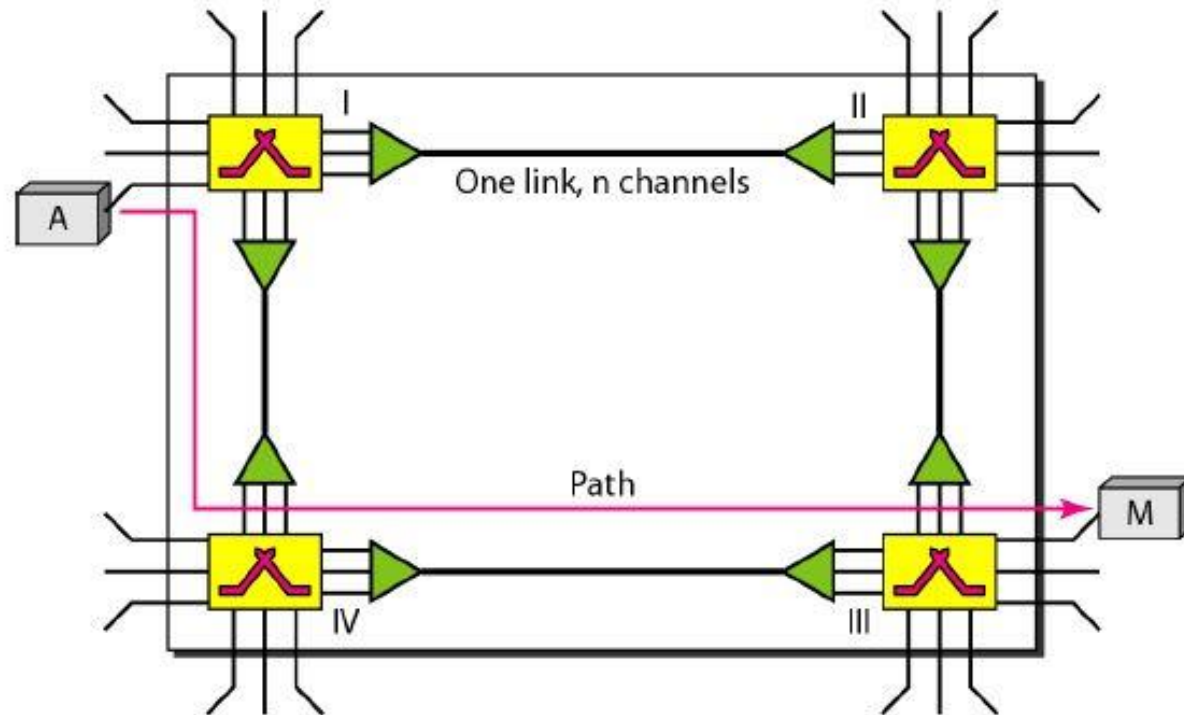


Switching and TCP/IP Layers

- Physical Layer - Circuit Switching
- Data-Link Layer - Packet Switching
- Network Layer - Packet Switching
- Application Layer - Message Switching

Circuit Switched Networks

- Made of a set of switches connected by physical links, in which each link is divided into n channels



Circuit Switched – Three Phases

- Setup Phase
- Data Transfer Phase
- Tear Down Phase

Circuit Switched - Setup Phase

- System A needs to request a connection to M that must be accepted by all switches as well as by M itself.
- Sets up a dedicated channel.
- An acknowledgment from system M needs to be sent in the opposite direction to system A.
 - Only after system A receives this acknowledgment is the connection established.

Circuit Switched - Data Transfer Phase

- After the dedicated path made of connected circuits (channels) is established, the data is transferred.

Circuit Switched - Tear Down Phase

- After all data have been transferred, the circuits are torn down.

Circuit Switched Networks

- Circuit switching happens in physical layer
- The resources need to be reserved during the setup phase.
 - The resources remain dedicated for the entire duration of data transfer until the teardown phase.

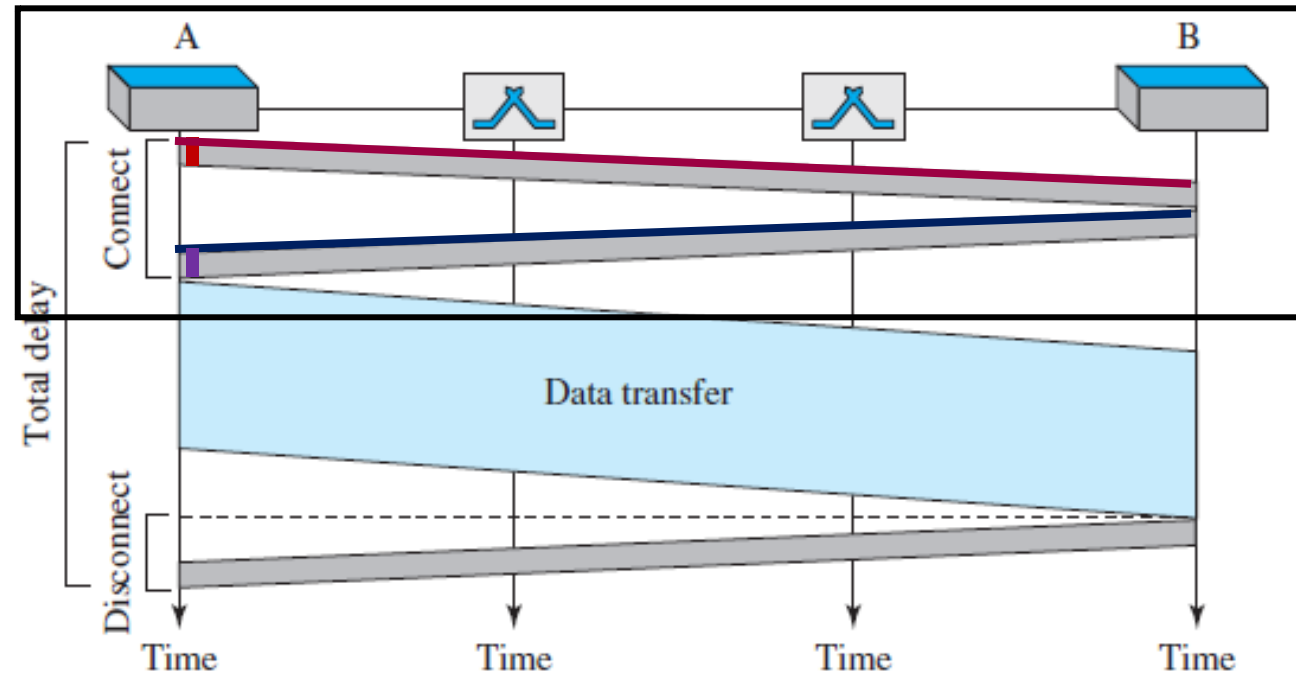
Efficiency

- Not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection.

Delay

- During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.

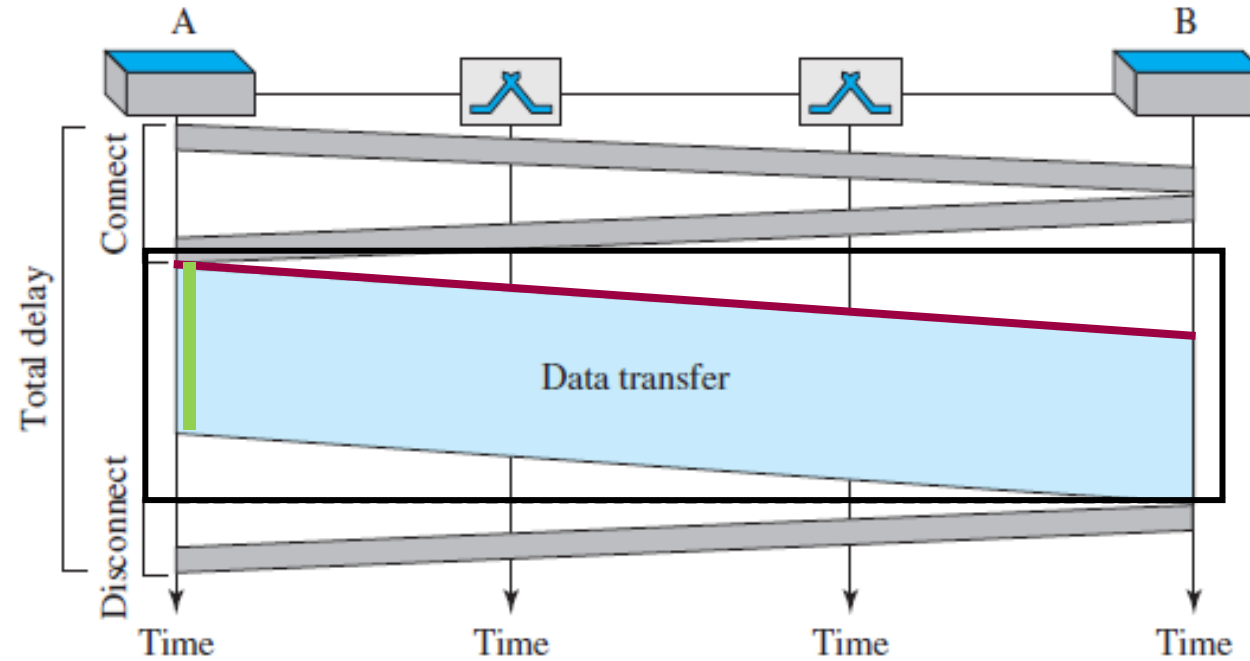
Setup Delay



Delay caused by setup = sum of four parts:

1. the propagation time of the source computer request (slope of first gray box)
2. the request signal transfer time (height of first gray box)
3. the propagation time of the acknowledgment from the destination (slope of second gray box)
4. the signal transfer time of the acknowledgment (height of second gray box)

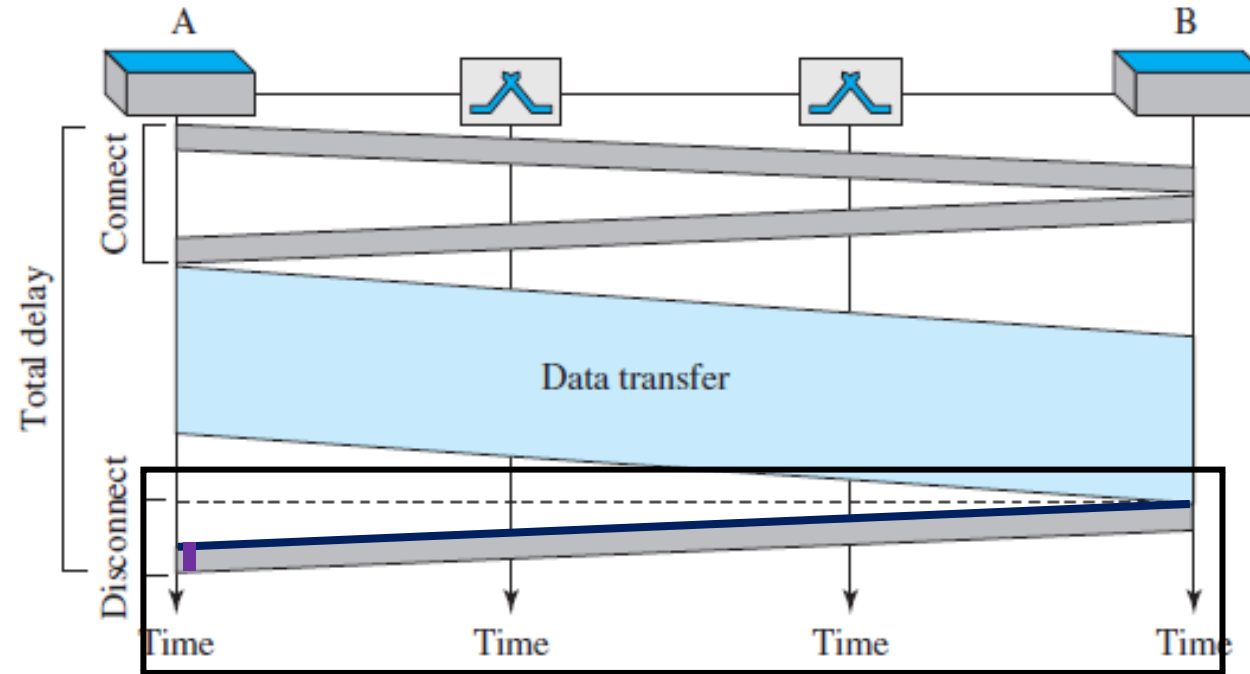
Transfer Delay



Delay caused by transfer = sum of two parts:

1. the propagation time (slope of blue box)
2. data transfer time (height of blue box)

Teardown Delay



Case given:
Receiver requests disconnection

Delay caused by teardown = sum of two parts:

1. the propagation time of tear down request (slope of third gray box)
2. tear down time (height of third gray box)

Packet Switching

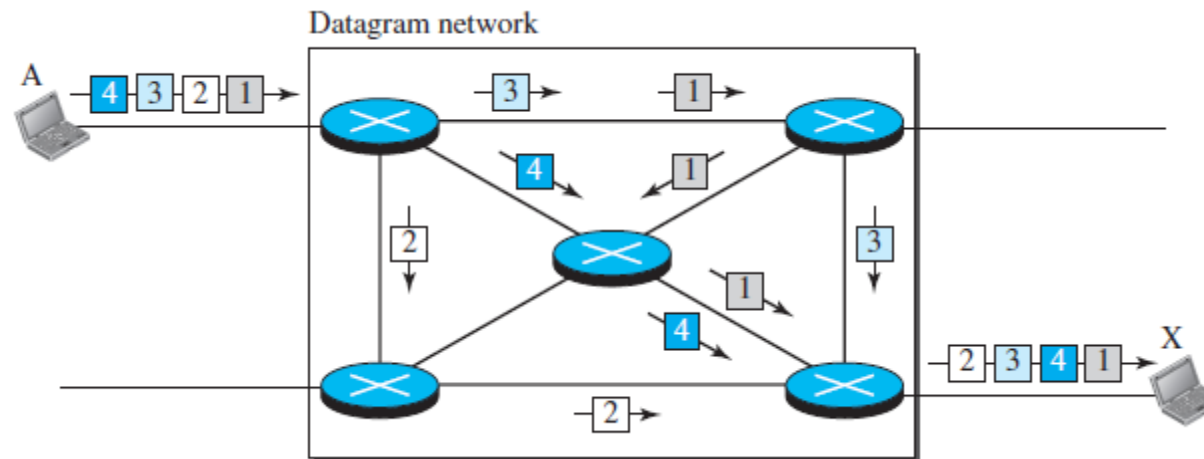
- Message needs to be divided into packets of fixed or variable size.
- There is no resource allocation for a packet. Resources are allocated on demand.
- The allocation is done on a first-come, first-served basis.

Packet Switched Network - Types

- Datagram networks
- Virtual circuit networks

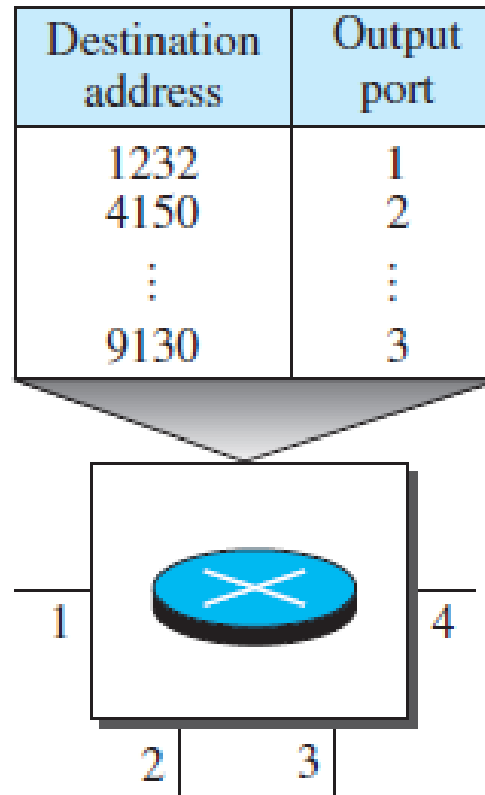
Datagram Networks

- Each packet is treated independently of all others.
 - Even if a packet is part of a multi packet transmission
- Packets in this approach are referred to as *datagrams*.
- Normally done at the network layer.



Routing Table

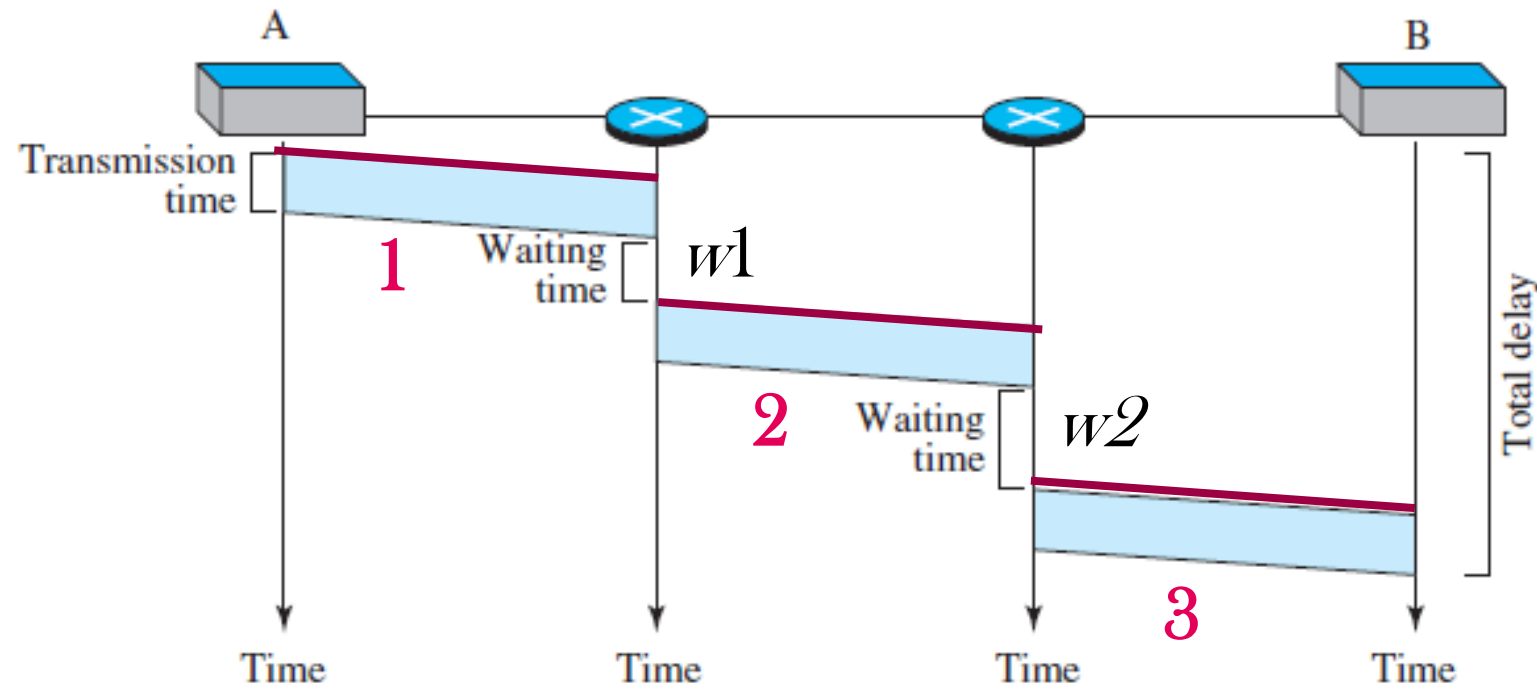
- A switch uses a routing table based on the destination address.



Efficiency

- The efficiency is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred.

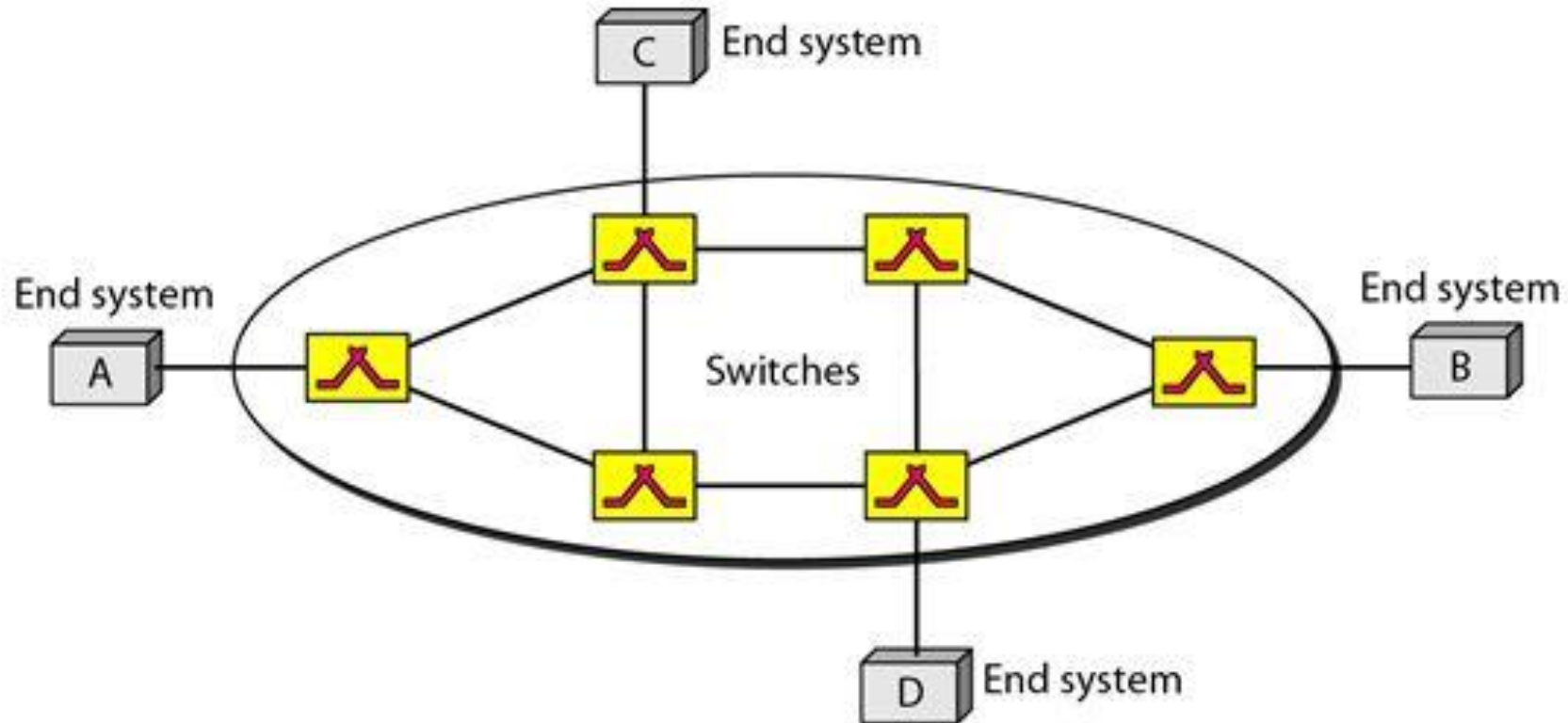
Delay



- There are **three transmission times ($3T$)**
- **three propagation delays** (slopes 3τ of the lines)
- and two waiting times ($w1 + w2$)

$$\text{Total delay} = 3T + 3\tau + w_1 + w_2$$

Virtual-Circuit Networks

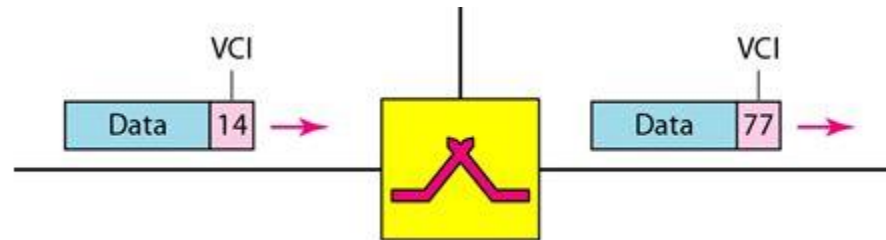


Virtual-Circuit Networks

- Cross between a circuit-switched network and a datagram network.
 - It has some characteristics of both.
- There are setup and teardown phases in addition to the data transfer phase.
- Resources can be allocated during the setup phase or on demand.
- Data are packetized and each packet carries an address in the header; the address in the header has local jurisdiction
- All packets follow the same path established during the connection.
- Normally implemented in the data-link layer.

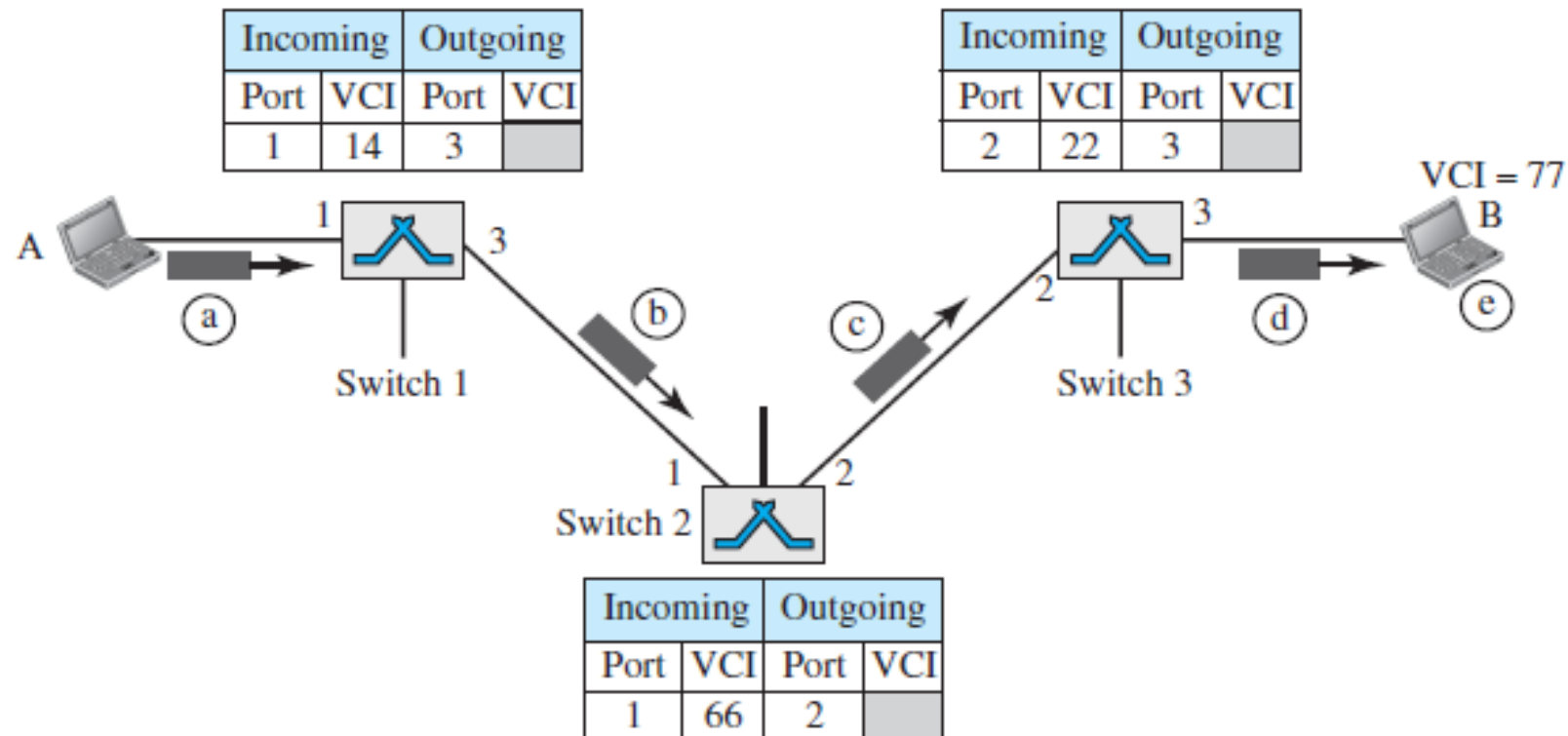
Addressing

- Global
 - A source or a destination needs to have a global address
- Local (virtual-circuit identifier)/Label
 - The one actually used for data transfer
 - A small number that has only switch scope
 - When a frame arrives at a switch, it has a VCI; when it leaves, it has a different VCI



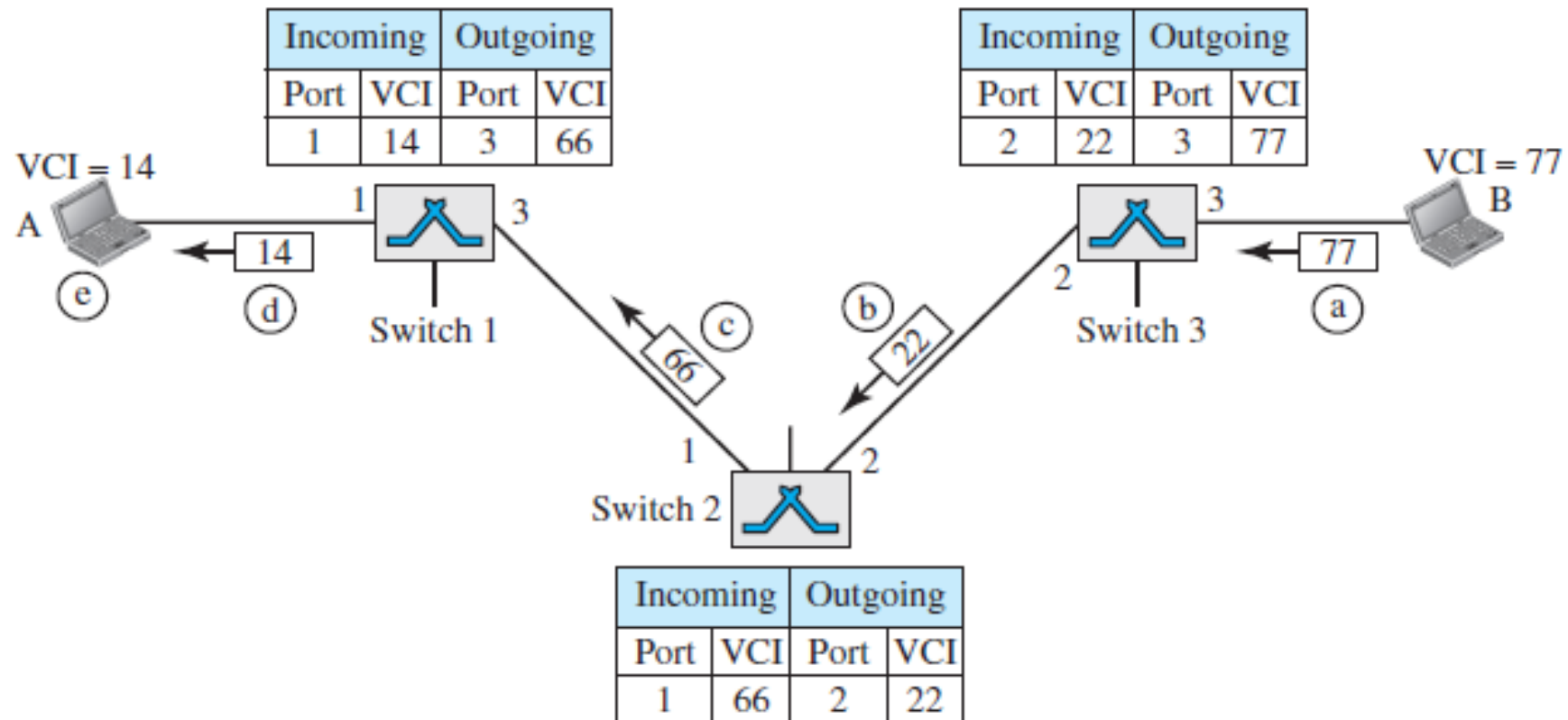
Three Phases

1. Setup Phase



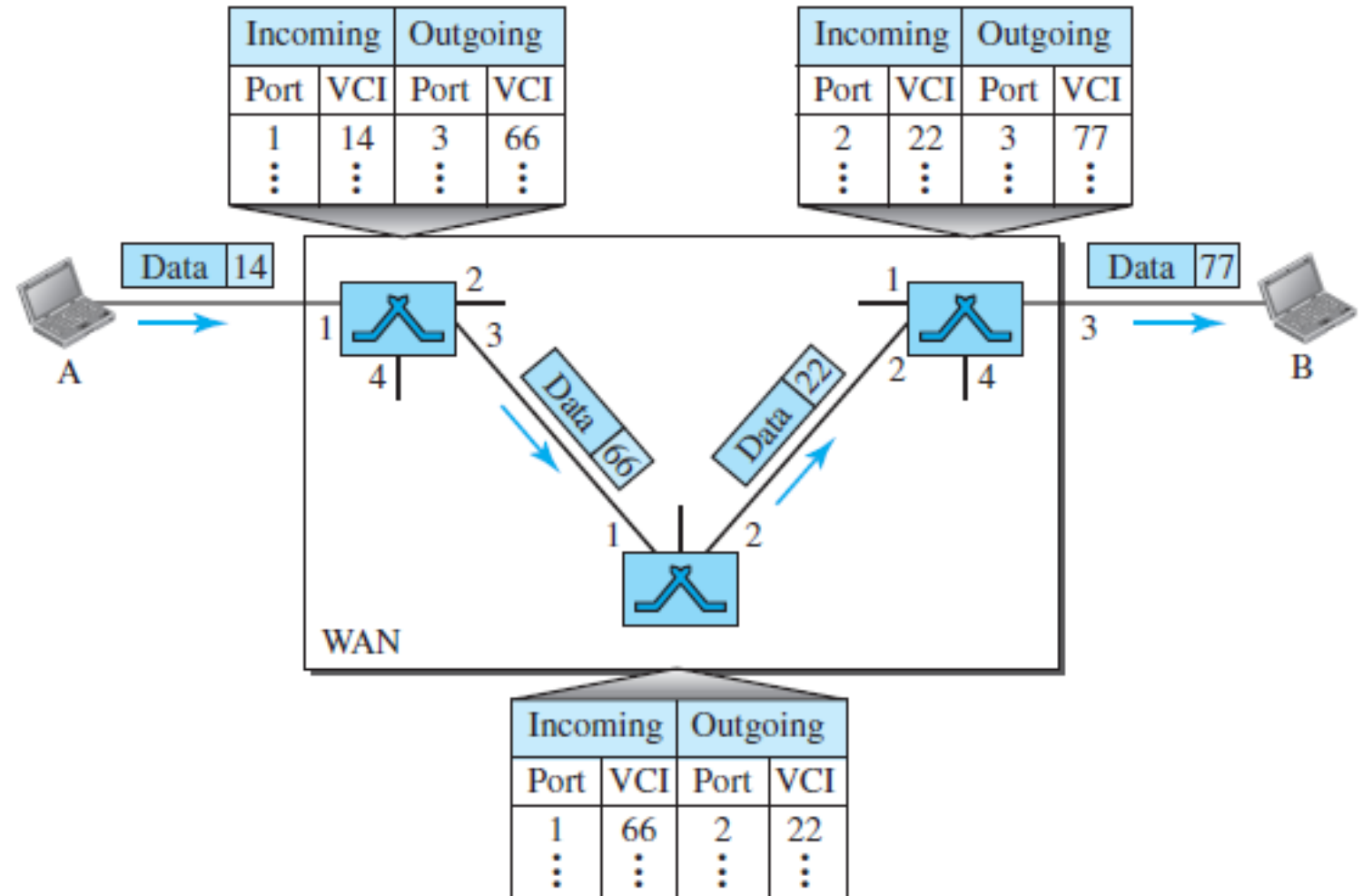
Three Phases

- Acknowledgment



Three Phases

2. Data-Transfer Phase



Three Phases

3. Teardown Phase

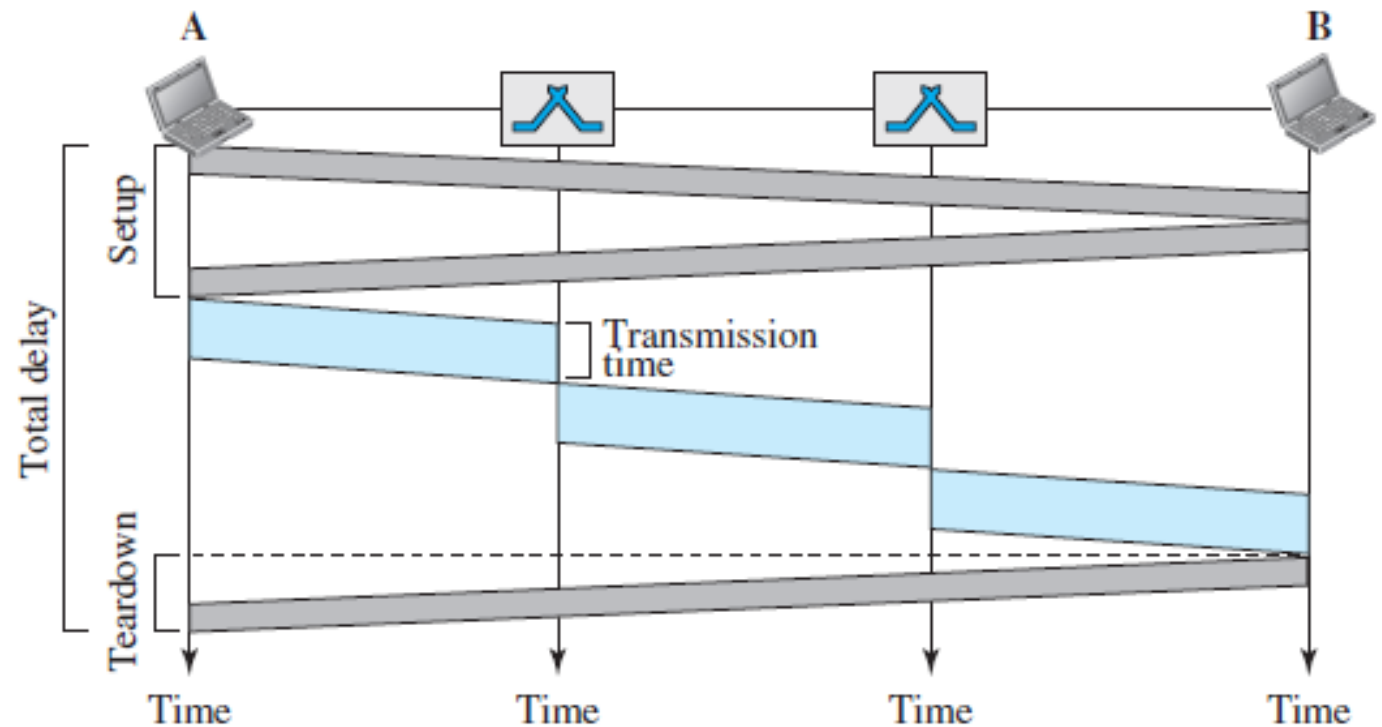
- Source A, after sending all frames to B, sends a special frame called a *teardown request*.
- Destination B responds with a teardown confirmation frame.
- All switches delete the corresponding entry from their tables.

Efficiency

- All packets belonging to the same source and destination travel the same path
- But the packets may arrive at the destination with different delays if resource allocation is on demand

Delay

- There is a one-time delay for setup and a one-time delay for teardown.
- If resources are allocated during the setup phase, there is no wait time for individual packets.



Total delay + $3T$ + 3τ + setup delay + teardown delay

Structure of Circuit Switch

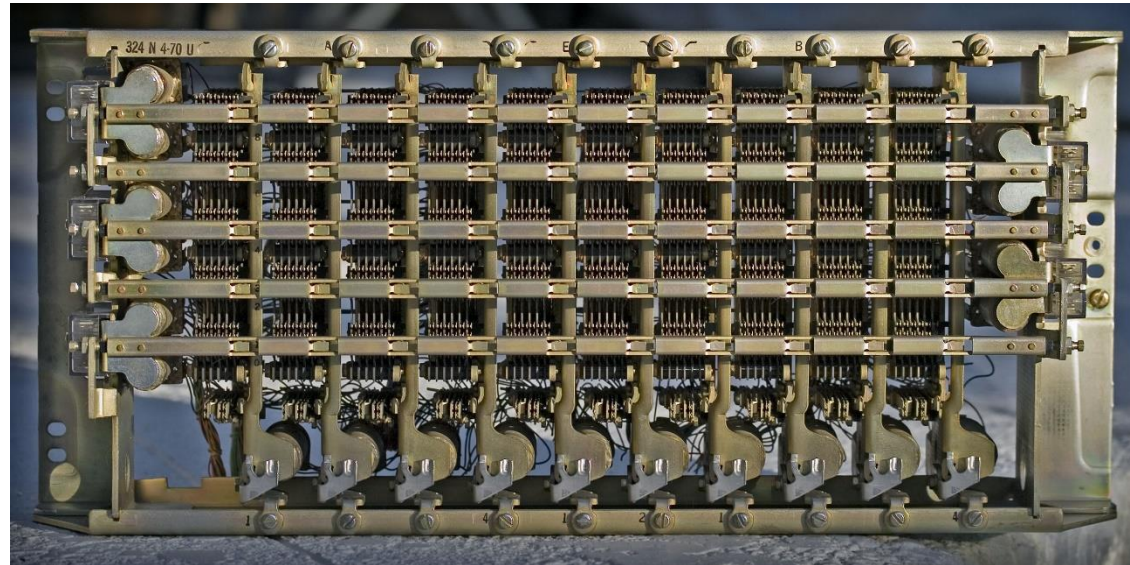
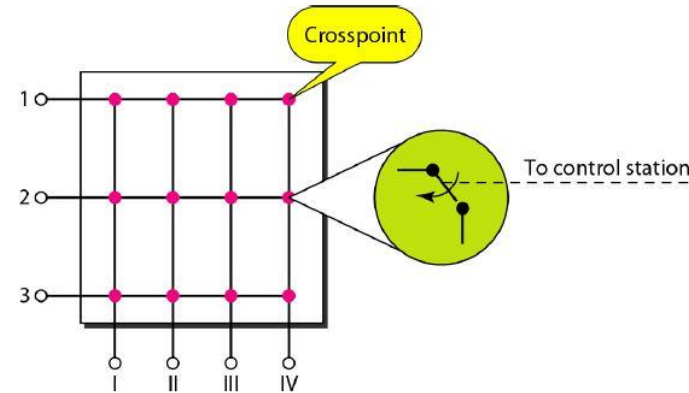
- Circuit switching today can use either of two technologies:
- the space-division switch
- (or)
- the time-division switch

Space-Division Switch

- The paths in the circuit are separated from one another spatially.
- This technology was originally designed for use in analog networks but is used currently in both analog and digital networks

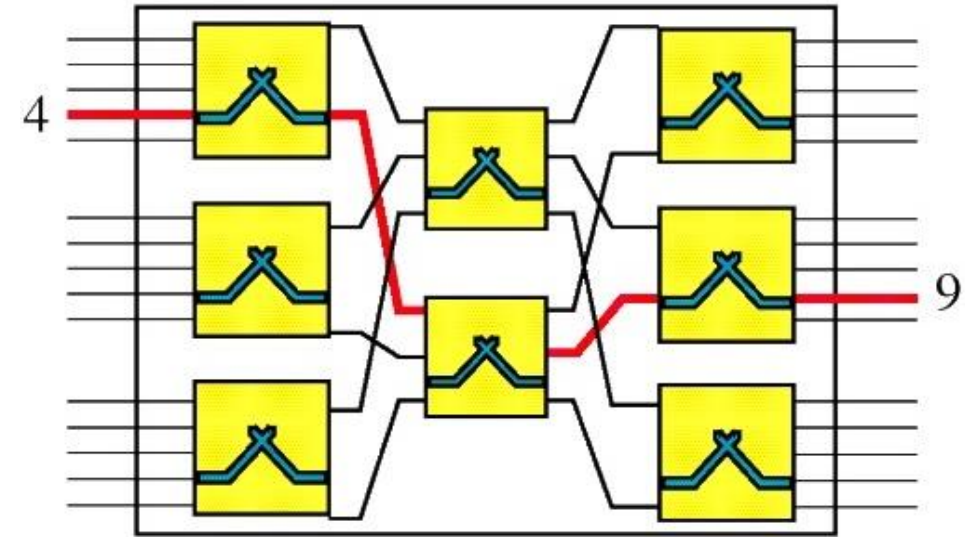
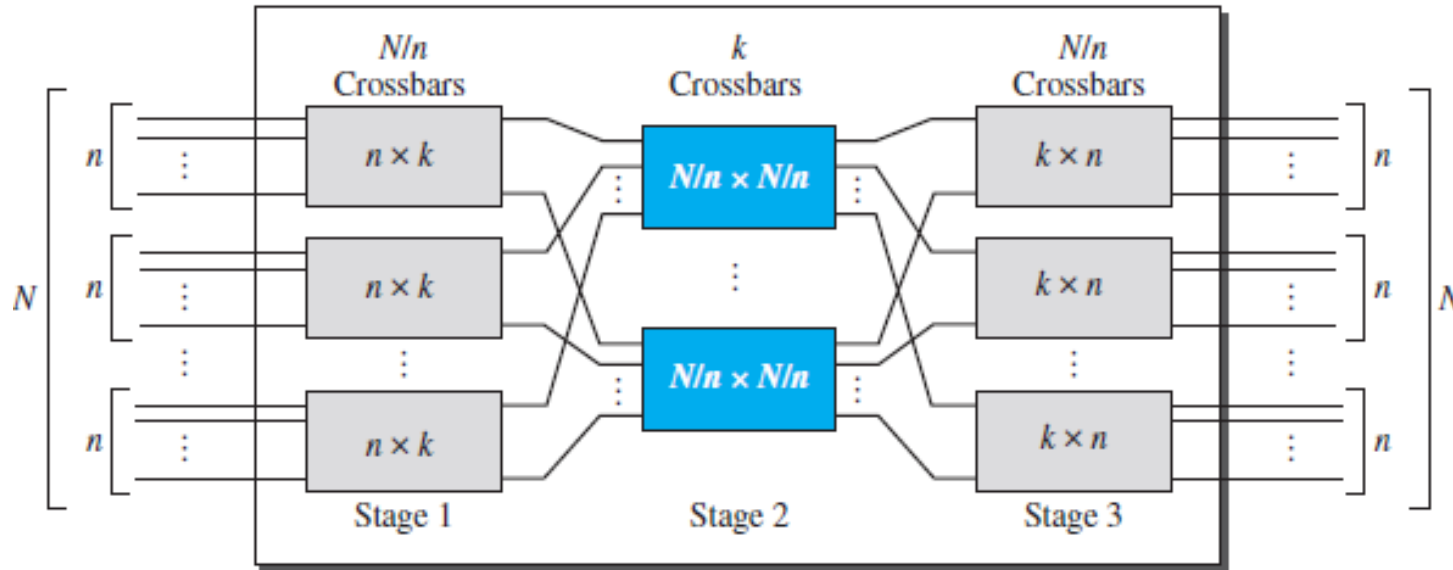
Space-Division Switch

- Cross-bar switch



Space-Division Switch

- Multistage switch



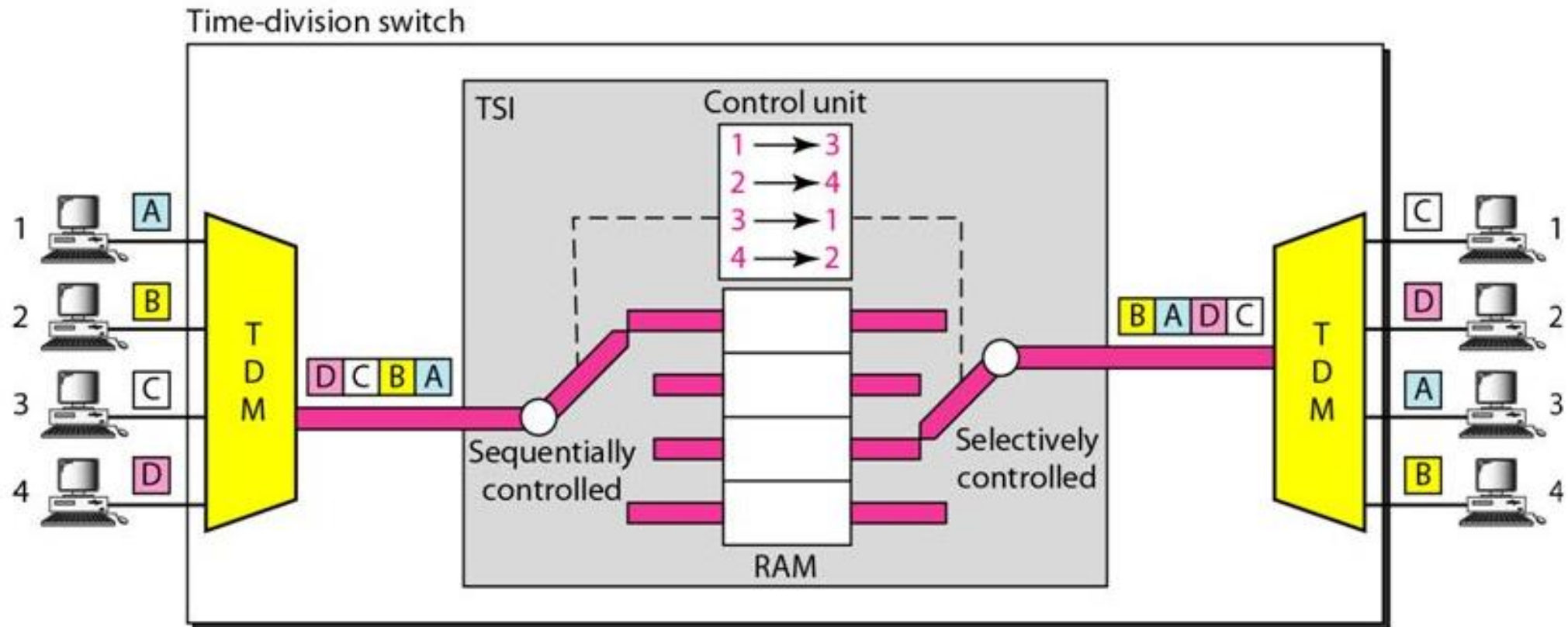
- Crosspoints:

$$\frac{N}{n} (n \times k) + k \left(\frac{N}{n} \times \frac{N}{n} \right) + \frac{N}{n} (k \times n) = 2kN + k \left(\frac{N}{n} \right)^2$$

Time-Division Switching

- **Time-division switching** uses time-division multiplexing (TDM) inside a switch.
- The most popular technology is called the **time-slot interchange (TSI)**

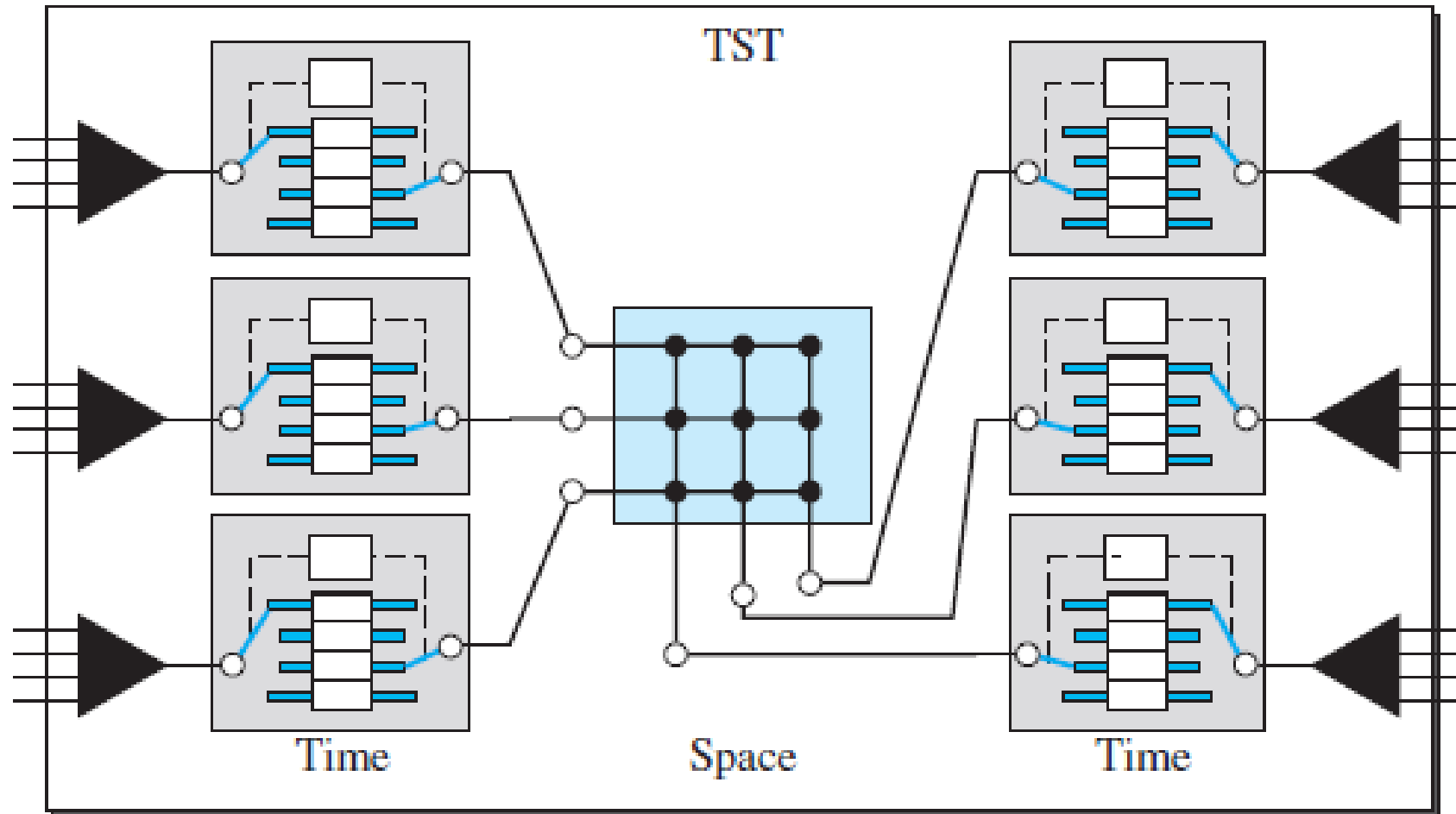
Time-Division Switching



Time- and Space-Division Switch Combinations

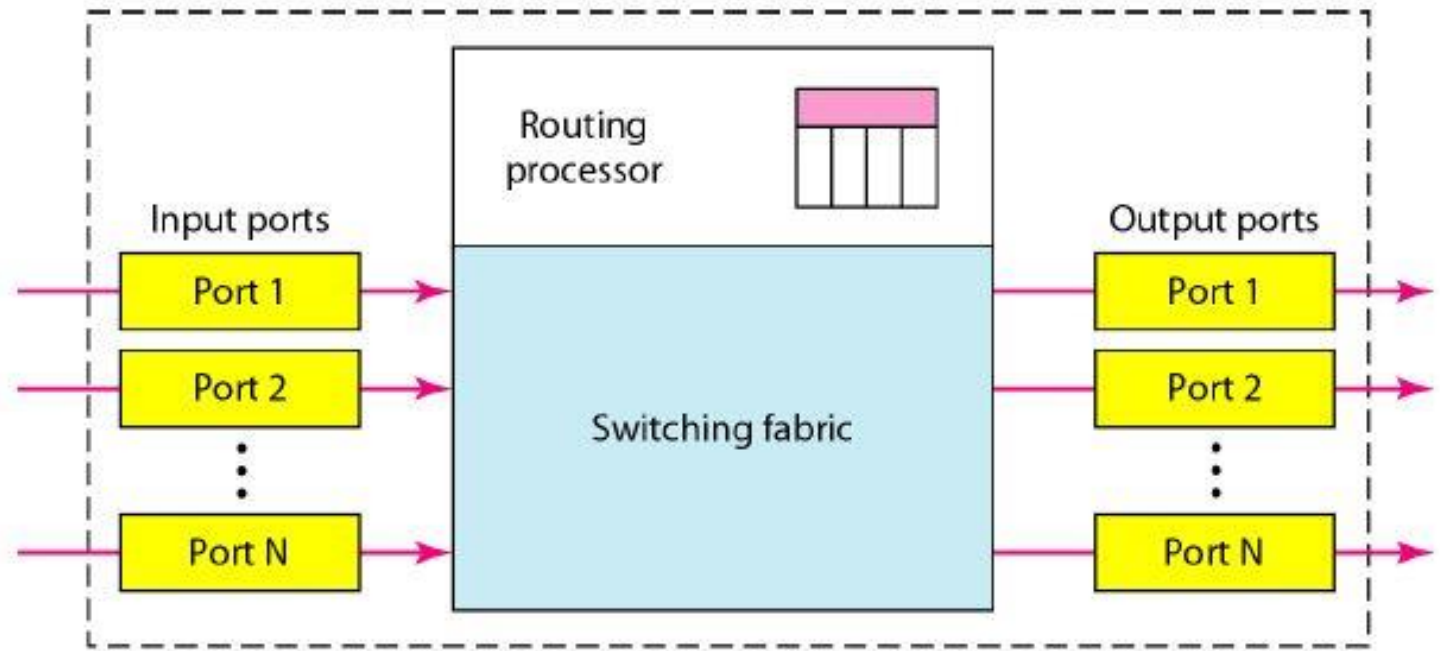
- The advantage of space-division switching is that it is instantaneous.
- Its disadvantage is the number of crosspoints required to make space-division switching acceptable in terms of blocking.
- The advantage of time-division switching is that it needs no crosspoints.
- Its disadvantage, in the case of TSI, is that processing each connection creates delays.
- Multistage switches can be designed as **time-space-time (TST)** switches.

Time- and Space-Division Switch Combinations



Structure of Packet Switch

- A packet switch has four components:
- Input ports
- Output ports
- Routing processor
- Switching fabric



Input Port

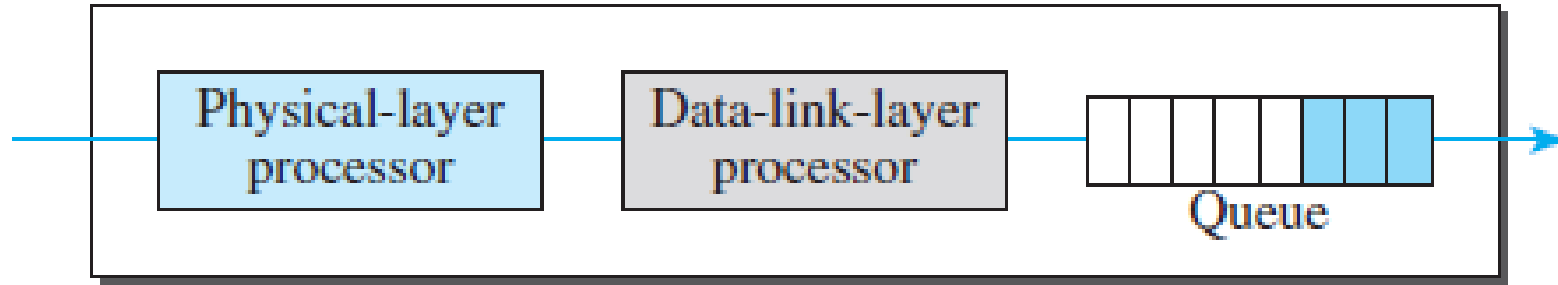
- Performs the physical and data-link functions of the packet switch.
- The bits are constructed from the received signal.
- The packet is decapsulated from the frame.
- Errors are detected and corrected.
- The packet is now ready to be routed by the network layer.

Output Port

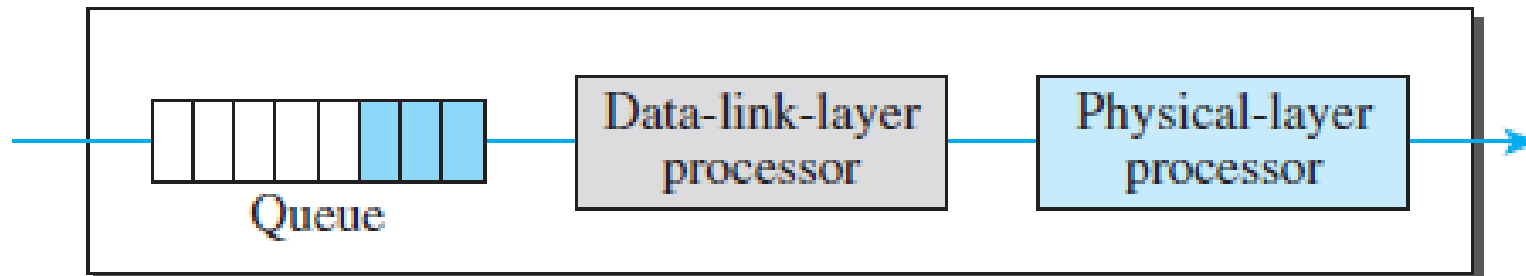
- The output port performs the same functions as the input port, but in the reverse order.

Input Port and Output Port

Input port

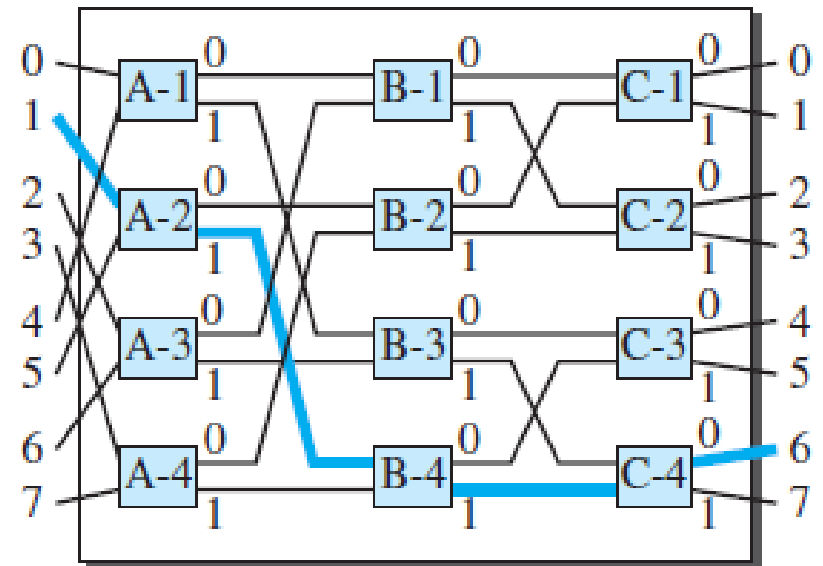
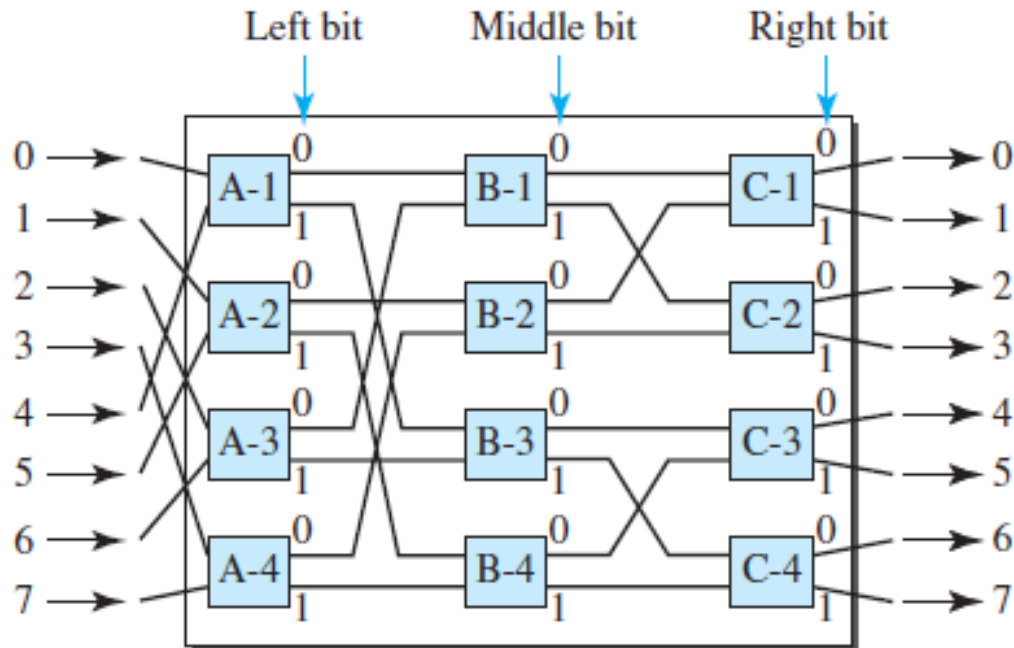


Output port



Switching Fabric

- Crossbar Switch
- Banyan Switch



a. Input 1 sending a cell to output 6 (110)

Batcher-Banyan Switch

- The problem with the banyan switch is the possibility of internal collision even when two packets are not heading for the same output port.
- Solution: Sort the arriving packets based on their destination port

